

# Who Speaks for the People?

## Electoral Reforms, Parliamentary Speech, and Legislative Behaviour in Belgium, 1880–1912

Riccardo Graziani

Jakob Vanschoonbeek

This version: October 4, 2026

### Abstract

Does extending the franchise change what a parliament attends to, and if so, does it work by changing the behaviour of the politicians already in office or by changing who holds office? We study this question in Belgium, which introduced plural male suffrage in 1894 and proportional representation in 1900, using a new dataset that links the plenary interventions of both chambers of parliament between 1880 and 1912 to the speaker's party and constituency, and to the bills that members introduced. The share of each intervention devoted to labour and social policy is measured by a transformer classifier trained for this corpus and validated against blind human coding. Because its eligibility census prevented working-class candidates from entering the Senate, the difference between the two chambers identifies the Chamber's differential response to electoral exposure, net of shocks common to both chambers, such as the strikes and unrest that produced the reforms. After enfranchisement, the Chamber's attention to the social question rose by 4.5 percentage points relative to the Senate's, on a base of 7.9, everywhere in the country and mainly among sitting deputies. Proportional representation produced no aggregate shift but a large effect in the coalfields, where it reopened seats the majoritarian rule had handed to the socialists. It was driven by the entry of new members. Social bills deposited by Catholic and Liberal members rose after 1894, mostly among Catholics, and again after 1900, among Liberals. Enfranchisement thus raised attention to the social question across the country, while proportional representation restored it in the coalfields through the members it elected. Which channel operated was set by the part of the electoral system each reform changed.

**Keywords:** democratisation, electoral reform, political representation, parliamentary speech, text as data

**JEL:** D72, N43, P16

# 1 Introduction

Representative government rests on the claim that those who hold office attend to the concerns of those they represent (Dahl, 1971; Pitkin, 1967). In the political economy of the franchise, extending the vote down the income distribution makes the median voter poorer and should therefore move policy towards redistribution (Meltzer & Richard, 1981). Since enfranchisement threatened redistribution, the literature has asked why European elites conceded the vote in the nineteenth century at all: because the alternative was worse (Acemoglu & Robinson, 2000, 2006; Aidt & Jensen, 2014; Boix, 2003) or because a wider electorate served ends of their own (Lizzeri & Persico, 2004). Yet the predicted redistribution did not always follow: franchise extensions raised public spending in some countries and periods but not in others (Aidt et al., 2006, 2010; Lindert, 2004). In contemporary democracies, policy and party platforms follow the preferences of the affluent more closely than those of other citizens. This holds in the United States (Bartels, 2008; Gilens, 2012; Gilens & Page, 2014) and in countries with far more compressed income distributions (Elsässer et al., 2021; Schakel & Burgoon, 2022). The literature on representation locates this shortfall in responsiveness at the link between the demands of constituents and the agenda of the assembly (Bevan & Jennings, 2014; Jones & Baumgartner, 2004; Miller & Stokes, 1963). Whether members of parliament change how they allocate their time when the electorate changes is rarely measured directly.

In Belgium in the 1880s the shortfall was large. The first economy on the European continent to industrialise, Belgium had a large industrial working class in the coal basins of Hainaut and Liège and the textile towns of Flanders (Vanthemsche & De Peuter, 2023; Witte et al., 2009). The franchise, however, was tied to a minimum of direct tax, and in 1892 it admitted 136,707 men in a population of about six million (Kollman et al., 2025; Stengers, 2004). Parliament was contested by two elite parties, Catholic and Liberal, divided over the role of the Church in public life and above all in schools (Lipset & Rokkan, 1967; Witte et al., 2016). The Belgian Workers' Party, founded in 1885, made the franchise its first demand. Its general strike of April 1893 ended within a week, when parliament conceded plural male suffrage (Brepoels, 2015; Polasky, 1992; Witte et al., 2016). The incumbents drafted its terms so that the concession would cost them as little as possible. They added supplementary votes for property and education and kept the eligibility census that the 1831 Constitution had set for the Senate, lowering it but still excluding working-class candidates (Gilissen, 1958; Luykx & Platel, 1985). The socialists nonetheless entered the Chamber in 1894, and in 1899 the incumbents turned to the electoral rule, adopting proportional representation in part to contain them (Ahmed, 2010; Emmenegger & Walter, 2019).

The social question, as contemporaries called it, was the condition of the industrial working class as a problem for the state: hours and wages, the work of women and

children, housing, and insurance against accident, sickness and old age (Flora & Heidenheimer, 1981; Lindert, 2004). It entered Belgian politics through the riots and strikes of March 1886, which the army put down and which produced the country’s first labour legislation (Deneckere, 1993; Strikwerda, 1988). The social question was central to the programme of the Workers’ Party (Bertrand, 1907; Polasky, 1992) and, after the encyclical *Rerum Novarum* of 1891, to that of a Christian-democratic wing of the Catholic party (Strikwerda, 1997; Witte et al., 2016).

What had not been measured was the attention the social question received in parliament. Belgium kept a verbatim record of every plenary sitting in both chambers. The share of floor speech that each representative devoted to the regulation of work, welfare and public assistance, and public health is therefore observable for every member and year, before and after each reform. We use this share to measure parliamentary attention to the social question, and its change across the reforms to measure how that attention responded. The share measures attention, not policy: in the literature on responsiveness, attention is the step between citizens’ priorities and what governments do (Bevan & Jennings, 2014; Jones & Baumgartner, 2004). Parliamentary speech has been used before to study electoral reform. Spirling (2016) shows that Britain’s Second Reform Act changed the form of speech in the House of Commons, and Osnabrügge et al. (2023) that New Zealand’s move to mixed-member proportional representation shifted its topics. We study how the content of speech responds to a franchise extension and to a change of electoral rule in the same parliament.

Belgium is unusually well suited to this question. The two reforms came six years apart, to the same parliament, and each changed a different part of the electoral system. Plural suffrage changed who could vote, everywhere and at once, but left the majoritarian electoral formula in place. Proportional representation replaced that formula without changing the electorate, and the change mattered only where the majoritarian rule had denied seats to a large minority (Boix, 1999; Cox, 1997; Stengers, 2004). From 1894 onward, senators were elected by the Chamber’s voters aged thirty or more, but the eligibility census for the Senate remained. Fewer than 1,700 men in a population above six million could stand for the Senate, and socialists struggled to win Senate seats even under plural suffrage (Luykx & Platel, 1985). The constitutional revision that introduced plural suffrage also made voting compulsory, so politicians could not count on the newly enfranchised staying away from the polls (Hooghe & Stiers, 2025; Witte et al., 2016). Socialist strength, finally, was geographically concentrated. Because the majoritarian formula awarded all of a district’s seats to a single list, whether by absolute majority or in a run-off, socialist victories shut the established parties out of the four coalfield districts, and proportional representation could change who was elected there. As in other industrial democracies, the socialist vote was concentrated where industrial workers lived (Rodden, 2019), and in Belgium it was strongest in the coal basins. Mining employ-

ment therefore marks where the majoritarian rule bound, and the 1896 industrial census records it for every district (Office du travail de Belgique, 1900–1902), three years before proportional representation was adopted.

We ask whether democratisation increased the attention parliament gave to the social question and, if so, through which channel. A reform can change the behaviour of incumbent members or the composition of parliament. Lee et al. (2004) call these the affect and elect channels and find, for the US House, that voters elect rather than affect policy. We call them adaptation and recomposition, and argue that which one operates depends on the part of the electoral system that a reform changes, and two reforms within the same political system can therefore work through different channels. In the framework of Section 3, a member’s electoral exposure to the social question is the product of who can vote, how the electoral formula allocates seats and who may stand, so a reform that moves one term changes attention only where the other two are open. Politician fixed effects and entry cohorts separate the two channels within a reform. Showing that the channel depends on what a reform changes, however, requires reforms to different parts of the same electoral system, and the Belgian sequence offers two such reforms.

We build a new dataset from the plenary record of both chambers: 553,958 interventions, of which 460,232 are dated 1880–1912 and can be attributed to a chamber (Table 1). Each intervention is linked to the speaker’s identity, party and constituency through the parliamentary archives’ own person identifiers. We add the 4,779 bills deposited in the same period, with their sponsor, subject and legislative fate, and the 1896 census measure of industrial structure for each constituency. We then measure attention with a French-language transformer model (Martin et al., 2020), fine-tuned on annotated Belgian parliamentary text, that assigns each intervention a probability distribution over six policy domains and a procedural residual. The classifier reads each intervention in context, so speeches on mine safety and on mining royalties fall in different domains although they share a vocabulary.<sup>1</sup>

To separate the reforms from everything else that changed at the same time, we compare the two chambers. The Chamber of Representatives and the Senate faced the same reforms, strikes and doctrinal shifts, and from 1894 their directly elected members were chosen by essentially the same voters. The difference that matters is who could stand for election. The reforms exposed deputies to socialist competition, while the Senate’s eligibility census, which excluded working-class candidates, left senators largely insulated from it. The change in the Chamber–Senate difference across each reform therefore removes the common shocks that a simple before-and-after comparison would attribute to the reform. It identifies the differential response of the exposed chamber. If the Senate’s response is non-negative, this is a lower bound on the Chamber’s own response.

---

<sup>1</sup>Section 5 describes the construction of the classifier, gives examples and reports its accuracy against blind human coding (0.86, with Cohen’s  $\kappa$  of 0.80; Table 4).

The estimation sample contains Catholic and Liberal members only. The socialists held no seats before 1894 and few in the Senate afterwards, so their speech offers neither a pre-reform baseline nor a counterpart in the comparison chamber, and including it would count the arrival of a new party as a change in attention. Nevertheless, socialist speech remains in the corpus and is used to characterise the agenda of each sitting day.

After plural suffrage the Chamber’s labour salience rose by 4.5 percentage points relative to the Senate’s (s.e. 2.1; wild-bootstrap  $p = 0.036$ ; Appendix Table A14), on a pre-reform base of 7.9. The effect is spread across the country: it remains positive when any single district is dropped and significant at the 5 per cent level in 38 of the 41 exclusions (Appendix Table A13). Most of it comes from adaptation. Among the members observed on both sides of the reform, labour salience rose from 7.9 to 15.3 per cent of substantive speech for the 179 deputies and from 8.3 to 12.0 for the 105 senators, a difference of 3.7 points; with politician fixed effects the estimate is 3.2 (s.e. 2.5; Table 9, Panel B). The deputies who entered in 1894 were more labour-oriented than the incumbents they joined, but they delivered only 13.5 per cent of Chamber speech in the majoritarian legislatures and cannot account for the aggregate change. Nor is the estimate driven by ministerial speech. The socialists who entered the Chamber in 1894 pushed labour onto its agenda, and part of the raw premium runs through their presence: conditioning on their share of the day’s words leaves 1.5 (s.e. 2.0; Table 18). Adaptation does not depend on that presence, however. On sitting days when members outside the two established parties did not raise labour, the estimate with politician fixed effects is 3.0 ( $t = 2.30$ ; same table).

Proportional representation produced no aggregate shift in the Chamber–Senate difference (1.5, standard error 3.1; Table 11) but a steep geographic gradient. The triple difference interacting mining intensity with the post-1900 and Chamber indicators is 9.3, with a two-sided randomization-inference  $p$ -value of 0.035 over 4,999 permutations of the district exposure map (Table 16). The post-1900 Chamber–Senate gap is  $-0.8$  outside the four coalfield districts and 13.4 inside them (lower panel of the same table). This effect is a result of recomposition. Deputies who entered at or after 1900 carry a mining gradient of 11.1, against 0.6 for the incumbents they joined on the same window (Appendix Table A22), and absorbing politician fixed effects turns the triple difference negative and insignificant ( $-3.5$ , s.e. 2.1; Table 11, column 6). By entry cohort, the gradient is flat for members who entered under the censitary and majoritarian rules and appears for those who entered under proportional representation, while socialist electoral strength grew steadily throughout. The gradient steps up with the change of rule (15.1, s.e. 3.2; Online Appendix Table OA18), although within-regime tests cannot rule out a smaller drift.

The coalfields link the two results. Plural suffrage handed the seats of Mons, Charleroi, Soignies and Liège to the socialists, and the coalfield share of Catholic and Liberal

deputies with recorded speech fell from 15.3 per cent in 1892 to 5.2 per cent in 1898 (Table 10). The 1894 effect is accordingly absent in the coalfields ( $-1.4$ , s.e.  $1.8$ , against  $6.2$  elsewhere), because the deputies who spoke had changed, not because individual deputies spoke less about labour. Proportional representation broke the socialist monopoly on coalfield seats, and the members elected there after 1900 were more labour-oriented than the members already sitting there. The second reform, adopted to contain the socialists, thus restored the established parties' attention to the social question in the coalfields, through the members it elected there.

Members' bills on the social question follow the same timing and the same partisan split as their speech. By the archive's own subject classification, labour and social bills deposited by Catholic and Liberal members rise from 5.2 a year before 1894 to 14.5 in 1894–99 and 16.5 after 1900 (Table 14), at an unchanged rate of success. The rise at 1894 comes from Catholic members and the further rise after 1900 from Liberals. This evidence is descriptive and identified differently from the speech results, so we treat it only as corroboration.

The partisan pattern differs across the two reforms, in line with the political history. The 1894 effect comes from Catholic members, since the party's Christian-democratic wing had the doctrine and the organisation to compete for working-class votes, whereas the Liberals, an urban and bourgeois party, had neither and lost heavily in 1894 (Delwit, 2017; Strikwerda, 1997). The 1900 effect is precisely estimated only for Liberals, the party to which proportional representation reopened the industrial districts from which it had been shut out. Appendix OA8 decomposes each party's change exactly into the contributions of continuing members, entrants and leavers.

The paper contributes to four literatures. Work on democratisation and redistribution explains why elites extend the franchise, who is elected afterwards (Berlinski & Dewan, 2011) and how this affects taxes and spending. It says less about whether sitting representatives change their priorities, which we observe in how they allocate their speech. Work on political selection and incentives (Besley, 2005; Gagliarducci & Nannicini, 2013; Gagliarducci et al., 2011; Lee et al., 2004) asks whether institutions act through the behaviour of sitting politicians or through who is selected. In our setting the same parliament gives one answer under one reform and the other under the next. Work on electoral systems studies why countries chose their rules and what those rules do (Boix, 1999; Cusack et al., 2007; Iversen & Soskice, 2006; Persson & Tabellini, 2003). Belgium changed the electorate and the rule that turns votes into seats in separate steps, and because the geographic concentration of the left made the majoritarian rule bind (Rodden, 2019), a census identifies the districts where the change of rule mattered.

Finally, for text-as-data work in political economy (Gennaro & Ash, 2022; Gentzkow et al., 2019; Grimmer & Stewart, 2013; Osnabrügge et al., 2023), we provide a new historical dataset: three decades of plenary interventions in both chambers of a national

parliament, linked to speakers and to members' bills and classified by policy domain with a classifier validated against blind human coding. The setting also supplies a comparison group that studies of a single chamber cannot use: an upper chamber that faced the same reforms and shocks as the lower chamber but not the same electoral competition for its seats. If the Senate's response is non-negative, the difference between the chambers gives a lower bound on the effect in the more exposed lower chamber.

Two limits on the interpretation apply throughout. First, we observe what representatives attended to, not what citizens wanted, because no survey measures public priorities in this period. The gap we document is therefore between a demand whose salience the historical record establishes and the attention parliament gave it. Second, the design measures how much more the Chamber responded than the Senate. Because senators responded too, this is less than the Chamber's whole response.

## 2 Historical Background

### 2.1 The social question and the pressure for inclusion

The first franchise under the Belgian Parliament was censitary. This had also been the case under Dutch rule before independence in 1830, and the Constitution of 1831 retained the principle: men could vote if they paid a minimum amount of direct tax (Gilissen, 1958; Stengers, 2004).

The industrial working class of the coal basins of Hainaut and Liège and the textile towns of Flanders had no political voice (Vanthemsche & De Peuter, 2023; Witte et al., 2009). National politics was dominated by competition between two elite parties, Catholic and Liberal. They were divided over the place of the Church in public life, above all in schooling (Lipset & Rokkan, 1967; Witte et al., 2016), and largely united in their attachment to a competitive, low-wage export economy (Witte et al., 2016). Parliament discussed the condition of the working class as the social question, but the men who discussed it did not answer to workers at the ballot box (Flora & Heidenheimer, 1981; Strikwerda, 1988).

By the last two decades of the nineteenth century, the working class had begun to organize and exert political pressure for better working conditions and greater political rights. In March 1886, riots and strikes spread across the Walloon industrial basin, from Liège to the Borinage, and were suppressed by the army (Deneckere, 1993). The unrest was followed by the first wave of Belgian social legislation, including a law in 1887 regulating the payment of wages and a law in 1889 restricting the employment of women and children (Stengers, 2004; Strikwerda, 1988; Witte et al., 2016). The Belgian Workers' Party (Parti Ouvrier Belge, POB) had been founded in 1885 as a federation of trade unions, cooperatives, and mutual societies. The party was reformist, and its

programme placed universal suffrage at the center of its political demands, reflecting the view that political representation would allow workers to pursue social reform through parliament (Bertrand, 1907; Polasky, 1992; Strikwerda, 1988). To press for an extension of the franchise, the socialists used the general strike as a political bargaining tool to put pressure on the government (Brepoels, 2015; Brepoels & Hoekx, 2019). In particular, the general strike of April 1893 followed the Chamber’s rejection of a proposal to extend the suffrage. It mobilized around 200,000 workers and ended within a week, as parliament adopted a compromise introducing universal male suffrage with plural voting (Polasky, 1992; Witte et al., 2016).

Pressure for improvements in working-class conditions also emerged from within the Catholic political camp. The encyclical *Rerum Novarum*, promulgated by Pope Leo XIII in May 1891, affirmed a role for the state in protecting workers and endorsed the formation of Catholic workers’ associations. In this context, the encyclical provided the Christian-democratic wing of the Belgian Catholic movement with a doctrinal basis for addressing the social question and competing for working-class support (Strikwerda, 1997; Witte et al., 2016). The *Ligue démocratique belge*, founded in 1891 to coordinate Catholic workers’ associations, subsequently entered electoral politics after the 1893 electoral reform (Strikwerda, 1997; Witte et al., 2016). From the early 1890s, therefore, the Catholic movement contained a Christian-democratic current with both the organizational capacity and the doctrine to address the social question, alongside a more conservative current that remained opposed to such policies (Strikwerda, 1997). A more radical Christian-democratic movement emerged independently of the Catholic Party in 1893 with the foundation of the Christian People’s Party, associated with the priest Adolf Daens, which contested elections from 1894 (Strikwerda, 1997; Witte et al., 2016). On the liberal and secular side, the Liberal Party, whose support was concentrated largely among urban and bourgeois constituencies, also contained a progressive current that favored extending the suffrage, but lacked a comparable organizational base among workers (Delwit, 2017; Witte et al., 2016).

## **2.2 The 1893 revision: plural suffrage, compulsory voting and the Senate**

The constitutional revision of 1893 gave the vote to every male citizen aged twenty-five, and the electorate rose from 136,707 to 1.37 million (Kollman et al., 2025; Witte et al., 2016). Nevertheless, two features limited the effect of the electoral reform. The first was plural voting. Fathers of families aged thirty-five or older who paid a minimum amount of personal tax received one supplementary vote. Besides, property owners and savers whose holdings exceeded a specified threshold were also entitled to one supplementary vote, while men with certain educational qualifications received two. No man could

cast more than three votes (Luykx & Platel, 1985, p. 195). The propertied classes thus retained a large share of the votes cast (Gilissen, 1958; Witte et al., 2016). The second was compulsory voting. Its authors feared that voters of the two established parties, particularly those who had already held the franchise before the reform, would abstain as the working class entered the electorate, leaving the field to the Workers' Party (Bouhon, 2018).

The reform was first applied at the general election of 14 October 1894 (Stengers, 2004). The Belgian Workers' Party entered the Chamber with 28 seats, ahead of the Liberals' 20, and won most of them in the four coalfield districts (Delwit, 2017; Van Molle, 1972; Witte et al., 2016). Until 1899 seats were allocated by a majoritarian list rule: a list that won an absolute majority of the votes in a district took every seat, and if no list did, a run-off between the two leading lists decided (Gilissen, 1958; Stengers, 2004). A socialist victory in Mons, Charleroi, Soignies, or Liège, whether outright or in the run-off, therefore returned an entirely socialist delegation, leaving Catholic and Liberal voters in these districts without a representative.

For the Senate, the 1893 revision left an important institutional difference. From 1894, directly elected senators were chosen by the same voters who elected deputies (Luykx & Platel, 1985, pp. 195–196)<sup>2</sup>. The revision nevertheless left the senatorial eligibility census largely intact. Candidates still had to pay a minimum amount of direct tax, lowered from 2,116 to 1,200 francs, or own real estate with a cadastral income of at least 12,000 francs. As a result, the pool of men eligible to stand for the Senate remained small, rising from 996 in 1890 to only 1,527 in 1900 and 1,637 in 1910, in a population of more than six million (Luykx & Platel, 1985, pp. 195–196). The reform also created provincial senators, elected by the provincial councils and exempt from the census, while retaining the senatorial mandate of eight years with staggered partial renewals (Luykx & Platel, 1985; Popelier, 2018). Luykx and Platel conclude from these figures that the pool of eligible candidates remained small, helping to explain why socialists struggled to win Senate seats even after the introduction of plural suffrage.

The Workers' Party won its first directly elected Senate seat in 1900, for Charleroi–Thuin, and the man it returned, Auguste Houzeau de Lehaie, was a former Liberal deputy, senator for Mons since 1897, who met the census (Dumoulin, 2024).<sup>3</sup> By 1910 the Senate held 11 socialist seats of 110, against 35 of 166 in the Chamber (Online Appendix Table OA1). The insulation lay on the supply side, in who could stand, and it eroded slowly. Section 3 uses this to predict when the Senate should stop serving as a control, and Section 8 shows that it does stop then.

Sharing an electorate also made the two chambers more alike in political composition,

<sup>2</sup>Except that Senate voters had to be at least thirty years old (Luykx & Platel, 1985, pp. 195–196)

<sup>3</sup>The Senate's historical dossier dates the party's first directly elected senator to 1904 (Sénat de Belgique, 2019). The lists of senators by legislature and Dumoulin (2024) place Houzeau de Lehaie among the members elected for the party in 1900.

and conflicts between them became rarer (Luykx & Platel, 1985, p. 196). This cuts against the design. A Senate that answers to the same voters can respond to them too, and if its response is non-negative the Chamber–Senate difference understates the Chamber’s own response (Section 7).

## 2.3 The 1899 law: proportional representation

Proportional representation followed in 1899 and was first applied in 1900. It was a Catholic–Liberal arrangement adopted in part to contain socialist growth and in part to rescue the Liberals, who under the majoritarian rule had been reduced to a rump (Ahmed, 2010; Barzachka, 2025; Dhondt, 2020; Emmenegger & Walter, 2019). The 1899 law replaced the majoritarian list rule with D’Hondt allocation in multi-member arrondissements and merged 41 pre-reform constituencies into 30 (Royaume de Belgique, 1899; Stengers, 2004). The Chamber elected in 1900 had 152 seats (Online Appendix Table OA1).

The reform changed nothing about who could vote. It changed the mapping from votes to seats. Under the majoritarian rule a large minority could lose every seat in a district (Cox, 1997), and for the question studied here that mattered most in the four coalfield districts, where the rule had turned socialist victories, outright or in the run-off, into socialist monopolies. After 1900 Catholic and Liberal lists could win seats there again, and the socialists could win seats in districts they had never carried. Section 6.2 documents both movements in the seat counts. Both reforms were instruments of elite management (Ahmed, 2010; Emmenegger & Walter, 2019). Since every law required the approval of both chambers, the Senate, partly insulated, remained a safeguard throughout (Gilissen, 1958; Popelier, 2018); a fuller account is in Online Appendix OA1.

## 3 A Framework: Three Primitives and Two Margins

This section sets out a simple framework that states which margin of behaviour each reform can move and when the Senate comparison should stop being informative as a control group. It assembles standard pieces: a legislator’s attention problem with a probabilistic-voting return to attention. We present it as an organising device rather than as a theoretical contribution. It generates signs and orderings, not magnitudes, and we use it only for the former.

### 3.1 The legislator’s problem

Consider an established-party legislator  $i$  (Catholic or Liberal) in chamber  $c \in \{C, S\}$ , elected in district  $d$ , sitting in legislature  $t$ . This is the population in the estimation sample; socialists are outside the model and enter it only through their vote share  $s_{dt}$ . The

legislator chooses  $a_{it} \in [0, 1]$ , the share of his floor speech devoted to labour and social policy. He has a type  $\alpha_i$ , the share he would choose absent any electoral concern. For example, for a conservative Catholic deputy whose politics were formed on the confessional and educational cleavage,  $\alpha_i$  is low.

The electoral environment a member of chamber  $c$  faces in district  $d$  at  $t$  is  $z_{cdt} = (n_{dt}^W, \tau_{dt}, e_{ct})$ : who votes, how votes become seats, and who may stand. The next subsection takes these in turn. Let  $\pi(a_{it}; z_{cdt})$  be the probability that the legislator keeps his seat. Over the range of attention members actually choose, we take it to be linear in  $a$ :

$$\pi(a_{it}; z_{cdt}) = \pi_0(z_{cdt}) + \lambda_{cdt} a_{it}, \quad \lambda_{cdt} \equiv \frac{\partial \pi}{\partial a}. \quad (1)$$

The first term,  $\pi_0$ , is how safe the seat is whatever the member says. The second is what his speech adds:  $\lambda_{cdt}$  is the increase in the probability of keeping the seat from one more unit of labour attention, and we call it the electoral return to labour attention.<sup>4</sup>

Office in chamber  $c$  is worth  $R_c$ , and departing from type costs  $\kappa_c/2$  per squared unit. The legislator maximises

$$U_{it} = R_c \pi(a_{it}; z_{cdt}) - \frac{\kappa_c}{2} (a_{it} - \alpha_i)^2. \quad (2)$$

The objective combines office and policy motivation in the tradition of Wittman (1983) and Calvert (1985). We borrow the objective and not the equilibrium results, since ours is a decision problem rather than a game. In those models policy preferences are defined over the policy eventually implemented, so the cost of moving is paid only if the candidate wins and depends on the opponent. Here  $\kappa_c$  is a direct cost, independent of the electoral outcome. In particular, it is what a member gives up by departing from the agenda that his own set. It is distinct from the electoral cost of trespassing, which is already in  $\lambda$ .

Since the direct cost is paid whether or not the member keeps his seat, it is subtracted from the objective rather than weighted by  $\pi$ . Substituting (1) into (2) and setting the derivative with respect to  $a$  to zero gives

$$a_{it}^* = \alpha_i + \frac{R_c}{\kappa_c} \lambda_{cdt}. \quad (3)$$

The objective is concave in  $a$ , so this is a maximum, and it is an internal solution as long as  $a^*$  lies in  $[0, 1]$ . A member speaks about labour as much as his type inclines him to, plus an amount proportional to the electoral return to doing so. The return  $\lambda$  is what the empirical exposure measures proxy.

---

<sup>4</sup>In the issue-competition literature, trespassing on an issue a rival owns is costly through the vote (Budge & Farlie, 1983; Petrocik, 1996). The cost is part of  $\pi$ , so  $\lambda$  is the return net of it.

### 3.2 Three primitives

We write the electoral return to labour attention as the product of three terms:

$$\lambda_{cdt} = \underbrace{\phi n_{dt}^W}_{\text{who can vote}} \times \underbrace{\tau_{dt}}_{\text{votes to seats}} \times \underbrace{e_{ct}}_{\text{who can stand}}. \quad (4)$$

The first factor,  $\phi n_{dt}^W$ , is the probabilistic-voting weight of Lindbeck and Weibull (1987) in the form used by Strömberg (2004), where a group’s weight is already a product of its size, the share of its members who are informed and vote, and its density of swing voters. Here  $n_{dt}^W$  is the share of the district’s votes cast by people who vote on labour, and  $\phi$  is the density of swing voters among them. Plural suffrage moves this factor, because it decides who may vote<sup>5</sup>.

Furthermore,  $e_{ct}$ , indicates whether a rival who could credibly claim the labour vote is eligible to stand in chamber  $c$ . In our model, a labour voter can punish neglect of labour only by switching to a candidate who owns the issue<sup>6</sup>. The Senate’s eligibility census set this term near zero. Since  $\lambda$  is a product of the three factors, follows that where  $e_{ct}$  is zero or near to zero, neither extending the franchise nor changing how seats are allocated moves the return

Finally, the term  $\tau_{dt}$  measures how a marginal vote for the establishment list translates into seats under the electoral formula. Under the majoritarian list rule, the list that won a district, either outright or in a run-off, took all its seats. We therefore model  $\tau$  as a step function of the socialist vote share  $s_{dt}$ . The value of the function is positive when the socialist vote is below the threshold that effectively prevents the establishment list from winning a seat, and zero above it. Proportional representation replaced this rule with D’Hondt allocation in multi-member districts, under which  $\tau$  varies smoothly with vote shares and stays positive far below a majority. Each element of the Belgian settlement acts mainly on one term: plural suffrage on  $n^W$ , proportional representation on  $\tau$  and the Senate’s eligibility census on  $e$ .

<sup>5</sup>In probabilistic-voting models with electoral districts this weight is already scaled by the seat return to a marginal vote (Persson & Tabellini, 2000, ch. 8).

<sup>6</sup>This third factor is our addition. Citizen-candidate models, which study single-winner elections, make the voting rule explicit: Osborne and Slivinski (1996) compare plurality with a two-round run-off, the single-member analogue of the Belgian rule before 1899, and Besley and Coate (1997) use plurality. They leave out who can stand, since they allow any citizen to run.

### 3.3 Two margins

Equation (3) gives two ways for the average  $a$  in a cell to change. Writing  $\bar{a}_{cdt}$  for the mean type among those seated,

$$\Delta \bar{a}_{cdt} = \underbrace{\Delta \bar{\alpha}_{cdt}}_{\text{recomposition}} + \underbrace{\frac{R_c}{\kappa_c} \Delta \lambda_{cdt}}_{\text{adaptation}}. \quad (5)$$

On one hand, the intensive margin changes  $a^*$  for a given  $\alpha_i$ . This is adaptation, which is the attention analogue of what Lee et al. (2004) call voters affecting policy, measured using roll-call votes.

On the other, the extensive margin changes the distribution of  $\alpha$  among those elected. This is recomposition, which corresponds to their electing channel.

This distinction maps directly onto our specifications. A regression with politician fixed effects captures only the intensive margin. The pooled regression captures both margins. Appendix OA8 provides an exact decomposition of each party's change.

### 3.4 What each reform moves

**The Senate: who can stand.** The eligibility census kept the pool of men who could stand for the Senate small and property-owning, so a candidate who could claim the labour vote was rarely on the ballot. Therefore,  $e_S$  was small, and after the Plural suffrage the Senate's return to labour attention rose less than the Chamber's. Writing  $\Delta$  for the change across a reform,

$$\Delta a_C - \Delta a_S = \frac{R_C}{\kappa_C} \Delta \lambda_C - \frac{R_S}{\kappa_S} \Delta \lambda_S \leq \frac{R_C}{\kappa_C} \Delta \lambda_C \quad \text{if } \Delta \lambda_S \geq 0. \quad (6)$$

What the bicameral comparison estimates is the change in the Chamber's electoral return to labour attention, scaled by  $R_C/\kappa_C$ <sup>7</sup>. The same term also indicates when this insulation should weaken. Therefore,  $e_S$  rises as socialists become able to win directly elected Senate seats. This happened first in 1900, with a single seat, and became more common from 1904 onward. For this reason, the upper chamber should begin responding to constituency pressure once socialists can win Senate seats. This suggests closing the estimation window for the second reform around that point.

**1894: who can vote.** Plural suffrage raised  $n^W$  in every district at once, from close to zero, since the censitary franchise had excluded almost all industrial workers, although the supplementary votes diluted the increase. From equations (3) and (4),  $\partial a_{it}^*/\partial n_{dt}^W > 0$

<sup>7</sup>Net of the corresponding change in the Senate; if the Senate's return did not fall, it is a lower bound on the change in the Chamber

for every seated member with  $\alpha_i$  fixed whose list could still win the district ( $\tau_{dt} > 0$ ). This implies an intensive-margin response, broad rather than concentrated, and within person. However, the same reform pushed  $s_{dt}$  past the winning threshold in the coalfield districts, driving  $\tau$  to zero for establishment lists and causing their members to lose their seats. The resulting change is therefore in who remains in the legislature. In the mining fields districts, the survivors are a selected group, who won in other dimensions outside labor. Thus, the framework predicts a compositional reversal in the coalfields, which should appear in the pooled triple difference but disappear under politician fixed effects<sup>8</sup>.

**1900: how votes become seats.** Proportional representation touches neither  $n^W$  nor  $e$ . It applied to the directly elected senators as well, so  $\tau$  moved in both chambers, but with  $e_S$  small the Senate's return moved little. Proportional representation lowers the threshold of exclusion in district  $d$ , the vote share above which a list is guaranteed a seat, from about one half to roughly  $q_d \approx 1/(M_d + 1)$ , where  $M_d$  is the district's magnitude (Cox, 1997; Lijphart & Gibberd, 1977; Rae et al., 1971). A list can therefore win a seat without winning the district, and the change binds only where the old rule had been binding. Therefore, sitting members can respond on the intensive margin only where  $\tau$  moved, which is where the old rule had handed the seats to the socialists<sup>9</sup>. For this reason, the reform works mainly through a recomposition mechanism.

Entry is a different object from the choice in equation (2), and needs its own probability. Let  $P(\alpha; q_d, n_d^W)$  be the chance that a candidate of type  $\alpha$  wins a seat, where the argument is a type rather than a choice. With an entry cost  $k$ , measured in the units of  $R_C$ , a potential candidate stands if  $R_C P(\alpha; q_d, n_d^W) > k$ . This is the entry condition of the citizen-candidate literature (Besley & Coate, 1997; Osborne & Slivinski, 1996), with one departure: there any citizen may run, whereas  $e_{ct}$  keeps a rival who could claim the labour vote off the Senate ballot. Nevertheless, we keep the assumption that makes those models citizen-candidate: a candidate cannot credibly promise to act against his type, so voters choosing among entrants choose among types, which is why  $P$  takes  $\alpha$  as its argument. Once seated, a member does depart from his type, as in equation (3), but not by promise. Thus, attention to topics is an action taken in office, which voters observe before the next election. Writing  $F_d(\alpha \mid q_d)$  for the distribution of seated types and  $E[\alpha \mid d, q_d]$  for its mean, lowering the threshold admits types who could not previously win:

$$\frac{\partial E[\alpha \mid d, q_d]}{\partial q_d} < 0, \quad \text{with magnitude increasing in } n_d^W. \quad (7)$$

Neither part of (7) follows from the threshold alone. Lowering  $q_d$  widens the set of types

<sup>8</sup>A within-person retreat from the issue, the dismissive strategy open to an established party where the owner dominated (Meguid, 2005), would instead survive the fixed effects.

<sup>9</sup>Only a few establishment members still sat there: their coalfield deputies had fallen from 15.3 per cent of the two parties' deputies in 1892 to 5.2 per cent in 1898 (Table 10).

who can win, but whether the new winners speak more about labour than those already seated depends on how they reach the lower quota. We assume that where labour voters are numerous, an establishment candidate reaches it by claiming part of their vote, and that this is easier the higher his  $\alpha$ . In particular, the gain in the chance of winning from a higher  $\alpha$  rises with the mass of labour voters,  $\partial^2 P / \partial \alpha \partial n_d^W > 0$ . This one assumption gives both parts of (7). When  $n_d^W$  is small, lowering the threshold admits types that are similar to those already seated. When  $n_d^W$  is large, it admits types with higher  $\alpha$ .

Equation (7) supplies the recomposition term of equation (5), which is otherwise an accounting identity:  $\Delta \bar{\alpha}_d \approx (\partial E[\alpha \mid d, q_d] / \partial q_d) \Delta q_d$ , and since proportional representation lowered the threshold,  $\Delta \bar{\alpha}_d > 0$  and larger where  $n_d^W$  is larger. The entrant mining gradient is the test of that prediction.<sup>10</sup>

**One rule, two margins.** The 1899 law enters the framework in two ways. First, for a list that already stands, it changes  $\tau$ , the number of seats a marginal vote is worth. Under the majoritarian rule, this was zero wherever the socialists won the district. Under D'Hondt, it is positive.

Second, for a newcomer, the law changes  $q_d$ , the vote share needed to secure a seat. This falls everywhere, from about one half to  $1/(M_d + 1)$ . The same law therefore affects both terms in equation (5).

The fall in  $q_d$  is largest where  $M_d$  is largest. Therefore, this provides a source of variation distinct from mining. However, on its own, it predicts nothing. Adding it to the regression also does not reduce the mining coefficient (Appendix A2).

This is what the framework predicts. A lower threshold changes who enters the Chambre, only where there are labour voters to attract. Therefore, the effect of the fall in  $q_d$  operates through  $n^W$ , not through  $q_d$  alone.

### 3.5 What the data could have contradicted

The framework has four implications that the estimates below could have contradicted. The 1894 response should be positive, and similar with and without politician fixed effects. The coalfield reversal at 1894 should disappear under politician fixed effects. After 1900, the aggregate Chamber–Senate difference should not move, the coalfield gradient should be carried by entrants rather than incumbents and should vanish under politician fixed effects, and it should be flat for cohorts selected under the earlier rules. And the Senate's

<sup>10</sup>Lizzeri and Persico (2001) and Persson and Tabellini (2000) predict the opposite sign, proportional representation making policy less targeted than winner-take-all. Both compare two parties, and Persson and Tabellini compare the composition of spending across national electoral systems; Belgian PR was applied in multi-member arrondissements with three parties competing for seats, so what changed was who could win locally, and our outcome is attention to a class rather than transfers to a place. The favoured minority in Myerson's sense is here a class that happens to be spatially concentrated.

mining gradient should turn positive from about 1904. Sections 8 and 9 take these in turn.

One limit of the framework belongs here rather than in the results. The complementarity behind equation (7) cannot be tested with these data, because we observe who was elected and not the pool of aspirants. The entrant gradient is consistent with it, but it is also consistent with a candidate pool that was already more labour-oriented in mining districts for reasons unconnected to the electoral rule. What speaks against that reading is the cohort pattern: a pool that was always more labour-oriented in those districts would have produced a gradient among the members elected there before 1900 as well, and it does not.

## 4 Data

We assemble a politician-level dataset that tracks parliamentary speech and lawmaking across both chambers of the Belgian parliament from 1880 to 1912. It combines four sources: biographical and career records for every member of either chamber, the digitised plenary proceedings of both chambers, the legislative metadata and texts of every bill deposited in the period, and the 1896 industrial census.

### 4.1 Politicians

The career records cover the 845 politicians who served in either chamber between 1880 and 1912. For each we record the chamber, party, constituency and legislature of every mandate, the legislature of first entry and of last activity, and mid-term entries and exits through death, resignation or by-election. Membership data come from the historical records of the Chamber (Belgian Chamber of Representatives, 2024b) and the Senate (Belgian Senate, 2024a), supplemented by Van Molle (1972). Party affiliation is recorded by legislature, so that a member who changed party carries the label of the party he sat for at the time.

The estimation sample contains Catholic and Liberal politicians only. This is the substantive restriction of the design: we ask how the established governing parties responded to the reforms, not how the parliament as a whole changed composition. Therefore, Socialist, Daensist and other members are excluded from the estimation sample, but they remain in the corpus. Section 11.4 uses their speech to characterise the agenda of each sitting day.

### 4.2 Parliamentary speech

The speech corpus covers all plenary debates in both chambers between 1880 and 1912. For the Chamber we use the digitised records of the Plenum.be project (Plenum.be, 2024).

For the Senate we collected the digitised proceedings from the Senate’s historical archives (Belgian Senate, 2024b) and applied our own optical character recognition pipeline to the scanned volumes. An online Appendix OA9 compares transcription quality across the two chambers. In both chambers the processed text is segmented into interventions through layout recognition and regular expressions that exploit the typographical conventions of the proceedings, and each intervention is linked to a speaker, party, chamber, date, legislature and constituency. An observation is a single uninterrupted intervention in a plenary session.

Table 1 traces the path from the digitised record to the estimation samples. We start from 553,958 interventions. These are all the interventions to which the classifier of Section 5 assigns a score. They span the whole digitised record, including sittings outside the study period, and come from speakers of every party. Of these, 460,232 were delivered in one of the two chambers between 1880 and 1912. In addition, Catholic and Liberal members account for 372,441 of them. Of the rest, 60,547 are socialist, 2,512 Daensist and 5,366 belong to other parties, almost all of them the Antwerp *Meetingpartij*. Finally, 19,718 carry no party label and are mostly the presiding officer.

From the Catholic and Liberal interventions we remove the 37,355 that a member delivered while presiding over a sitting: calling speakers, putting questions to the vote and closing debates. This is procedure rather than a choice of subject. Most of it belongs to Catholic members, because the Catholic party held the presidency of both chambers for most of the period. We also correct speaker identities and party labels that the archive records inconsistently (Online Appendix OA10.2). These corrections remove a further 674 on net, so the two steps together account for the 38,029 rows of Table 1. The result is the analysis sample of 334,412 interventions and the two estimation samples are windows on it. The 1894 analysis uses 1880–1899 (208,325 interventions) and the 1900 analysis uses 1894–1906 (119,448). Both cover 41 arrondissements.<sup>11</sup>

---

<sup>11</sup>The 1899 law folded 41 constituencies into 30, through nine merged constituencies: Tournai–Ath, Tongeren–Maaseik, Huy–Waremmes, Arlon–Marche–Bastogne, Neufchâteau–Virton, Dinant–Philippeville, Ghent–Eeklo, Roeselare–Tielt and Veurne–Diksmuide–Ostend. None of the four coal-field districts was merged, and the proceedings continue to record all four separately. Clustering at the post-1900 constituency rather than the arrondissement leaves the post-1900 estimate unchanged at 9.25 with 30 clusters (s.e. 2.39), and assigning each merged constituency the mean mining exposure of its components gives 9.28 (s.e. 2.40; Online Appendix Table OA15), so the merger is a question of clustering rather than of measurement.

Table 1: From the digitised record to the estimation samples

|                                               | Interventions | Words      | Speakers |
|-----------------------------------------------|---------------|------------|----------|
| Scored corpus, all sessions and speakers      | 553,958       |            |          |
| dated 1880–1912 with a chamber                | 460,232       |            |          |
| Plenary record, 1880–1912, both chambers      | 460,584       | 30,086,418 | 1,206    |
| Catholic and Liberal members                  | 372,441       | 24,744,993 | 1,024    |
| Socialist members                             | 60,547        | 2,850,374  | 124      |
| Daensist members                              | 2,512         | 215,796    | 24       |
| Other members                                 | 5,366         | 283,448    | 18       |
| Presiding officer or unassigned speaker       | 19,718        | 1,991,807  |          |
| Catholic and Liberal rows removed in cleaning | 38,029        | 2,473,148  |          |
| Analysis sample                               | 334,412       | 22,271,845 | 957      |
| Panel A, 1880–1899                            | 208,325       | 14,803,490 |          |
| Panel B, 1894–1906                            | 119,448       | 7,494,342  |          |

An intervention is a single uninterrupted speech in a plenary sitting, and a speaker is a distinct person reference in the parliamentary archive; that is not the same object as a documented mandate, and Online Appendix OA10 explains why the reference count exceeds the number of politicians reported in Section 4. Words are counted on the lemmatised text, which is the length the regressions use as a weight. The rows are computed independently of one another rather than by merging the sources on a key: the first two come from the file of classifier predictions, the plenary record and the speaker groups from the corpus, and the analysis sample and its two panels from the estimation file. The record holds a few hundred interventions more than the classifier scored, which is why the two window figures differ; the speaker groups are counted on the record and add to the row above them. The line for interventions removed in cleaning is accordingly the difference between the two rows it sits between, and covers interventions of the presiding officer recorded under a member’s name together with the party-label repairs described in Section 4.

The corpus is overwhelmingly French. Dutch never reaches one per cent of interventions in any legislature (Online Appendix Table OA4). Its parliamentary use expanded only after the De Vriendt–Coremans law of 1898, but in plenary debate the change is barely visible before 1912. The classifier is a French-language model, and at this share differential performance across languages cannot generate the results.

### 4.3 Legislative records

The law-level dataset combines the legislative metadata of the Chamber’s Genesis database (Belgian Chamber of Representatives, 2024a) with the full text of the legislative documents of both chambers. Genesis is the historical part of the Chamber’s legislative file

and goes back to the nineteenth century. For every parliamentary document it records the sponsor and the subject tags assigned by the Chamber’s own documentation service. It also records the date of deposit, the dates of the votes in each chamber, and the date of promulgation. The database holds 4,779 proposals deposited between 1880 and 1912. Of these, 2,250 carry a promulgation date. A separate file of adopted acts contains 1,965 laws promulgated within the same window.<sup>12</sup> A proposal counts as a member’s bill when its sponsor is a named member rather than the government. The analysis of Section 10 applies the same party restriction as the speech analysis and keeps the 2,041 proposals introduced by Catholic and Liberal members. Government bills are kept as a comparison group and never pooled with members’ bills. Section 5 defines the legislative outcome.

## 4.4 Industrial structure

The 1896 Belgian industrial census reports employment by sector for every arrondissement (Office du travail de Belgique, 1900–1902). Our measure of a district’s exposure to the social question is the share of mining-sector workers in total arrondissement employment, standardised to mean zero and unit variance. The census was taken on 31 October 1896, between the two reforms. Therefore, the measure is predetermined with respect to proportional representation, and for plural suffrage we rely on the fact that a district’s mining employment share is a slow-moving feature of its economy, set by the coal seams. The Census comes from an administrative source that has nothing to do with parliamentary activity, and Section 6.2 shows that it tracks the geography of socialist electoral strength closely. We do not use it as an instrument. Mining intensity plausibly affects labour speech through constituency composition and industrial conflict as well as through electoral competition, and we make no exclusion restriction. We use it as a predetermined proxy for where the electoral rule bound. On this measure Mons (3.89 standard deviations above the mean), Charleroi (3.15), Soignies (1.88) and Liège (1.87) are the four districts whose mining employment share exceeds a quarter. Therefore, we refer to them as the coalfields districts (Appendix Table A5).

## 4.5 Summary statistics

Table 2 reports summary statistics for the analysis sample: the number of interventions, words and distinct speakers by chamber and reform window; the distribution of intervention length; and the distribution of the labour salience measure defined in the next section, both across interventions and across politicians.

---

<sup>12</sup>The two populations differ because a proposal deposited in the window may be enacted after it, and an act promulgated in the window may have been deposited before it.

Table 2: Summary statistics, Catholic and Liberal members, 1880–1912

| <i>Panel A. The analysis sample by chamber and window</i>       |               |           |             |           |
|-----------------------------------------------------------------|---------------|-----------|-------------|-----------|
|                                                                 | Interventions | Words     | Politicians | Districts |
| Chamber, 1880–93                                                | 142,773       | 9,023,224 | 340         | 41        |
| Chamber, 1894–99                                                | 47,166        | 2,763,529 | 240         | 41        |
| Chamber, 1900–12                                                | 101,902       | 5,049,411 | 315         | 40        |
| Senate, 1880–93                                                 | 14,181        | 2,270,288 | 263         | 38        |
| Senate, 1894–99                                                 | 4,784         | 746,449   | 142         | 38        |
| Senate, 1900–12                                                 | 23,606        | 2,418,944 | 312         | 39        |
| <i>Panel B. Words per intervention</i>                          |               |           |             |           |
|                                                                 | Mean          | Median    | p10         | p90       |
| Chamber                                                         | 57.8          | 14        | 2           | 152       |
| Senate                                                          | 128.2         | 31        | 3           | 348       |
| <i>Panel C. Labour salience, per cent of substantive speech</i> |               |           |             |           |
|                                                                 | Mean          | S.d.      | Median      | Above 50  |
| Chamber, per intervention                                       | 12.01         | 23.32     | 1.96        | 0.052     |
| Senate, per intervention                                        | 10.27         | 21.35     | 1.92        | 0.051     |
| Chamber, per politician-legislature                             | 13.32         | 15.52     | 8.21        | 0.042     |
| Senate, per politician-legislature                              | 10.99         | 14.90     | 5.55        | 0.035     |

Catholic and Liberal members, 1880–1912. Words are the intervention length used as the regression weight. Labour salience is the measure of equation (8); the last column of Panel C is the share of units scoring above 50. Politicians are distinct persons on the archive’s person identifier. That count is of person references in the corpus rather than of documented mandates, and Online Appendix OA10 explains why it exceeds the number of politicians reported in Section 4.

## 5 Measuring Attention to the Social Question

### 5.1 What we measure

Every intervention is assigned a probability distribution over seven categories: six substantive policy domains, of which labour and social policy is one, and a procedural residual for material with no policy content (motions of order, communications, votes, fragments produced by the segmentation). Table 3 lists the seven categories and what each covers. Online Appendix OA3 gives the topics behind each category and the prompt used to label the training sample.

This paper measures the share of a politician’s floor speech that concerns labour and

social policy. We call this labour salience. An intervention is included if it concerns the regulation of work (hours, wages, safety, the employment of women and children, strikes, and unions), welfare and public assistance (poor relief, pensions, accident and sickness insurance, and workers' housing), or public health. It does not count when the subject is the funding of Catholic schools, the tariff on coal, the concession of a mine, or a railway line through a mining district. What separates the two is whether the speech engages the condition of workers as a problem for the state.

Subjects that border on the social question are kept outside the labour domain. Education sits with religion rather than with social policy, so that school funding stays out of it. Mining and industry sit with the economy, so that debates on mining regulation and industrial policy stay outside the outcome even though they are the natural neighbours of labour legislation. Besides, the pay, status and pensions of the state's own employees count as governance, so that a debate on the wages of railwaymen or of the road menders of the public works department does not enter the social question even when it uses the vocabulary of work.

Table 3: The six policy domains and the procedural residual

| Category                     | Content                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------|
| Labour and social policy     | Regulation of work; welfare and public assistance; public health                            |
| Confessional                 | Church–state relations and secularism; education                                            |
| Economic and fiscal          | Budget; taxation and customs; finance and money; commerce; industry and mining; agriculture |
| State and governance         | Administration; electoral and constitutional questions; justice; policing; citizenship      |
| Foreign and defence          | Colonial and foreign affairs; the army                                                      |
| Territory and infrastructure | Public works and transport; land use and housing; environment; communications               |
| Procedural residual          | Procedure, ceremony and fragments without policy content                                    |

*Notes:* The classifier assigns every intervention a probability for each of the seven categories. The right-hand column summarises what each category covers; Online Appendix Table OA2 gives the full list of topics.

Appendix Table OA6 shows how the measure works using three interventions from the corpus. The first is assigned almost entirely to labour and social policy. The second is split between labour and the economy. The third uses the vocabulary of the coalfields but concerns the mining industry rather than the miners.

The middle case shows that the measure is continuous. An intervention that is half about workers' conditions and two-fifths about the tax system scores 50.

## 5.2 How the classifier was built

The classifier is a fine-tuned CamemBERT model (Martin et al., 2020), a transformer pretrained on a large corpus of French text and adapted here to assigning parliamentary interventions to policy domains<sup>13</sup>.

**Categories.** The categories come from the corpus itself. The proceedings record every debate under a heading that names the item on the agenda, and the heading is a good summary of what the debate is about. We grouped the headings into topics with an unsupervised clustering procedure and checked the topics by hand. A large language model then grouped the topics into the six policy domains and the residual of Table 3, and we checked that grouping by hand as well. Online Appendix OA3 lists the topics under each category.

**Labelled examples.** No hand-labelled corpus of nineteenth-century Belgian parliamentary speech exists, and labelling by hand at the scale a transformer requires was not feasible. The training labels were generated by a large language model (GPT-4o, snapshot `gpt-4o-2024-08-06`, at temperature zero with a fixed seed). The prompt, reproduced in Online Appendix OA3.3, gives the model the list of topics, a definition of each, and rules for the boundaries most likely to be contested in this corpus, such as labour against industry, or a budget debate against the policy area it concerns. The model returns one primary topic for each intervention, and we map that topic to its category in Table 3. The category is the label the classifier is trained on.

**The labelled sample.** The interventions sent for labelling are a probability sample of 7,000 drawn from the corpus, with a minimum number from each chamber and period so that the Senate, which produced far fewer interventions, is represented. A further 1,500 interventions were labelled from the chamber-and-period combinations with the fewest interventions, to give the model more examples of them. These are used to train the model and never to evaluate it.

**Training and scoring.** The model assigns each intervention a probability distribution over the seven categories of Table 3. Interventions longer than the model’s input window are cut into overlapping chunks. Thus, each chunk is scored and the chunk probabilities are averaged. The trained model is then run over all 553,958 interventions and returns, for each, seven probabilities that sum to one.

---

<sup>13</sup>The usual alternative is a keyword dictionary. Online Appendix OA17 explains why a dictionary fails in this corpus and reports a dictionary measure estimated on identical rows. The two measures agree on the sign of the main results and disagree on their robustness.

### 5.3 Validation

Agreement between the classifier and the labels it was trained on measures fidelity to the annotator, not accuracy. Therefore, a classifier can reproduce its annotator perfectly and both can be wrong. For this reason, we hand-coded a blind subsample of 250 interventions drawn from the probability sample. The coding sheet was shuffled and contained no model output. The chamber and period columns were also hidden, so the coder could not condition their coding on the dimensions used in the design. Table 4 reports how closely the classifier and the language-model annotations agree with this standard.

Table 4: Validation against blind human coding

|                          | Classifier | Training annotation |
|--------------------------|------------|---------------------|
| Accuracy                 | 0.86       | 0.84                |
| Cohen’s $\kappa$         | 0.80       | 0.77                |
| Macro F1                 | 0.823      | 0.774               |
| Hand-coded interventions |            | 250                 |

*Notes:* Agreement between each source and a human coder on 250 interventions drawn at random from the probability sample, coded blind to chamber, period and model output. The classifier is the fine-tuned CamemBERT model. The training annotation is the language model label the classifier was trained on.

An accuracy of 0.86 with  $\kappa = 0.80$  indicates substantial agreement under the conventional interpretation (Landis & Koch, 1977). It is also in line with the standards typically used for content analysis.

The classifier agrees with the human coder at least as closely as its own annotator does: 0.86 versus 0.84 for accuracy, and 0.80 versus 0.77 for  $\kappa$  (Table 4). Training on machine labels therefore does not appear to have compounded the annotator’s errors<sup>14</sup>.

### 5.4 The outcome variable

The labour salience of intervention  $s$  is the probability the classifier assigns to the labour and social-policy domain, expressed as a share of the probability assigned to the six substantive domains and multiplied by 100:

$$y_s = 100 \times \frac{p_s^{\text{LAB}}}{\sum_{k \in \mathcal{K}} p_s^k}, \quad \mathcal{K} = \{\text{six substantive domains}\}. \quad (8)$$

We use the probability share rather than the most likely category. This preserves

<sup>14</sup>The 250 interventions do not allow us to estimate reliable accuracy rates for each of the four chamber-by-period cells. Online Appendix OA3 discusses what can and cannot be inferred at this level. Appendix A1 instead addresses the concern that matters for our estimand: whether classifier performance differs across cells.

information from interventions that mix domains instead of assigning them to a single category at a cutoff.

We exclude the procedural residual from the denominator, but keep these interventions in the sample. The modal class is procedural for 56.1 per cent of the interventions dated 1880–1912 (Appendix Table A1), which is unsurprising for a plenary record. Rescaling the remaining content keeps the measure interpretable. Dropping these interventions would instead remove observations unevenly across the cells we compare<sup>15</sup>.

Regressions are weighted by the number of words in the intervention, so that the estimand is the share of parliamentary language, rather than of parliamentary interventions, devoted to the social question. A short interjection and an hour-long speech are not equal contributions to the record<sup>16</sup>.

## 5.5 The legislative outcome

For legislation we use a different and simpler measure, and it passes through no model of ours. The Genesis archive tags each bill with its subject, and we count a bill as addressing the social question if its tags include *Labor & Work*, *Welfare & Social Protection* or *Health*. Health is included so that the legislative outcome covers the same object as the speech outcome, whose labour domain contains public health. *Public Sector Personnel* is excluded because it concerns the pay and status of state employees, which the speech taxonomy places under governance. We measure the extensive margin only, whether a bill addresses the social question at all, because the length of a statute is set by drafting convention rather than by the attention behind it.

The unit is the proposal, dated by its deposit date. A proposal is classified as a private member’s bill when the archive’s author field does not name the government.<sup>17</sup>

We keep proposals and adopted acts as separate populations. A file filtered by promulgation date, which only enacted proposals have, would count the same acts twice under two dates.

The two populations also differ in subject. Among all proposals deposited in the window, failed proposals address the social question more often than enacted ones. A promulgation-date filter would therefore select against the subject we study.

<sup>15</sup>Every intervention therefore remains in the sample. As a check, dividing by all seven categories gives 3.94 (s.e. 2.03) for 1894 and 9.01 (s.e. 2.40) for 1900, compared with baselines of 4.46 and 9.25 (Table A2).

<sup>16</sup>The choice matters for the 1894 estimate. Unweighted at the level of the intervention it falls to 1.76 (s.e. 1.28), because an unweighted regression counts a procedural interjection and an hour-long speech equally and the Chamber after 1894 produced many more of the former. Aggregating to the politician-by-legislature level, which removes the question by construction, returns 4.21 (s.e. 1.97). The 1900 estimate is insensitive to the choice: 4.82 unweighted and 9.45 on the member panel, against 9.25 in the baseline (Appendix Table A2).

<sup>17</sup>The alternative rule, which treats a missing constituency as a government bill, misclassifies members’ proposals in this window because the archive does not record their constituency. The author field is unambiguous, so we use it throughout.

The pattern is reversed for the Catholic and Liberal members' bills used in the analysis, where enacted bills address the social question more often than failed ones. This is consistent with the unchanged success rate reported in Section 10.

## 6 Stylized Facts

Four facts are visible in the data without any regression. First, how the reforms changed the composition of the two chambers. Second, where these changes occurred. Third, how attention to the social question changed over time in each chamber. Fourth, how members who entered after each reform differed from those already in parliament.

### 6.1 Seats

The two reforms changed the Chamber's composition without changing which party governed. Under the censitary franchise the Chamber was divided between the Catholic and Liberal parties, with a Catholic majority from 1884 that survived both reforms and lasted until 1914. On one hand, plural suffrage replaced the Liberals with the socialists as the second-largest party. Under the majoritarian list rule, Liberal minorities in industrial districts won almost no seats, while socialist victories gave them the district's full delegation. On the other, proportional representation restored the Liberals to a share of seats proportionate to their votes and reduced the socialist delegation in the coalfields, again without touching the Catholic majority (Luykx & Platel, 1985; Witte et al., 2016). The Senate followed the dynamic of the low room of the Parliament with a lag. The Workers' Party won its first directly elected Senate seat only in 1900, a single one, and in 1910 it held 11 socialist seats of 110 against 35 of 166 in the Chamber (Online Appendix Table OA1).

### 6.2 Where: the coalfields

The effect of the reforms also relate to the geography of the coal basins. Table 5 shows the number of socialist members returned by each arrondissement to its mining intensity. Before 1894 there were no socialist members, so there is nothing to explain. Under plural suffrage and the majoritarian rule, one standard deviation of mining intensity predicts 1.63 additional socialist members (s.e. 0.23), and mining intensity alone accounts for 0.82 of the cross-sectional variance. Furthermore, under proportional representation the coefficient is 1.23 (s.e. 0.35) and the  $R^2$  0.50. Panel D gives the same fact as means. Under the majoritarian rule, the socialists held an average of 5.62 seats in each of the four coalfield districts. This was 68 per cent of the seats in those districts, compared with just 0.18 seats on average elsewhere. Under proportional representation their coalfield

presence fell, to 5.00 seats and 51 per cent of the delegation, while outside the coalfields it tripled. Proportional representation broke the socialist monopoly where the majoritarian rule had created it and spread socialist representation across the country.

Table 5: Mining intensity and socialist representation in the Chamber

|                                                                     | Coefficient                          | Std. err.    | <i>t</i>     | <i>R</i> <sup>2</sup> |
|---------------------------------------------------------------------|--------------------------------------|--------------|--------------|-----------------------|
| <i>Panel A. Socialist members per district, by electoral regime</i> |                                      |              |              |                       |
| Censitary, to 1892                                                  | no socialist members in any district |              |              |                       |
| Majoritarian, 1894–98                                               | 1.630                                | (0.234)      | 6.95         | 0.816                 |
| Proportional, 1900–                                                 | 1.234                                | (0.348)      | 3.55         | 0.498                 |
| <i>Panel B. Pooled, legislature fixed effects</i>                   |                                      |              |              |                       |
| Count of socialist members                                          | 0.697                                | (0.171)      | 4.08         |                       |
| Socialist share of members                                          | 0.081                                | (0.007)      | 11.18        |                       |
| <i>Panel C. Pooled, 1894 onward</i>                                 |                                      |              |              |                       |
| Count of socialist members                                          | 1.323                                | (0.325)      | 4.07         |                       |
| Socialist share of members                                          | 0.153                                | (0.014)      | 11.16        |                       |
| <i>Panel D. Socialist members per district, means</i>               |                                      |              |              |                       |
|                                                                     | Censitary                            | Majoritarian | Proportional |                       |
| Coalfield districts                                                 | 0.00                                 | 5.62         | 5.00         |                       |
| Elsewhere                                                           | 0.00                                 | 0.18         | 0.55         |                       |
| Coalfield socialist share                                           | 0.00                                 | 0.68         | 0.51         |                       |

*Notes:* Panels A to C regress the number (or share) of socialist members of the Chamber returned by an arrondissement on its standardised 1896 mining employment share; standard errors in parentheses. Members are counted as socialist from the legislature of October 1894, the first at which the Belgian Workers' Party won seats. Panel D reports means by regime for the four coalfield districts (Mons, Charleroi, Soignies and Liège) and the other 37.

The same shift can be seen in the presence of established parties in the coalfields.

Table 10 reports two measures for each legislature. The first is the share of Catholic and Liberal deputies with recorded speech who represented a coalfield district. The second is the share of Chamber speech by Catholic and Liberal members that came from those districts.

Before plural suffrage, the deputy share stayed close to one in seven. The word share ranged from one tenth to one fifth, partly because a few long interventions could change it from one legislature to the next.

After plural suffrage, both shares fell sharply. The deputy share fell from 15.3 per cent in 1892 to 5.2 per cent in 1898. The word share fell from 10.3 per cent in 1892 to 3.2 per cent in 1896 (Table 10).

After proportional representation, both recovered. By 1906, the shares had reached 14.1 and 12.5 per cent (Table 10).

Plural suffrage therefore reduced the presence of established parties in the districts where the social question was most acute. Proportional representation brought them back.

### 6.3 Attention to the social question, by chamber

Figure 1 plots labour salience among Catholic and Liberal members in each chamber, legislature by legislature, weighted by words. The two chambers move together through the censitary period, from about 5 per cent of substantive speech in the early 1880s to 12 in the Chamber and 15 in the Senate in 1888, and back to 6 to 8 in 1892. Furthermore, the 1888 peak follows the industrial unrest of 1886 and the first social laws, and it is present in both houses. They diverge after 1894. Chamber labour salience averages 7.9 per cent of substantive speech before 1894 and 16.2 in the majoritarian legislatures, while the Senate rises from 8.1 to 11.8. Both houses move, but the Chamber moves more (Table 6). In addition, the gap persists through the early proportional legislatures, with the Chamber at 18.2 and the Senate at 10.4 in 1900–06, and narrows from 1906.

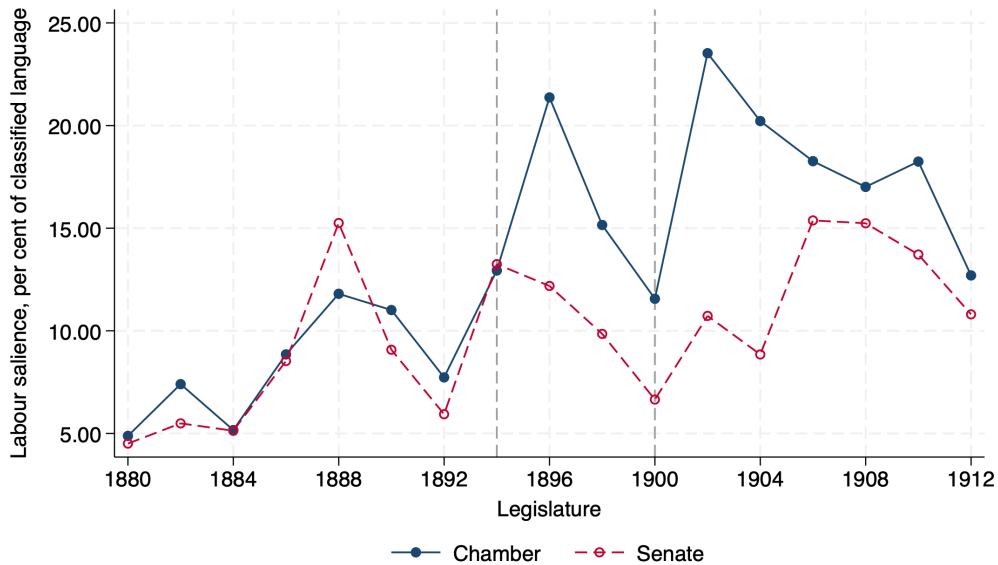


Figure 1: Labour salience by chamber and legislature, Catholic and Liberal members. Each point is the word-weighted mean of the labour salience measure of equation (8) over the interventions of the legislature. Vertical lines mark 1894 and 1900.

Table 6: Labour salience by chamber and period

| Panel                                    | Period    | Chamber |         | Senate |        |
|------------------------------------------|-----------|---------|---------|--------|--------|
|                                          |           | Mean    | Obs.    | Mean   | Obs.   |
| A. 1894 cutoff                           | 1880–1893 | 7.89    | 142,392 | 8.07   | 14,131 |
|                                          | 1894–1899 | 16.18   | 47,049  | 11.75  | 4,753  |
| B. 1900 cutoff                           | 1894–1899 | 16.18   | 47,049  | 11.75  | 4,753  |
|                                          | 1900–1906 | 18.18   | 56,684  | 10.43  | 10,962 |
| <i>Panel B, coalfield districts only</i> |           |         |         |        |        |
|                                          | 1894–1899 | 12.71   | 1,525   | 16.06  | 1,455  |
|                                          | 1900–1906 | 34.88   | 7,273   | 13.65  | 2,522  |

Labour-bucket share of classified policy language, in percentage points, weighted by intervention length. The coalfields are Mons, Charleroi, Soignies and Liège. These are the bases against which the estimates in Tables 8 and 11 should be read.

Inside and outside the coalfields the picture differs (Figure 2). Under the majoritarian rule the few Catholic and Liberal deputies who still sat for coalfield districts spoke about labour less than the senators from the same districts, 12.7 against 16.1 per cent of substantive speech. After 1900 the coalfield Chamber figure rises to 34.9 while the coalfield Senate figure falls to 13.7 (Table 6, lower panel), so the gap opens from both sides. Outside the coalfields the post-1900 movement is small in both houses.

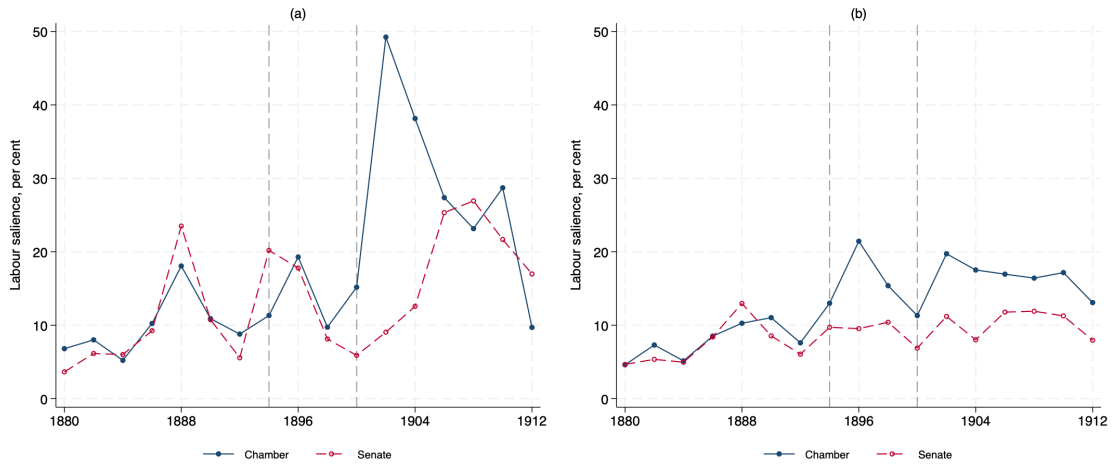


Figure 2: Labour salience by chamber, legislature and geography, Catholic and Liberal members. Panel (a) is the four coalfield districts, Mons, Charleroi, Soignies and Liège; panel (b) is the other thirty-seven. Each point is the word-weighted mean of the measure of equation (8) over the interventions of the legislature. Vertical lines mark 1894 and 1900. The coalfield cells are thin after 1894, when the established parties held few seats there: 1,525 Chamber interventions in the majoritarian legislatures against 45,524 elsewhere (Table 6), which is why the Chamber series in panel (a) moves sharply from one legislature to the next.

## 6.4 Incumbents and entrants

Table 7 divides the Chamber in the two post-reform windows into three groups: members who were already sitting, members who entered at the election that opened the window, and members who entered later.

The 1894 entrants were more labour-oriented than the incumbents they joined: 18.9 versus 15.3 per cent (Table 7). However, they produced only 13.5 per cent of Chamber speech in the majoritarian legislatures. The rise in labour salience after 1894 therefore cannot be explained by these entrants.

The 1900 entrants were different. They produced 22.3 per cent of Chamber speech in the proportional legislatures. Their labour salience was 26.9 per cent, compared with 15.7 per cent for the members they joined (Table 7). The coalfield gradient in Section 9 appears among these entrants.

We cannot make the same comparison for the censitary legislatures. A member’s entry is defined as the first legislature in which the corpus observes him. Since the corpus begins in 1880, every member sitting that year is recorded as an entrant. The table therefore focuses on the two post-reform windows, where this distinction is meaningful.

Table 7: Incumbents and entry cohorts in the Chamber

| Cohort                                                 | Members | Words     | Share of window (%) | Labour salience |
|--------------------------------------------------------|---------|-----------|---------------------|-----------------|
| <i>Panel A. The majoritarian legislatures, 1894–98</i> |         |           |                     |                 |
| Entered before 1894                                    | 165     | 2,222,959 | 80.4                | 15.31           |
| Entered in 1894                                        | 39      | 374,364   | 13.5                | 18.94           |
| Entered after 1894                                     | 36      | 166,206   | 6.0                 | 21.66           |
| <i>Panel B. The proportional legislatures, 1900–06</i> |         |           |                     |                 |
| Entered before 1900                                    | 165     | 2,097,491 | 74.2                | 15.65           |
| Entered in 1900                                        | 36      | 629,689   | 22.3                | 26.88           |
| Entered after 1900                                     | 40      | 98,307    | 3.5                 | 16.43           |

Chamber, Catholic and Liberal members. A cohort is defined by the first legislature in which a member is observed, on the parliamentary archive’s person identifier. Words are the intervention lengths used as regression weights, and the share is of all Catholic and Liberal Chamber words in the window. Labour salience is the word-weighted mean of the measure of equation (8) over the cohort’s interventions in the window. The censitary legislatures cannot be split this way: the corpus begins in 1880, so every member sitting in that year is recorded as an entrant.

## 7 Empirical Strategy

The analysis proceeds in four steps. First, we ask whether each reform changed attention to the social question more in the exposed chamber than in the more insulated one. This

is the difference in exposure.

Second, we ask whether the response was larger in the coalfields, where the socialists had their electoral base and where the majoritarian rule had the strongest effect. This is the triple difference.

Third, when we find an effect, we ask where it came from. Did sitting members change their behaviour, or did the composition of the chamber change? This is adaptation versus recomposition.

Finally, we ask whether the legislative record moved in the same direction.

This section presents the identification strategy, the estimating equations, and the inference.

## 7.1 The bicameral comparison

Electoral reforms are not adopted in politically neutral conditions. Belgian plural suffrage followed the general strike of 1893, while proportional representation followed a decade of socialist growth. A simple comparison of parliamentary speech before and after either reform would therefore attribute to the reform changes that may instead reflect the political events that preceded it (Deneckere, 1993; Polasky, 1992). To see the problem, write the labour salience of a member of chamber  $c \in \{C, S\}$  in legislature  $t$  as

$$y_{Ct} = \alpha_C + \beta_C \text{Post}_t + \varepsilon_{Ct}, \quad y_{St} = \alpha_S + \beta_S \text{Post}_t + \varepsilon_{St}, \quad (9)$$

where  $\text{Post}_t$  indicates the legislatures after the reform and the error terms contain the strikes, the industrial conflict and the doctrinal shifts of the period. A before-and-after estimate of  $\beta_C$  is biased because these events are correlated with  $\text{Post}_t$ , and year fixed effects cannot help, since a national reform is collinear with them. The bicameral structure of parliament provides a solution. The Chamber of Representatives and the Senate faced the same reforms and the same political events. However, the reforms changed their electoral incentives to different degrees. From 1894, both chambers answered to the same voters. What differed was who could challenge their members. The Senate's eligibility census kept working-class candidates off the ballot throughout the period (Section 2). Write  $\varepsilon_{ct} = \eta_t + u_{ct}$ , with  $\eta_t$  the shocks common to both houses. If the house-specific remainder is mean-independent of the reform,  $E[u_{Ct} - u_{St} \mid \text{Post}_t] = 0$ , then differencing the two equations gives

$$y_{Ct} - y_{St} = (\alpha_C - \alpha_S) + (\beta_C - \beta_S) \text{Post}_t, \quad (10)$$

and the change in the Chamber–Senate difference across the reform identifies  $\beta_C - \beta_S$ , the differential response of the exposed chamber, net of everything that hit both. In the framework of Section 3 this is equation (6): the change in the Chamber's electoral return

to labour attention, scaled by  $R_C/\kappa_C$ , net of the corresponding change in the Senate.

The design identifies the differential response of the Chamber relative to the Senate under asymmetric electoral exposure. Therefore, it does not identify the aggregate effect of the reform on the legislature, because the Senate also responds to some degree, and Section 8 shows by how much. When the Senate response is non-negative,  $\beta_C - \beta_S$  is a lower bound on  $\beta_C$ .

A non-negative Senate response is plausible because directly elected senators also faced the plural-suffrage electorate from 1894. The design, however, cannot test this restriction. It is needed only to interpret the Chamber–Senate difference as a lower bound on the Chamber effect. The differential response itself is identified without it.

The design also does not by itself explain why the Chamber–Senate difference changed. The two chambers differed in composition, speaking practices, and legislative role, as well as in electoral exposure. The Chamber indicator absorbs these differences as long as they remain constant over time. If one of them changed at the time of a reform for reasons unrelated to the reform, the design would attribute that change to the reform.

For 1894, there is no third dimension of comparison. The result therefore relies on direct tests of specific alternatives. Section 8 examines turnover and ministerial speech, Section 11.4 examines the socialist agenda, and Section 11.3 examines shocks affecting the Chamber as a whole.

For 1900, the triple difference removes any change in the Chamber–Senate difference that is common across districts. To bias  $\theta$ , an alternative explanation must therefore vary with mining exposure. Section 9 examines the two main candidates: turnover that follows the calendar rather than the electoral rule, and ministerial speech.

Provincially elected senators were also insulated on the demand side. They answered to provincial councils and had no electoral constituency. We exclude them from the estimation sample. The control group therefore consists of senators who did face an electorate, rather than senators who did not (Online Appendix OA12 compares the two groups).

Both treatments are also bundles of institutional changes. The 1893 revision combined plural suffrage with compulsory voting. The 1899 law combined D’Hondt allocation with district mergers. The coefficients therefore capture the reform packages as enacted.

We assign directly elected senators to the constituency recorded in the proceedings, as we do for deputies. For merged constituencies, we assign them to a component *arrondissement* of the 1896 census map. Online Appendix OA11 describes this rule and shows that the results do not depend on it.

## 7.2 Estimating equations

**The 1894 margin.** For plural suffrage we estimate, on interventions  $s$  by politician  $i$  in year  $t$  over 1880–1899,

$$y_{ist} = \beta (\text{Chamber}_i \times \text{Post1894}_t) + \gamma \text{Chamber}_i + \rho \text{Post1894}_t + \delta_{d(i)} + \phi_{p(i)} + \tau_t + \varepsilon_{ist}, \quad (11)$$

with arrondissement ( $\delta$ ), party ( $\phi$ ) and year ( $\tau$ ) fixed effects, observations weighted by the number of words in the intervention, and standard errors clustered by arrondissement. The coefficient of interest is  $\beta$ , the estimate of  $\beta_C - \beta_S$  in equation (10).

**The 1900 margin.** For proportional representation the prediction is a gradient rather than a shift, so we interact the design with the predetermined mining measure  $\text{Mine}_d$  over 1894–1906:

$$\begin{aligned} y_{ist} = & \theta (\text{Mine}_{d(i)} \times \text{Post1900}_t \times \text{Chamber}_i) \\ & + \lambda_1 (\text{Mine}_{d(i)} \times \text{Post1900}_t) + \lambda_2 (\text{Mine}_{d(i)} \times \text{Chamber}_i) \\ & + \lambda_3 (\text{Post1900}_t \times \text{Chamber}_i) + \mu \text{Mine}_{d(i)} + \gamma \text{Chamber}_i + \rho \text{Post1900}_t \\ & + \delta_{d(i)} + \phi_{p(i)} + \tau_t + \varepsilon_{ist}. \end{aligned} \quad (12)$$

All main effects and pairwise interactions are included. The coefficient of interest is  $\theta$ . It measures the change after 1900 in the mining gradient of the Chamber–Senate difference.

The estimate of  $\lambda_3$  from the same regression and the estimate of  $\beta$  from equation (11), using the 1900 cutoff, measure the aggregate Chamber–Senate shift. The framework predicts no such shift.

We also estimate the same triple difference over 1880–1899, using the 1894 cutoff. This captures the coalfield heterogeneity of the first reform.

The window for the second reform closes in 1906 for the reason given in Section 3. Directly elected socialist senators number four from 1904, and the Senate’s mining gradient turns positive from that legislature, so the control group is already contaminated at the end of the window and further contaminated beyond it. Closing earlier would therefore give a larger estimate, not a smaller one, and every legislature added past 1906 puts more treated observations into the control group. Section 11 reports the estimate under every closing legislature<sup>18</sup>.

<sup>18</sup>Constituency codes enter in two forms: the difference-in-differences specifications use the parliamentary archive’s own constituency coding, and the triple-difference specifications use a harmonised coding built to merge the 1896 census, which reports employment by administrative arrondissement rather than by the archive’s constituency labels. Both partition the sample into 41 units.

### 7.3 Adaptation and recomposition

Equation (3) distinguishes two margins. The intensive margin is sitting members changing  $a^*$  at fixed type. The extensive margin is a change in the distribution of types among those elected. We separate the two in three ways.

First, we estimate each specification with and without politician fixed effects. With politician fixed effects, identification comes only from within-person variation. Therefore, changes in who sits and speaks are absorbed. Hence, the difference between the pooled and fixed-effects estimates captures composition, together with any difference in response between politicians observed on both sides of the reform and the rest.

Second, we split politicians by entry cohort. Incumbents are those first observed before the reform, while entrants first appear at or after it. Since entrants have no pre-reform observations, their estimate is therefore a cross-sectional gradient within the post-reform period. We compare it with the incumbents' cross-sectional gradient over the same period.

Third, the sequence of two reforms lets us relate recomposition to the electoral rule rather than simply to time. We estimate the mining gradient among entrants separately for cohorts entering under the censitary regime, under plural suffrage with majoritarian elections, and under proportional representation. If the gradient is driven by the electoral rule, it should be flat under the first two regimes and emerge under proportional representation. If it instead reflects the secular growth of socialist pressure, it should increase across all three cohorts.

### 7.4 Inference

Standard errors are clustered by arrondissement, giving 41 clusters.

For the 1894 reform, treatment varies within every arrondissement. The wild cluster bootstrap with the null imposed is therefore informative, and we report it.

Inference is more difficult for the 1900 reform. The exposure measure is constant within arrondissement, and three districts account for about three quarters of the leverage on the triple interaction. The effective number of clusters is therefore only about three. We report cluster-robust  $t$ -statistics, but we do not rely on them for inference on this margin.

Instead, we use randomization inference. We permute the district exposure map across the 41 arrondissements and re-estimate the statistic while keeping the sample, weights, and fixed effects unchanged (Fisher, 1935; MacKinnon & Webb, 2020; Young, 2019). We apply the same procedure to every split-sample coefficient reported in the paper.

Section 11 reports the permutation distributions and the leave-one-district-out exercises. It also addresses a separate limitation of the 1894 design. The treatment is a single policy change applied to a single chamber, which no variance estimator can solve. We

therefore address this concern with direct tests of the specific alternative explanations.

## 8 Plural Suffrage: Adaptation, Everywhere

### 8.1 The average effect

Table 8 reports estimates of equation (11). Plural suffrage increased the Chamber’s labour salience relative to the Senate by 4.46 percentage points (s.e. 2.10,  $t = 2.12$ ).

Adding indicators for members who first appear or last appear within one legislature of the reform leaves the estimate unchanged at 4.46 (s.e. 2.11; Table 8, column 2). This check ensures that the result is not driven by members entering or leaving around the cutoff.

With politician fixed effects, the estimate falls to 2.92 (s.e. 2.51,  $t = 1.16$ ; Table 8, column 3). It is smaller and less precise, but measures the same reform contrast using only within-person variation. This specification absorbs politician, party, arrondissement, and year effects<sup>19</sup>.

Table 6 reports the levels behind these coefficients. Before 1894, labour salience averages 7.89 per cent of substantive speech in the Chamber and 8.07 per cent in the Senate. After 1894, the corresponding figures are 16.18 and 11.75 per cent.

The raw difference in differences from these four means (Table 6) is 4.61 percentage points. The regression estimate, with fixed effects and length weights, is 4.46.

Both chambers therefore increase their attention to labour. Labour salience roughly doubles in the Chamber and rises by almost half in the Senate. The result is not that the Chamber responded while the Senate did not. It is that the Chamber responded more.

Relative to the Chamber’s pre-reform level, the differential effect is large: 4.46 percentage points is more than half of the baseline. If the Senate response is non-negative, this estimate is also a lower bound on the Chamber’s own response. The standard errors are wide, however, and Section 11 returns to inference.

---

<sup>19</sup>All specifications with politician fixed effects use the archive identifier described in Section 4.

Table 8: Plural suffrage and labour salience, 1880–1899

|                                            | Difference in differences |                    |                  | Triple difference    |                      |                  |
|--------------------------------------------|---------------------------|--------------------|------------------|----------------------|----------------------|------------------|
|                                            | (1)                       | (2)                | (3)              | (4)                  | (5)                  | (6)              |
| Chamber $\times$ Post-1894                 | 4.455**<br>(2.100)        | 4.455**<br>(2.108) | 2.917<br>(2.507) |                      |                      |                  |
| Mining $\times$ Post-1894 $\times$ Chamber |                           |                    |                  | -3.238***<br>(1.031) | -3.333***<br>(0.970) | 0.189<br>(1.360) |
| Arrondissement, party, year FE             | Yes                       | Yes                | Yes              | Yes                  | Yes                  | Yes              |
| Entry/exit controls                        | No                        | Yes                | No               | No                   | Yes                  | No               |
| Politician FE                              | No                        | No                 | Yes              | No                   | No                   | Yes              |
| Observations                               | 208,325                   | 208,325            | 208,190          | 208,325              | 208,325              | 208,190          |

Outcome: labour-bucket share of classified policy language, in percentage points. Sample 1880–1899. Observations weighted by intervention length. Standard errors in parentheses, clustered by arrondissement (41 clusters); columns 1–3 use the raw constituency coding, columns 4–6 the harmonised coding. Mining is the standardised mining share of the arrondissement in the 1896 industrial census, a continuous measure defined for all 41 arrondissements. Politician fixed effects are defined on the parliamentary archive’s person identifier. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The framework predicts that an increase in  $n^W$  should raise the return to labour attention across districts, rather than only in a few constituencies. The data are consistent with this prediction.

When we exclude one arrondissement at a time, the estimate remains positive in all 41 cases. It is significant at the 5 per cent level in 38 of them and ranges from 2.71 to 5.58, with a standard deviation of 0.37 (Appendix Table A13, Panel D)<sup>20</sup>.

## 8.2 The 1894 channel is adaptation

Having established the aggregate effect, we now ask how it was produced. Did sitting members change their behaviour, or did the Chamber change because new members replaced old ones?

Table 9 first restricts the sample to incumbents—politicians first observed before 1894 and still sitting after the reform. The estimate is 3.88 (s.e. 1.81,  $t = 2.14$ ), close to the aggregate estimate of 4.46. Most of the Chamber–Senate shift therefore remains when entry after the reform is excluded.

A stricter test follows exactly the same politicians before and after the reform. Among the 179 Chamber deputies observed in both periods, labour salience rises from 7.9 per cent under the censitary regime to 15.3 per cent under plural suffrage. Among the 105

<sup>20</sup>The three exclusions that fall below the 5 per cent threshold are Alost, Brussels, and Leuven, with  $t$ -statistics of 1.77, 1.89, and 1.91.

senators observed in both periods, it rises from 8.3 to 12.0 per cent. Both chambers move, but deputies move more: by 7.4 percentage points compared with 3.7 for senators (Table 9, Panel B).

The corresponding within-person difference in differences is 3.24 (s.e. 2.52,  $t = 1.28$ ; Table 9, Panel B). Adding politician fixed effects within the incumbent sample gives 3.19 (s.e. 2.55; Panel A of the same table). These estimates use only variation within politicians and therefore ask more of the data than the aggregate specification. They are less precise, but their magnitudes are consistent with the 3.7-point difference in the raw changes. We therefore treat them as evidence on the mechanism rather than as separate tests of whether the aggregate effect exists.

The composition channel points in the same direction but is much smaller. Deputies entering in 1894 are more labour-oriented than the incumbents they joined: 18.9 versus 15.3 points (Table 7). Their post-reform Chamber–Senate gap is 4.95 (s.e. 2.50; Table 9, Panel A). However, they account for only 13.5 per cent of Chamber speech in the majoritarian legislatures. Entry alone therefore cannot explain the aggregate shift.<sup>21</sup>

Taken together, the evidence points to adaptation as the main channel after 1894. The rise in labour attention is already present among politicians who were sitting before the reform and remained afterward. Changes in composition reinforce that movement, but account for only a small part of it.

---

<sup>21</sup>Entry cohorts are defined using the archive person identifier adopted in Section 4.

Table 9: Adaptation versus composition at the 1894 margin

|                                                             | Estimate  | Std. err. | Obs.    |
|-------------------------------------------------------------|-----------|-----------|---------|
| <i>Panel A. Who moves</i>                                   |           |           |         |
| All politicians                                             | 4.455**   | (2.100)   | 208,325 |
| Incumbents only                                             | 3.881**   | (1.810)   | 162,255 |
| Incumbents, politician FE                                   | 3.193     | (2.551)   | 162,209 |
| Entrants, Chamber gap                                       | 4.949*    | (2.497)   | 9,299   |
| <i>Panel B. The same people, against their own baseline</i> |           |           |         |
| Deputies on both sides, mean 1880–93                        | 7.90      |           | 108,436 |
| Deputies on both sides, mean 1894–99                        | 15.31     |           | 37,975  |
| Senators on both sides, mean 1880–93                        | 8.29      |           | 9,847   |
| Senators on both sides, mean 1894–99                        | 12.03     |           | 4,096   |
| Within-deputy, Chamber                                      | 6.368***  | (2.071)   | 146,410 |
| Within-senator, Senate                                      | -0.826    | (1.523)   | 13,919  |
| Within-person difference in differences                     | 3.236     | (2.524)   | 160,354 |
| <i>Panel C. Geography</i>                                   |           |           |         |
| Outside the coalfields                                      | 6.189***  | (2.271)   | 179,517 |
| Inside the coalfields                                       | -1.373    | (1.841)   | 28,808  |
| Difference (interacted)                                     | -8.339*** | (2.762)   | 208,325 |
| Inside the coalfields, joint model                          | -2.270    | (1.812)   | 208,325 |
| <i>Panel D. Party</i>                                       |           |           |         |
| Catholic                                                    | 7.187**   | (2.787)   | 124,236 |
| Liberal                                                     | -2.004    | (1.830)   | 84,086  |

Panel B restricts to the 179 Chamber deputies and the 105 senators observed both before and after the reform, on the parliamentary archive’s person identifier. Its first four rows are word-weighted means of the outcome, with the number of interventions in the last column; the three rows below them absorb politician fixed effects. Row samples in Panel B are not additive: singleton groups are dropped within each estimation sample, so the pooled within-person specification retains observations that are singletons in the chamber-specific rows. Entrants have no pre-reform observations, so their row is a post-reform cross-section and not a difference. In Panel C the coalfields are the four arrondissements with the highest mining share in the 1896 industrial census—Mons, Charleroi, Soignies and Liège—entered as an indicator. The first two rows are split samples, each with its own year and party effects; the last two come from one regression that interacts the coalfield indicator with the full design, and the inside effect is the sum of its two coefficients. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Section 11 examines two alternative explanations for the Chamber premium. Neither overturns the result.

First, ministers speak before both chambers and account for 15.9 per cent of recorded words. Removing every intervention delivered while the speaker held ministerial office leaves an estimate of 3.08 (s.e. 1.41; Online Appendix Table OA17).

Second, the socialist deputies who entered the Chamber in 1894 may have pushed labour onto its agenda. We therefore restrict the sample to sitting days on which they did not speak about labour. The within-politician estimate is 3.02, with a  $t$ -statistic of 2.30 (Table 18). It is as large as the baseline within-politician estimate of 2.92 and more precisely estimated, and we treat it as the cleanest estimate of adaptation (Section 11.4).

### 8.3 Timing, pre-trends and the closing of the gap

Figure 3 plots the Chamber–Senate difference by legislature. The difference is unchanged in the 1894 legislature itself (−0.10; Appendix Table A18) and opens in 1896 (9.59,  $t = 2.12$ ) and 1898 (3.59,  $t = 2.19$ ). Moving the treatment cutoff to 1896 raises the estimate to 6.36 (s.e. 1.85,  $t = 3.43$ ), and excluding the 1894 legislature entirely gives 6.58 (s.e. 2.06,  $t = 3.20$ ; Appendix Table A13, Panel C). The plural-suffrage election was held on 14 October 1894 and the resulting legislature’s first full session ran into 1895, so the 1894 legislature is short (963,271 recorded Chamber words against 1.4 million in 1892; Appendix Table A18); a behavioural response that takes a session to appear is what adaptation would look like. We describe the effect as a divergence beginning with the first full post-reform session and report the alternative cutoffs rather than choosing among them.

The identifying assumption is that the Chamber–Senate difference would have evolved in parallel absent the reform. Interacting the Chamber indicator with a linear trend over the pre-1894 period gives −0.022 (s.e. 0.079; Appendix Table A13, Panels A and B). Placebo reforms at 1886, 1888 and 1890, estimated on the pre-1894 sample, give −1.16 (s.e. 0.90), −0.97 (s.e. 0.95) and 1.96 (s.e. 1.14). Therefore, none is significant at the 5 per cent level, although the 1890 placebo has a  $t$ -statistic of 1.7.

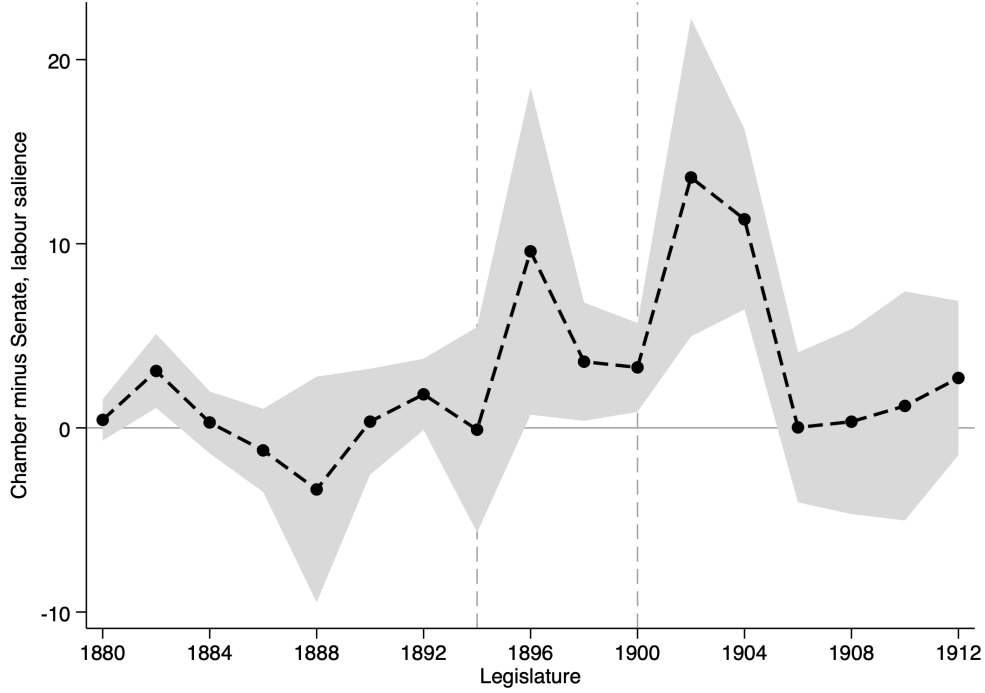


Figure 3: Chamber–Senate difference in labour salience, by legislature. Each point is the coefficient on the Chamber indicator estimated within a legislature, with arrondissement and party fixed effects and observations weighted by intervention length; the shaded band is the 95 per cent confidence interval. Vertical lines mark the first legislatures elected under plural suffrage (1894) and under proportional representation (1900).

The figure also shows the Chamber–Senate gap narrowing to about two points from 1906 onward. This convergence comes from the Senate catching up, not from the Chamber falling back.

Chamber labour salience remains high: 11.6 in 1900, 23.5 in 1902, 20.2 in 1904, and 18.3 in both 1906 and 1910 (Appendix Table A18). The Senate instead rises from 6.7 in 1900 and 10.7 in 1902 to 15.4 in 1906 and 15.2 in 1908.

This pattern is consistent with the framework. Once socialists can win directly elected Senate seats, the eligibility term  $e_S$  rises. The first socialist Senate seat appears in 1900, with further gains from 1904. The Senate was therefore never fully insulated. Instead, its response was delayed and weaker.

The narrowing Chamber–Senate gap should not be read as the Chamber effect disappearing. Rather, the control group becomes less insulated and begins to catch up. From that point, the comparison becomes less informative. This is also why the estimation window for the second reform ends when it does.

## 8.4 The coalfield reversal

The framework predicts a weaker 1894 response in districts where plural suffrage handed the seats to the socialists. The establishment members who remained there were a selected

group, so the effect could even turn negative. This is what we find.

Estimated separately, the Chamber–Senate effect is 6.19 (s.e. 2.27) outside the four coalfield districts and  $-1.37$  (s.e. 1.84) inside them (Table 9, Panel C). In the joint specification, which interacts the coalfield indicator with the full design, the difference between the two areas is  $-8.34$  (s.e. 2.76,  $t = -3.02$ ). The implied effect inside the coalfields is  $-2.27$  (s.e. 1.81; same panel).

The continuous mining measure gives the same pattern. At mean exposure, the 1894 effect is 4.76 (s.e. 1.93; Appendix Table A8, column 4), while the triple interaction with mining exposure is  $-3.24$  (s.e. 1.03,  $t = -3.14$ ; Table 8, column 4).

The reversal is compositional, as the framework predicts. When we add politician fixed effects, the mining-interacted coefficient moves from  $-3.24$  to 0.19 (s.e. 1.36; column 6). The same deputies in the coalfields therefore did not reduce their labour salience after 1894. Instead, the aggregate changed because the set of deputies speaking from these districts changed, consistent with the displacement documented in Table 10.

The reversal also does not reflect a pre-existing coalfield trend. In the censitary period alone (1880–1892), the mining  $\times$  Chamber interaction with a linear trend is 0.006 (s.e. 0.053). Placebo cutoffs in 1886, 1888, and 1890 give estimates of  $-0.08$ ,  $-0.27$ , and 0.26, none statistically significant (Appendix Table A11).

The coalfield gradient is therefore flat before the reform and turns negative in 1894. In particular, the unrest of 1886 does not appear to have opened the gap earlier.

Table 10: Establishment presence in the coalfields, by legislature

| Legislature | Coalfield share of deputies | Coalfield share of words |
|-------------|-----------------------------|--------------------------|
| 1880        | 14.3%                       | 12.1%                    |
| 1884        | 14.2%                       | 15.7%                    |
| 1888        | 15.5%                       | 19.6%                    |
| 1892        | 15.3%                       | 10.3%                    |
| 1894        | 10.9%                       | 3.7%                     |
| 1896        | 10.7%                       | 3.2%                     |
| 1898        | 5.2%                        | 4.1%                     |
| 1900        | 10.1%                       | 6.3%                     |
| 1902        | 10.3%                       | 12.9%                    |
| 1904        | 12.8%                       | 13.1%                    |
| 1906        | 14.1%                       | 12.5%                    |

*Notes:* The first column is the coalfield share of the distinct Catholic and Liberal deputies with recorded speech in the Chamber; the second is their share of Chamber speech. Coalfields are Mons, Charleroi, Soignies and Liège. Shares rather than counts, because a member who sits in more than one legislature is counted in each, and a legislature interrupted by by-elections returns more distinct members than it has seats. The blank lines mark the first legislatures elected under plural suffrage and under proportional representation.

## 8.5 By party

The 1894 effect is concentrated among Catholic members. Estimated separately by party, the Chamber–Senate effect is 7.19 for Catholics (s.e. 2.79,  $t = 2.58$ ) and  $-2.00$  for Liberals (s.e. 1.83,  $t = -1.09$ ; Table 9, Panel D).

This pattern is consistent with the political history described in Section 2. The Christian-democratic wing of the Catholic party entered the Chamber in force in 1894. The party also had an organisation and a political tradition that allowed it to compete for working-class votes, especially in Flanders. The Liberals, whose support was more urban and bourgeois and who lost heavily in 1894, were less well positioned to do so.

Appendix OA8 decomposes each party’s change into a within-member component and a turnover component.

## 9 Proportional Representation: Recomposition, in the Coalfields

### 9.1 No aggregate effect, a large local one

Proportional representation did not produce a detectable change in the aggregate Chamber–Senate difference. The coefficient is 1.48 (s.e. 3.06; Table 11, column 1)<sup>22</sup>.

The framework instead predicts that the effect should vary with the pre-existing electoral distortion, and this is what we find. The triple difference in equation (12) is 9.25 (s.e. 2.35,  $t = 3.94$ ). Adding turnover controls gives 7.50 (s.e. 1.83; Table 11, columns 4 and 5).

The same pattern appears when we split the sample geographically. After 1900, the Chamber–Senate difference is  $-0.82$  (s.e. 2.68) outside the coalfields and 13.44 (s.e. 2.07) inside them (Table 11, lower panel). The second reform therefore produces no detectable additional shift outside the coalfields, while the increase is concentrated in the districts where the majoritarian rule had been most distortive.<sup>23</sup>

In particular, this is not because proportional representation failed to expand socialist representation outside the coalfields. Socialist representation in non mining areas tripled after 1900 (Table 5). Yet Catholic and Liberal deputies in those districts show no comparable increase in labour attention. At the same time, Table 6 shows that aggregate Chamber labour salience remains around 18 per cent after 1900 rather than falling back.

The second reform therefore adds a geographically concentrated effect on top of the higher level of labour attention already reached after 1894.

Besides, table 6 shows what lies behind the gradient. Before 1900, labour salience in the four coalfield districts averages 12.71 in the Chamber and 16.06 in the Senate. Established-party deputies from the coalfields therefore spoke less about labour than senators from the same districts. This is the displacement documented in Section 8.4, now visible in levels.

After 1900, the Chamber figure rises to 34.88, while the Senate figure is 13.65 (Table 6). The Chamber level therefore nearly triples, while the Senate level falls slightly. Part of the triple difference thus comes from the Senate decline. The pre-1900 coalfield cell is also relatively thin, with 1,525 interventions compared with 7,273 after 1900.

---

<sup>22</sup>This result is not sensitive to the choice of the closing legislature. Starting in 1894 and ending the sample in each legislature from 1902 to 1912 gives estimates of 3.68, 3.53, 1.48,  $-1.40$ ,  $-2.00$ , and  $-1.86$ . Standard errors remain close to three, and no  $t$ -statistic exceeds 1.2 (Appendix Table A19).

<sup>23</sup>These are split-sample estimates, with year and party effects estimated separately within each group. In a joint specification that interacts the coalfield indicator with the full design, the coalfield difference is 24.03 (s.e. 4.35) and the implied effect inside the coalfields is 23.07 (s.e. 3.45; both in the lower panel of Table 11). The joint model gives a larger estimate because the year effects are pinned down mainly by the 37 non-coalfield districts rather than by four thin coalfield cells. We use the split-sample estimates throughout because they can be recovered directly from Table 11. The inside-coalfield estimate is identified from four districts, and its randomization-inference  $p$ -value is 0.055 (Section 11.2).

Finally, a long difference comparing the censitary period directly with the proportional-representation period gives 3.12 (s.e. 3.01) with the continuous mining measure and 12.75 (s.e. 7.47) with the coalfield indicator (Appendix Table A16). Both estimates are positive but imprecise. For this reason, we can claim that proportional representation changed who represented the coalfields and was accompanied by a sharp increase in the labour attention of their Chamber representatives. The data do not establish that labour attention in the coalfields ultimately rose above its censitary-period level.

---

<sup>24</sup>The size of the increase should also be interpreted relative to the depressed post-1894 baseline. Since plural suffrage displaced established-party deputies from the coalfields, part of the post-1900 increase represents recovery: in 1894–99 the Chamber’s mining gradient lies 4.53 points below the Senate’s (s.e. 1.09; Mining  $\times$  Chamber in Appendix Table A9, column 4).

Table 11: Proportional representation and labour salience, 1894–1906

|                                                                                   | Difference in differences |                  |                  | Triple difference   |                     |                   |
|-----------------------------------------------------------------------------------|---------------------------|------------------|------------------|---------------------|---------------------|-------------------|
|                                                                                   | (1)                       | (2)              | (3)              | (4)                 | (5)                 | (6)               |
| Chamber $\times$ Post-1900                                                        | 1.475<br>(3.058)          | 0.935<br>(2.792) | 2.788<br>(2.386) |                     |                     |                   |
| Mining $\times$ Post-1900 $\times$ Chamber                                        |                           |                  |                  | 9.253***<br>(2.351) | 7.503***<br>(1.832) | -3.474<br>(2.136) |
| Arrondissement, party, year FE                                                    | Yes                       | Yes              | Yes              | Yes                 | Yes                 | Yes               |
| Entry/exit controls                                                               | No                        | Yes              | No               | No                  | Yes                 | No                |
| Politician FE                                                                     | No                        | No               | Yes              | No                  | No                  | Yes               |
| Observations                                                                      | 119,448                   | 119,448          | 119,341          | 119,448             | 119,448             | 119,341           |
| <i>Where the effect is: Chamber <math>\times</math> Post-1900 by geography</i>    |                           |                  |                  |                     |                     |                   |
| Outside the coalfields                                                            | -0.817                    |                  | (2.675)          |                     | 106,673             |                   |
| Inside the coalfields                                                             | 13.441***                 |                  | (2.071)          |                     | 12,775              |                   |
| Difference (interacted)                                                           | 24.030***                 |                  | (4.351)          |                     | 119,448             |                   |
| Inside the coalfields, joint model                                                | 23.072***                 |                  | (3.450)          |                     | 119,448             |                   |
| <i>By party: Mining <math>\times</math> Post-1900 <math>\times</math> Chamber</i> |                           |                  |                  |                     |                     |                   |
| Catholic                                                                          | 7.436                     |                  | (12.344)         |                     | 86,578              |                   |
| Liberal                                                                           | 5.850***                  |                  | (1.675)          |                     | 32,867              |                   |

Sample 1894–1906. The window closes in 1906 because the Senate begins to respond to mining exposure from that legislature, when directly elected socialist senators first sit in numbers. Mining is the standardised mining share of the arrondissement in the 1896 industrial census, a continuous measure defined for all 41 arrondissements; it enters columns 4–6 and the party panel. The coalfields are the four arrondissements with the highest mining share—Mons, Charleroi, Soignies and Liège—and enter the geography panel only, as an indicator. Politician fixed effects are defined on the parliamentary archive’s person identifier. In the geography panel the first two rows are split samples, each with its own year and party effects; the last two come from one regression that interacts the coalfield indicator with the full design. The party panel re-estimates column 4 separately for each party, on that party’s members in all 41 arrondissements. The Catholic sample is therefore large, but its standard error reflects the near-absence of Catholic deputies from the coalfields after 1894, documented in Table 10: the districts where mining exposure is high are those where they no longer sit. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Figure 4 plots the mining gradient within each chamber, legislature by legislature. The Chamber gradient is flat and close to zero through the censitary and majoritarian periods, jumps after 1900 and peaks in 1902–1904, when the deputies elected under the new rule are sitting. The Senate gradient is small and unstable through the majoritarian and early proportional legislatures (4.7 in 1894, 3.1 in 1896, then  $-0.9$ ,  $0.2$  and  $-1.1$  from 1898 to 1902; Appendix Table A18). From 1904 it turns positive and stays there, reaching 6.0 in 1906. That is the contamination the estimation window is chosen against,

and it arrives when the framework implies it should.

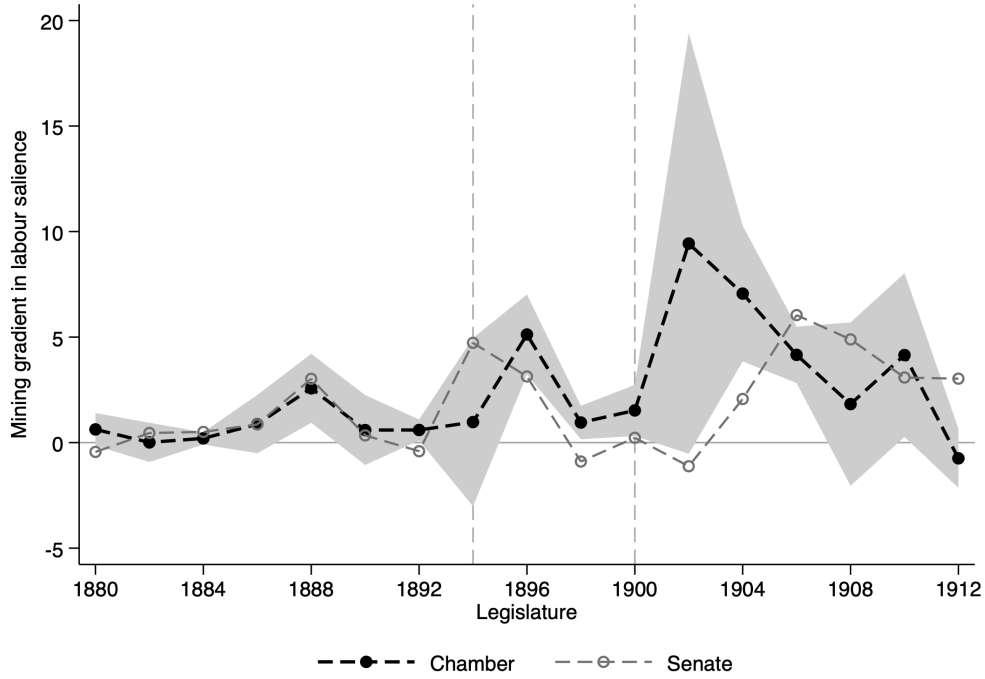


Figure 4: Mining gradient in labour salience, by legislature and chamber. Each point is the coefficient on standardised mining share estimated within a chamber and legislature, with party fixed effects and observations weighted by intervention length; the shaded band is the 95 per cent confidence interval for the Chamber. Vertical lines mark 1894 and 1900.

The sign of the gradient reverses across electoral regimes. Using the 1894 cutoff, the triple difference is  $-3.24$  (s.e. 1.03). With turnover controls, it is  $-3.33$  (s.e. 0.97; Table 8, columns 4 and 5).

The mining gradient is therefore negative after 1894 under majoritarian rules. After 1900, under proportional representation, it becomes positive.

The exposure measure is fixed and predetermined. District fixed effects are included throughout. The main institutional difference between the two reform periods is the electoral rule.

Table 5 shows the corresponding change in representation. Under the majoritarian rule, socialists held 68 per cent of coalfield seats. Under proportional representation, their share fell to 51 per cent. Catholic and Liberal candidates returned to the seats that the socialists no longer held. The next subsection examines these entrants.

## 9.2 The channel in 1900 is recomposition

Table 12 splits the post-1900 sample by entry cohort. The mining gradient is concentrated among entrants. Deputies entering in 1900 or later have a gradient of 11.07 (s.e. 1.83,  $t = 6.04$ ), the most precisely estimated coefficient in the paper.

The comparable gradient among incumbents is much smaller. Estimated over the same post-1900 period, with the same weights and fixed effects, it is 0.63 (s.e. 0.73); Appendix Table A22). The difference between entrants and incumbents is 8.59 (s.e. 1.61; Appendix Table A22). The post-1900 mining gradient is therefore concentrated among new Catholic and Liberal deputies.

The within-politician estimates point in the same direction. Among incumbents, adding politician fixed effects gives  $-3.99$  (s.e. 2.19). In the full Panel B sample, the estimate is  $-3.47$  (s.e. 2.14; Table 11, column 6). Neither estimate is statistically distinguishable from zero. Therefore, there is no evidence that sitting members increased their own labour salience after the reform.

The pooled triple difference, by contrast, is 9.25 (Table 11, column 4). Taken together, these results point to recomposition as the main channel after 1900. The members carrying the new mining gradient are the entrants, not the incumbents<sup>25</sup>.

Table 12: Recomposition at the 1900 margin

|                               | Estimate  | Std. err. | Obs.    |
|-------------------------------|-----------|-----------|---------|
| Incumbents, triple difference | 4.490**   | (1.739)   | 100,174 |
| Incumbents, politician FE     | -3.989*   | (2.187)   | 100,136 |
| Entrants, mining gradient     | 11.073*** | (1.833)   | 14,182  |

Incumbents are politicians first observed before 1900 and still sitting after; entrants are first observed in 1900 or later and therefore have no pre-reform period, so their row is a cross-sectional gradient. All three rows use the continuous mining share of Table 11, defined for all 41 arrondissements, and not the coalfield indicator: the first two report the Mining  $\times$  Post-1900  $\times$  Chamber coefficient, the third the Mining  $\times$  Chamber coefficient among entrants. Politician fixed effects are defined on the parliamentary archive's person identifier. The sign difference between the pooled incumbent estimate and the within-incumbent estimate is not a contradiction: across incumbents the mining gradient is positive, within each incumbent it is not. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

About half of the entrant gradient comes from members who later held cabinet office. We report this in the here because it is part of the mechanism, not simply a robustness check. Ministers defend legislation before the Chamber, so some of their labour attention may reflect their portfolio rather than electoral incentives alone.

We therefore remove every intervention delivered while the speaker held cabinet office. The entrant gradient falls from 11.07 to 4.95 (s.e. 1.42), while the triple difference falls to 4.79 (s.e. 1.43). Both estimates remain positive and are more precise than the baseline

<sup>25</sup>This does not mean that incumbents could not have adapted. Few establishment deputies survived in the coalfields after 1894, so a within-person test there has limited power. The stronger result is compositional: the post-1900 gradient appears among new members, while no comparable gradient appears among incumbents.

estimates<sup>26</sup>.

The result is concentrated in the Industry and Labour portfolio. Excluding only the nine members who held that portfolio gives an estimate of 5.24. Cabinet membership in general therefore does not account for the difference. Online Appendix OA13 reports the full set of results.

The post-1900 result is therefore still one of recomposition, but part of that recomposition operates through office holding. Some of the entrants who carried the coalfield gradient also became responsible for labour policy. The gradient should therefore not be interpreted as arising entirely from backbench speech or as independent of who entered the cabinet.

### 9.3 The gradient tracks the rule, not the calendar

Table 13 reports the mining gradient among entrants under each electoral regime. For cohorts entering under the censitary franchise, the gradient is  $-0.58$  (s.e. 1.08). For those entering under plural-suffrage majoritarian rules, it is  $-2.11$  (s.e. 2.54). For entrants under proportional representation, it rises to 11.07 (s.e. 1.83).

Socialist electoral strength increased across all three periods. The entrant gradient did not. It appears only under proportional representation.

The electoral reform of 1899 allowed Catholic and Liberal candidates to win seats again in the coalfields. Therefore, the entrants elected under the new rule are more concentrated in mining districts, and it is among this cohort that the positive mining gradient appears.

Table 13: Entrant mining gradient by entry regime

| Entry regime            | Electoral rule       | Estimate  | Std. err. | Obs.   |
|-------------------------|----------------------|-----------|-----------|--------|
| Censitary, 1882–1892    | Restricted franchise | -0.580    | (1.083)   | 46,734 |
| Majoritarian, 1894–1898 | Plural suffrage      | -2.114    | (2.544)   | 9,299  |
| Proportional, 1900–     | PR                   | 11.073*** | (1.833)   | 14,182 |

Mining  $\times$  Chamber gradient among politicians whose first observed legislature falls in each regime, estimated within that regime. Socialist electoral strength grew steadily across all three periods; the entrant gradient did not. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### 9.4 By party

The 1900 effect is precisely estimated only for Liberals. The triple difference is 5.85 for Liberals (s.e. 1.68,  $t = 3.49$ ) and 7.44 for Catholics (s.e. 12.34; Table 11, lower panel). The Catholic estimate is therefore too imprecise to be informative.

<sup>26</sup>Under randomization inference, the corresponding  $p$ -values are 0.104 and 0.057 (Table 17 and Section 11.2).

This imprecision reflects the weak Catholic presence in the coalfields after 1894. There is little mining-interacted variation left within the Catholic subsample, consistent with the displacement documented in Table 10.

The Liberal pattern is clearer. Liberals retained an industrial-bourgeois base in Liège, and proportional representation gave them a route back into coalfield constituencies. The entrant gradient is 12.46 (s.e. 1.14) for Liberals and 2.45 (s.e. 2.69) for Catholics (Online Appendix Table OA13).

## 10 Legislation

Parliament can talk about the social question without legislating on it. Therefore, a result confined to speech would therefore say little about whether political attention translated into legislative action.

This section asks whether the legislative record moved in the same direction, in the same period. The identification is different from the speech analysis. Thus, we treat the results as corroborating evidence, not as a second estimate of the same effect.

A law belongs to parliament as a whole, not to one chamber. For this reason, there is no insulated control group. Instead, the analysis compares legislative initiative across the three electoral regimes.

In particular, this comparison cannot separate the reforms from the political events that preceded them. The legislative series, for example, rises after the unrest of 1886 as well as after 1894 (Online Appendix OA15)<sup>27</sup>.

The legislative evidence can answer a narrower question. Did the composition of legislative initiative change in the direction suggested by the speech results? The answer is yes.

The outcome is the extensive-margin measure defined in Section 5. As in the speech analysis, we apply the same party restriction. The sample contains 2,041 bills introduced by Catholic and Liberal members (Appendix Table A23). Since Government bills adoption rate is about four fifths, compared with about one quarter for members' bills, we kept the two groups separate, because pooling the two would obscure important differences.

### 10.1 Initiative

Table 14 reports bills per year and the share of each sponsor's proposals devoted to labour and social policy.

The results show that, for Catholic and Liberal members, the main increase occurs after 1894. Before the reform the party members introduced 5.2 labour and social bills per year, in the period 1894–99 14.5, and 16.5 after 1900. As a share of their own

---

<sup>27</sup>This is precisely why the main speech analysis does not rely on a simple time-series comparison.

proposals, the corresponding figures are 0.126, 0.211, and 0.205 (Table 14). The largest shift therefore occurs in 1894, not in 1900.

On the other side, Government bills follow a different pattern. Labour and social bills rise from 3.0 per year before 1894 to 5.8 in the period 1894–99, then fall to 3.4 after 1900 (Table 14). The increase is therefore concentrated in the plural-suffrage period.

Besides, for the period after the proportional representation reform the party split is more informative. On one hand, Catholic members move from 3.0 labour bills per year before 1894 to 10.7 in 1894–99, and then remain at 10.2 after 1900. On the other, Liberal members move from 2.2 to 3.8 and then to 6.3 (Table 14).

The legislative record therefore mirrors the party pattern in speech. The 1894 increase is concentrated among Catholics, while the additional increase after 1900 is concentrated among Liberals (Figure 5).

It is also interesting looking at what does not change throughout time. Members' labour bills were promulgated at 0.288, 0.264 and 0.265 across the three periods, and government bills at 0.833, 0.743 and 0.818 (Table 14). Neither rate moves, and the gap between them is large. In fact, a bill carried by the cabinet was about three times as likely to become law as a bill on the same subject carried by a member.

Table 14: Legislative initiative and enactment on the social question

|                                                                        | 1880–1893 | 1894–1899 | 1900–1912 |
|------------------------------------------------------------------------|-----------|-----------|-----------|
| <i>Panel A. Labour and social bills deposited per year</i>             |           |           |           |
| Catholic and Liberal members                                           | 5.2       | 14.5      | 16.5      |
| Government                                                             | 3.0       | 5.8       | 3.4       |
| of which Catholic                                                      | 3.0       | 10.7      | 10.2      |
| of which Liberal                                                       | 2.2       | 3.8       | 6.3       |
| <i>Panel B. Labour bills as a share of the sponsor's own proposals</i> |           |           |           |
| Catholic and Liberal members                                           | 0.126     | 0.211     | 0.205     |
| Government                                                             | 0.043     | 0.078     | 0.061     |
| <i>Panel C. Share of labour bills promulgated</i>                      |           |           |           |
| Catholic and Liberal members                                           | 0.288     | 0.264     | 0.265     |
| Government                                                             | 0.833     | 0.743     | 0.818     |
| <i>Labour bills deposited</i>                                          |           |           |           |
| Catholic and Liberal members                                           | 73        | 87        | 215       |
| Government                                                             | 42        | 35        | 44        |

Proposals dated by deposit, 1880–1912, on the definition of Table 15. Counts are divided by the calendar years in each window (14, 6 and 13), because the three periods are of different length. A bill counts as promulgated if the archive records a date of law. Socialist and other members are outside the sample throughout, so the rise in Panel A is the established parties multiplying their own initiative rather than a new party arriving.

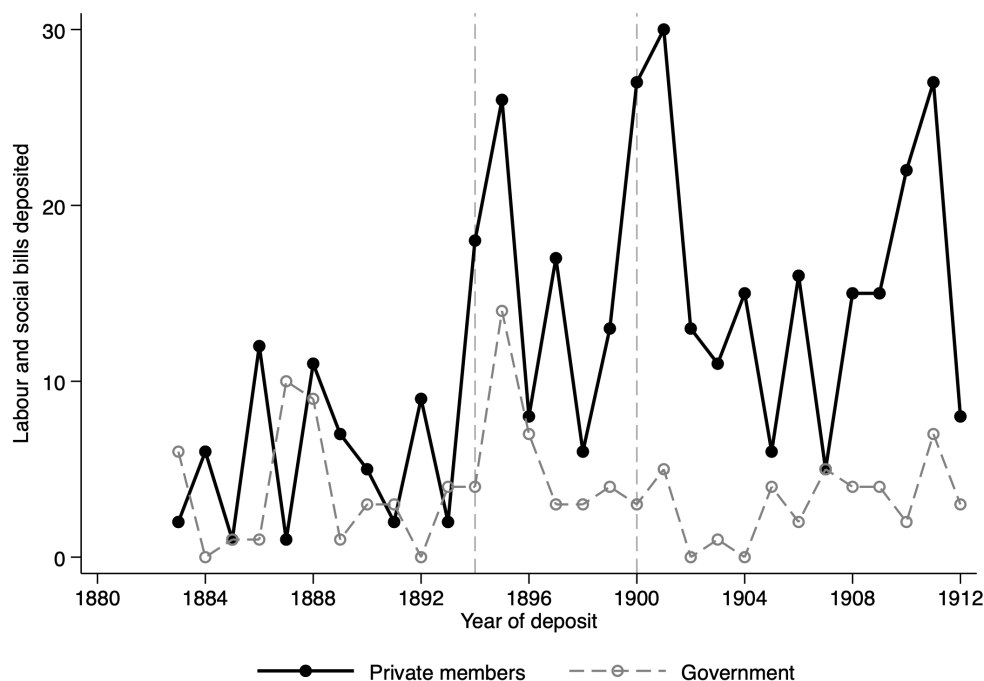


Figure 5: Labour and social bills deposited, by sponsor. A bill counts if the Genesis archive assigns it the subject tags *Labor & Work*, *Welfare & Social Protection* or *Health*. Bills are dated by deposit and every bill deposited is counted, whether or not it became law. Private members are Catholic and Liberal members; socialist and other members are outside the sample throughout, so the rise is the established parties multiplying their own initiative rather than a new party arriving. Vertical lines mark 1894 and 1900.

## 10.2 Where the bills died

A natural interpretation is that the Senate blocked members' labour bills. The data do not support this.

Table 15 follows each labour bill introduced by a Catholic or Liberal member to the furthest stage recorded in the archive. Of 375 bills, 263 never reached a vote in the Chamber. Eleven were voted on in the Chamber but went no further. The remaining 101 passed both houses and were promulgated.

Once a labour bill reached a Chamber vote, it was more likely to be promulgated than a bill on another subject. The promulgation rates were 0.84 versus 0.73 before 1894, 1.00 versus 0.90 in 1894–99, and 0.89 versus 0.63 after 1900 (Appendix Table A23). The Senate was therefore not the main obstacle. Most labour bills failed before reaching a Chamber vote.

This high failure rate is also not specific to labour. Across all other subjects, 69.5 per cent of members' bills failed, compared with 70.1 per cent of labour bills. With period fixed effects, the difference is  $-0.009$  (s.e. 0.026; Appendix Table A23). Therefore, failure was the normal fate of a private member's bill in this parliament.

What changes after 1900 is the relative success of labour bills. Their promulgation

rate remains broadly stable across the three periods. On the contrary, for bills on other subjects, the rate falls from 0.30 before 1894 to 0.30 in 1894–99 and 0.14 after 1900 (Appendix Table A23). Private members’ bills became much less likely to succeed after proportional representation, but labour bills did not follow that decline.

The labour-minus-other difference in promulgation rates is 0.058 (s.e. 0.025) with period fixed effects. In the post-1900 period alone, it is 0.121 (s.e. 0.032). The earlier stages show the same pattern. After 1900, the share of bills that never reached a Chamber vote rises from 0.67 to 0.77 for other subjects, but falls from 0.74 to 0.70 for labour (Appendix Table A23, columns 1–2).

Statistical inference is more straightforward here than in the speech analysis. The standard errors are heteroskedasticity-robust and use variation across 2,041 bills, rather than relying on district clusters, and the comparison lacks a control group.

Therefore, the legislative evidence supports a narrower conclusion. After 1900, labour bills did not share the decline experienced by other private members’ bills.

Table 15: Stage reached by labour bills deposited by members

| Furthest stage recorded               | Bills | Share |
|---------------------------------------|-------|-------|
| No vote in the Chamber                | 263   | 0.701 |
| Voted in the Chamber, not promulgated | 11    | 0.029 |
| Promulgated                           | 101   | 0.269 |
| All labour bills deposited by members | 375   | 1.000 |

Labour and social bills deposited by Catholic and Liberal members, 1880–1912, by the furthest stage the Genesis archive records. A bill counts as promulgated if the archive carries a date of law. Appendix Table A23 applies the same classification to members’ bills on every other subject.

## 11 Inference and Robustness

This section presents the robustness checks that matter for interpreting the results. The first set concerns inference, and examines the concentration of the 1900 estimate, the permutation tests, and the fact that the 1894 treatment is a single policy change applied to one chamber.

The second set examines possible alternative explanations. These include the socialist agenda, the choice of estimation window, placebo cutoffs, and additional checks reported in the appendices.

## 11.1 Concentration, effective clusters and leave-one-out

Standard errors are clustered at the arrondissement level, giving 41 clusters. For the 1900 reform, however, this number overstates the amount of independent identifying variation.

The triple difference is identified by the interaction between mining exposure, the post-1900 indicator, and the Chamber indicator. That variation is highly concentrated. Liège accounts for 56.0 per cent of the partial leverage, Charleroi for 13.2 per cent, and Mons for 5.6 per cent (Appendix Table A15). Together, these three districts provide about three quarters of the identifying variation.

The effective number of clusters is therefore only 2.95. We compute it using the single-coefficient measure of Carter et al. (2017): the inverse of the sum of squared partial leverages after the fixed effects are partialled out (MacKinnon et al., 2023). Appendix Table A15 reports the calculation.

Cluster-robust inference relies on a large-number-of-clusters approximation. An effective cluster count of about three is far from that setting. We therefore report conventional cluster-robust  $t$ -statistics for the 1900 estimates, but do not rely on them for inference.

The treated cell is also highly concentrated. Liège accounts for 65.2 per cent of its words, followed by Mons with 16.0 per cent, Charleroi with 10.8, and Soignies with 8.0 (Appendix Table A15).

This concentration should not be confused with exposure. Liège is only the fourth most mining-intensive district in the census measure, at 1.87 standard deviations above the mean. Mons is at 3.89 and Charleroi at 3.15 (Appendix Table A5). Liège matters because it is large, not because it is the most exposed district.

The post-1900 estimate remains positive in all 41 leave-one-district-out exercises. It is significant at the 5 per cent level in 40 of them and ranges from 3.20 to 10.09 (Online Appendix Table OA20). Liège is the important exception. Dropping it leaves an estimate of 3.20 (s.e. 2.54,  $t = 1.26$ ).

Dropping the two largest contributors together gives 5.22 (s.e. 1.90). This is not a stricter test than dropping Liège alone, because removing the second district by itself raises the estimate. Dropping all four coalfield districts gives 15.97 (s.e. 11.59; Appendix Table A15). At that point, identification comes only from differences among low-exposure districts, and the estimate becomes too imprecise to be informative.

The conclusion is therefore specific. The post-1900 coalfield effect is concentrated in Liège. It is supported by randomization inference, but it is not statistically precise once Liège is removed.

The Liberal entrant result is more fragile still. The coalfield entrant cell contains only four Liberal members. Paul Trassenster of Liège accounts for 35 per cent of their recorded words. Dropping him changes the Liberal entrant gradient from 12.17 to  $-4.46$  (Online Appendix Table OA12). This is a reversal of sign, not simply a loss of precision.

We therefore do not rely on the Liberal entrant coefficient on its own. Online Appendix OA13 reports the four members, the leave-one-member-out results, and shows that none entered on a joint list with the socialists.

## 11.2 Randomization inference

With only three effective clusters, we rely on randomization inference. We permute the mining exposure measure across arrondissements and re-estimate the statistic (Fisher, 1935; MacKinnon & Webb, 2020; Young, 2019).

In each permutation, every arrondissement is assigned the mining exposure of another district. The outcome, sample, weights, and fixed effects remain unchanged. Table 16 reports the results for the two main coefficients.

Table 16: Randomization inference, 4,999 permutations of the district exposure map

|                              | Observed $t$ | One-sided $p$ | Two-sided $p$ | Null s.d. |
|------------------------------|--------------|---------------|---------------|-----------|
| Triple difference, post-1900 | 3.935        | 0.021         | 0.035         | 1.729     |
| Entrant mining gradient      | 6.040        | 0.033         | 0.033         | 2.345     |

*Notes:* The exposure map is permuted across arrondissements; everything else is held fixed. Null means are  $-0.07$  and  $0.10$  respectively. The null 95th percentiles are 2.785 and 4.550, the null maxima 8.343 and 9.106.

For the post-1900 triple difference, the one-sided randomization  $p$ -value is 0.021 and the two-sided value is 0.035. For the entrant gradient, the corresponding values are 0.033 and 0.033. Both headline results therefore reject at conventional levels without relying on a large-number-of-clusters approximation.

The permutation distribution is wide. For the triple difference, its standard deviation is 1.729 and its maximum is 8.343. This happens because assigning high mining exposure to a large district can generate a large coefficient. We therefore treat this as a demanding randomization-style test, not as exact randomization inference. Mining exposure was not randomly assigned, and we do not have an assignment mechanism that would justify an exact test.

Table 17 applies the same procedure to the split-sample results from Section 9. The results depend on how much geographic variation identifies each coefficient. Statistics using all 41 districts pass the test. These include the jointly estimated coalfield gap ( $p = 0.021$ ), the entrant gradient ( $p = 0.037$ ), the coalfield triple difference using the binary exposure measure ( $p = 0.016$ ), and the Liberal entrant gradient ( $p = 0.029$ ).

Results identified from the four coalfield districts alone are weaker. The split-sample coalfield gap of 13.44 has a randomization  $p$ -value of 0.055, despite a cluster-robust  $t$ -statistic of 6.49. The two specifications that remove cabinet speech give  $p$ -values of 0.057

and 0.104. With an effective number of clusters close to three, the difference between conventional and randomization-based inference is therefore substantial.

The Liberal entrant result requires an additional caution. The permutation test changes the exposure map but keeps the members in the sample fixed. Its  $p$ -value of 0.029 therefore tests whether the geographic pattern is unusual under reassigned exposure. It does not address the separate fact that the Liberal coalfield entrant cell contains only four members and is sensitive to the presence of one of them. The randomization test therefore supports the geography of this coefficient, not the robustness of its composition.

Table 17: The split estimates under randomization inference

|                                                     | Estimate | Cluster $ t $ | RI    |
|-----------------------------------------------------|----------|---------------|-------|
| Post-1900 gap inside the coalfields, split sample   | 13.44    | 6.49          | 0.055 |
| The same, implied by joint estimation               | 23.07    | 6.69          | 0.021 |
| Entrant mining gradient                             | 11.07    | 6.04          | 0.037 |
| Triple difference, ministerial speech excluded      | 4.79     | 3.34          | 0.057 |
| Entrant gradient, ministerial speech excluded       | 4.95     | 3.49          | 0.104 |
| Entrant mining gradient, Liberals only              | 12.46    | 10.93         | 0.029 |
| Coalfield triple difference, binary exposure map    | 24.03    | 5.52          | 0.016 |
| Liberal entrant gradient (12.46)                    | 10.93    | 0.022         | 0.022 |
| Coalfield gap, joint estimation, binary map (24.03) | 5.52     | 0.010         | 0.017 |

The wild cluster bootstrap with the null imposed is informative for the 1894 reform (Cameron et al., 2008; Roodman et al., 2019). Treatment varies within every arrondissement, and the test rejects the null ( $p = 0.036$  with Webb weights; Appendix Table A14).

The same test is less informative for the 1900 reform. Mining exposure is constant within arrondissement, and the identifying variation is concentrated in about three effective clusters. In this setting, the wild cluster bootstrap is conservative rather than simply imprecise (MacKinnon & Webb, 2018). Appendix A8 reports both sets of results.

### 11.3 One treated group

The 1894 design has a further inference problem that is separate from the number of clusters. The treatment is a single policy change applied to one treated group: the Chamber, at one date.

Clustering by arrondissement allows for arbitrary dependence within districts. It does not solve the problem of a shock that affected the entire Chamber at the same time as the reform. Such a shock could come from changes in standing orders, transcription practices, the use of the Chamber by ministers, or the creation of the labour administration in November 1894. With only one treated group, no variance estimator can separate such a

shock from the treatment (Bertrand et al., 2004; Conley & Taber, 2011; Donald & Lang, 2007; Ferman & Pinto, 2019).

We address this concern with three direct checks. First, the agenda decomposition removes the most concrete Chamber-specific alternative: the presence of socialist deputies on the floor. The adaptation estimate remains.

Second, we remove all ministerial speech. The estimate remains 3.08 (s.e. 1.41), compared with 4.46 in the baseline. Allowing ministers to have their own post-reform shift instead leaves the Chamber–Post interaction at 4.15 (s.e. 1.79), while the minister-by-post term is close to zero. Online Appendix Table OA17 reports these estimates. Online Appendix OA13 also explains why the creation of the Office du travail cannot be separated from the reform by timing alone.

Third, the coalfield reversal provides variation within the Chamber. A shock common to the whole Chamber would not naturally produce an effect of 6.2 outside the coalfields and  $-1.4$  inside them (Table 9, Panel C).

One possible additional strategy is to use variation across arrondissements in the growth of the electorate between 1892 and 1894. We do not rely on it because that measure fails its own pre-reform placebo. Online Appendix OA16 reports the exercise without drawing a conclusion.

These checks do not turn the design into one with many treated groups. We therefore interpret the 1894 result narrowly. It is a differential response consistent with electoral adaptation, supported by the Senate comparison and by direct tests of the main alternative explanations. It cannot rule out every Chamber-specific shock that may have occurred in 1894.

## 11.4 Adaptation, or a reply to the socialists on the floor?

Adaptation to a new electorate is not the only possible explanation for the Chamber premium. Twenty-eight socialist deputies entered the Chamber in 1894, while socialist representation in the Senate remained limited before 1904. Therefore, socialist deputies could shift only the Chamber’s agenda toward labour. For this reason, Catholic and Liberal deputies might then speak more about labour simply because they were responding to socialist interventions, rather than because their electoral incentives had changed.

The corpus allows us to separate these two mechanisms. Every intervention is dated and linked to a speaker, including speakers outside the estimation sample. Therefore, we measure the labour content of speech delivered each day by members outside the Catholic and Liberal sample. This provides a measure of the day’s agenda that does not depend on what Catholic and Liberal deputies themselves chose to discuss.

Table 18 shows that the agenda did change. The labour content of speech by members outside the two established parties is 7.17 points higher after 1894 in the Chamber than

in the Senate ( $t = 4.39$ ). Before the reform that speech comes from the few members the archive assigns to neither party. After the reform, speeches comes largely from socialists. Therefore The estimate measures socialist entry itself, which shifted what the Chamber discussed.

The adaptation result does not depend on this agenda shift. On days when outside members did not speak about labour, the Chamber–Senate difference is 2.49 ( $t = 3.53$ ). Adding politician fixed effects gives 3.02 ( $t = 2.30$ ). The corresponding estimate using all sitting days is 2.92 ( $t = 1.16$ ; Table 18). The point estimate is therefore almost unchanged, while the standard error falls substantially.

Three qualifications matter. First, the agenda is itself affected by the reform. The estimate of 3.02 should therefore be interpreted as conditional on the observed agenda, not as the effect net of agenda-setting.

Second, the restriction changes the composition of the four cells unevenly. It retains about half of the words in three cells, but only 14 per cent in the censitary Senate. On 363 of its 492 sitting days, no member outside the estimation sample spoke at all. The pre-reform Senate baseline therefore contains only 2,142 interventions (Appendix Table A21).

Third, socialist presence explains part of the raw Chamber premium. Controlling for the socialist share of words spoken each day reduces the estimate to 1.51 (s.e. 1.97). We treat this as a decomposition, not as a correction, because socialist presence is itself a consequence of the reform.

A cleaner comparison would use only days on which no socialist spoke. After 1894, however, such days are rare. Socialists spoke on 85 per cent of Chamber sitting days in 1894–99. They also accounted for 19.9 per cent of Chamber words, compared with 8.6 per cent in the Senate (Appendix Table A21). Socialist presence was therefore much stronger in the Chamber, both in frequency and in intensity.

Table 18: The agenda channel and the content channel at the 1894 margin

|                                                 | Estimate | Std. err. | Obs.    |
|-------------------------------------------------|----------|-----------|---------|
| Baseline                                        | 4.455**  | (2.100)   | 208,325 |
| within politician                               | 2.917    | (2.507)   | 208,190 |
| <i>What the day was about</i>                   |          |           |         |
| Labour content of others' speech that day       | 7.165*** | (1.634)   | 193,088 |
| <i>On days others did not make about labour</i> |          |           |         |
| Labour salience                                 | 2.492*** | (0.705)   | 97,841  |
| within politician                               | 3.020**  | (1.310)   | 97,737  |
| <i>Mediation, not a robustness check</i>        |          |           |         |
| Controlling for the socialist share of the day  | 1.507    | (1.971)   | 208,325 |

All rows are the Chamber  $\times$  Post-1894 coefficient on 1880–1899. A chamber-day is classified by the labour content of speech by members outside the estimation sample—socialists, Daensists and others—so it does not depend on what the deputy himself said. The two within-politician rows absorb arrondissement, party, year and politician effects, the last on the parliamentary archive's person identifier, as in column 3 of Table 8. The final row conditions on the socialist share of the day's words, which is a consequence of the reform rather than a confounder: it decomposes the Chamber premium and does not correct it. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 11.5 Windows, placebos and the Senate

Table 19 reports the main identification checks for the post-1900 result.

First, there is no evidence of a pre-trend in the mining gradient. Over the 1894–1898 pre-period, the interaction between mining exposure, the Chamber indicator, and a linear trend is 0.14 (s.e. 0.72). Therefore, mining gradient in the Chamber–Senate difference is flat before proportional representation.

The simpler Chamber–Senate difference shows a modest upward pre-trend. The Chamber-by-trend interaction, without the mining dimension, is 1.52 (s.e. 0.88). We report it alongside the triple-difference pre-trend.

Placebo cutoffs are also close to zero. Moving the treatment date to 1896 gives  $-0.76$  (s.e. 2.42), while a cutoff at 1898 gives 0.54 (s.e. 3.01). Both placebo tests use only the 1894–1898 period, so observations on both sides of the placebo cutoff precede the actual reform.<sup>28</sup>

<sup>28</sup>This restriction is important. If the 1896 placebo is estimated on the full 1894–1906 sample, the actual post-1900 treatment period falls into the placebo post-period. The specification would then partly recover the true treatment effect under a different cutoff rather than provide a genuine placebo.

Table 19: Identification checks

|                                                 | Estimate  | Std. err. | Obs.    |
|-------------------------------------------------|-----------|-----------|---------|
| <i>Panel A. Pre-trends, 1894–1898</i>           |           |           |         |
| Mining $\times$ Chamber $\times$ trend          | 0.144     | (0.717)   | 51,802  |
| Chamber $\times$ trend                          | 1.520*    | (0.881)   | 51,802  |
| <i>Panel B. Placebo cutoffs, 1894–1898 only</i> |           |           |         |
| Triple difference, 1896                         | -0.761    | (2.417)   | 51,802  |
| Triple difference, 1898                         | 0.538     | (3.010)   | 51,802  |
| <i>Panel C. Estimation window</i>               |           |           |         |
| 1896–1902                                       | 11.955*** | (3.340)   | 66,741  |
| 1894–1904                                       | 11.156*** | (2.469)   | 104,756 |
| 1894–1906                                       | 9.253***  | (2.351)   | 119,448 |
| 1894–1908                                       | 8.665***  | (2.738)   | 138,077 |
| 1894–1912                                       | 7.437**   | (2.812)   | 176,985 |
| Donut, excluding 1900                           | 9.489***  | (2.827)   | 101,940 |

Panel A is the identifying assumption for the post-1900 triple difference. Panel B places both sides of the placebo cutoff before the reform: run on the full 1894–1906 panel a cutoff at 1896 would put the entire treatment period in the post group. Panel C shows the estimate attenuating as the window extends past 1906, which is when the Senate begins to respond to mining exposure. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The estimation window is a choice, and we report how the results change with it. Appendix Table A7 gives the estimate for every closing legislature. It is 11.61 through 1902, 11.16 through 1904, 9.25 through 1906, 8.67 through 1908, 8.81 through 1910, and 7.44 through 1912. All estimates are significant at the 5 per cent level or better.

An institutional rule would close the window in 1902. This is the last legislature before the number of directly elected socialist senators rises from one to four. That window gives the largest estimate. We instead use 1906 as a more conservative choice.

The estimate becomes smaller as the window expands. The reason is that the Senate gradually becomes less insulated. Its mining gradient is flat or negative through 1902, but turns positive from 1904 onward (Figure 4). From that point, adding later legislatures makes the Senate a less suitable control group and reduces the Chamber–Senate contrast.

## 11.6 Further checks

Five further sets of checks are reported in the appendices.

*Exposure measures.* The result is similar with alternative measures of industrial exposure. A binary high-mining indicator gives 24.03 (s.e. 4.35). A binary heavy-industry

indicator gives 26.96 (s.e. 3.01), while the continuous heavy-industry measure gives 4.37 (s.e. 2.19). Binary measures produce larger coefficients because they discard variation within groups. We therefore use the continuous mining measure as the more conservative baseline (Appendix Table A4).

We also test whether district magnitude itself explains the result. The fall in the exclusion threshold under D'Hondt,  $0.5 - 1/(M_d + 1)$ , gives 1.21 (s.e. 1.90) on its own. When included with mining exposure, its coefficient falls to 0.34, while the mining coefficient rises to 10.02 (Appendix Table A6). District magnitude therefore does not explain the mining gradient. Four other electoral measures give the same conclusion (Online Appendix OA6).

*Party-specific trends.* Adding party-by-year fixed effects leaves the mining-interacted estimates almost unchanged. The post-1900 estimate moves from 9.25 to 9.11, while the post-1894 estimate moves from  $-3.24$  to  $-2.79$ . The 1894 difference in differences becomes 3.89, with a  $t$ -statistic of 1.72 (Appendix Table A11).

*Domain specificity.* We estimate the same triple difference for each of the six substantive domains. The increase appears only for labour. The largest positive coefficient in any other domain is 0.20, with a  $t$ -statistic of 0.10 (Appendix Table A12).

*Language and transcription.* Restricting the sample to French-language interventions leaves the estimates essentially unchanged. Measures of transcription noise also show no systematic relationship with the design (Online Appendices OA5 and OA9).

*Venue.* Ministers speak before both houses, so differences in ministerial speech could affect the Chamber–Senate comparison. Adjusting for this does not reduce the estimates. Across the alternative specifications, the coefficients instead rise by at most 0.7 points (Online Appendix Table OA10).

## 12 Conclusions

Belgium expanded the right to vote without this implying a change in who was governing the country. The franchise reform of 1893 was adopted under the pressure of a general strike, and proportional representation followed six years later. Both reforms were designed by Catholic and Liberal elites seeking to limit the political consequences of democratisation. These parties largely succeeded in preserving political control: the Catholic party retained its majority until 1914. Nevertheless, this paper asks whether representation could change even when control of parliament did not. Besides, it asks through which channel such change occurred.

The answer differs across the two reforms. Plural suffrage expanded the electorate to which deputies answered. After 1894, the Chamber's attention to labour and social policy rose by 4.5 percentage points relative to the Senate, more than half its pre-reform level. The increase was broad across districts and was already present among politicians

who remained in office. On days when socialist deputies did not speak about labour, the within-politician estimate is still 3.0 points, with a  $t$ -statistic of 2.30. The main channel after 1894 was therefore adaptation: existing representatives changed what they talked about.

Proportional representation worked through a different margin. It did not change who could vote. It changed how votes translated into seats. There is no detectable aggregate Chamber–Senate shift after 1900. Instead, the response is concentrated in the coalfields, where the majoritarian rule had most distorted representation. The mining gradient rises by 9.3 points and is supported by randomization inference. The gradient appears among Catholic and Liberal entrants, while no comparable gradient appears among incumbents. Recomposition was therefore the main channel after 1900.

The two results are linked by what plural suffrage had done to the coalfields. Under the majoritarian rule, socialist victories displaced Catholic and Liberal deputies from the districts where the social question was most acute. Proportional representation allowed these parties to return. The members who entered under the new rule were also those who carried the new mining gradient in labour attention. In the language of Lee et al. (2004), the Belgian sequence displays both margins within the same political system: the first reform mainly changed the behaviour of politicians already in office, while the second mainly changed who held office. The contrast suggests that the channel through which democratisation changes representation depends on the margin of the electoral system that a reform alters.

The legislative record provides independent evidence in the same direction. Labour and social bills introduced by Catholic and Liberal members nearly tripled after enfranchisement, while their rate of success remained unchanged. The increase after 1894 was concentrated among Catholic members. The additional increase after 1900 came from Liberals. These results come from a different source and do not depend on the construction of the speech measure.

Three limitations bound these claims. First, the design identifies differential responses under asymmetric electoral exposure. It does not identify the aggregate effect of either reform on parliament as a whole. Second, the 1900 estimate is geographically concentrated. One district carries about two thirds of the weight in the treated cell. Inference therefore relies on randomization tests rather than cluster-robust  $t$ -statistics alone, and the coefficients estimated on the coalfields alone do not cross the 5 per cent threshold under permutation even when their cluster-robust  $t$ -statistics exceed six. Third, we observe representatives' allocation of attention, not citizens' priorities. The paper therefore identifies changes in parliamentary attention to the social question rather than the size of the underlying gap between citizen demand and parliamentary priorities.

One further scope condition matters for the first reform. The 1893 revision combined plural suffrage with compulsory voting. The design cannot separate the two components.

We therefore cannot say whether the same response would arise under voluntary voting, especially where turnout is lowest among newly enfranchised groups.

The broader implication is that electoral reform can change representation without changing who governs. But the mechanism depends on what the reform changes. Expanding the electorate can change the behaviour of politicians already in office. Changing the translation of votes into seats can instead change who represents the affected constituencies. The Belgian sequence shows both margins in the same parliament, six years apart.

## References

- Acemoglu, D., & Robinson, J. A. (2000). Why did the West extend the franchise? democracy, inequality, and growth in historical perspective. *The Quarterly Journal of Economics*, 115(4), 1167–1199. <https://doi.org/10.1162/003355300555042>
- Acemoglu, D., & Robinson, J. A. (2006). *Economic origins of dictatorship and democracy*. Cambridge University Press. <https://doi.org/10.1017/cbo9780511510809>
- Ahmed, A. (2010). Reading history forward: The origins of electoral systems in European democracies. *Comparative Political Studies*, 43(8-9), 1059–1088. <https://doi.org/10.1177/0010414010370436>
- Aidt, T. S., Daunton, M., & Dutta, J. (2010). The retrenchment hypothesis and the extension of the franchise in England and Wales. *The Economic Journal*, 120(547), 990–1020. <https://doi.org/10.1111/j.1468-0297.2009.02344.x>
- Aidt, T. S., Dutta, J., & Loukoianova, E. (2006). Democracy comes to Europe: Franchise extension and fiscal outcomes 1830–1938. *European Economic Review*, 50(2), 249–283. <https://doi.org/10.1016/j.euroecorev.2004.07.004>
- Aidt, T. S., & Jensen, P. S. (2014). Workers of the world, unite! franchise extensions and the threat of revolution in Europe, 1820–1938. *European Economic Review*, 72, 52–75. <https://doi.org/10.1016/j.euroecorev.2014.08.001>
- Bartels, L. M. (2008). *Unequal democracy: The political economy of the new Gilded Age*. Princeton University Press.
- Barzachka, N. (2025). Using process-tracing to evaluate competing accounts of proportional representation in Belgium. *Politics of the Low Countries*, 5(2), 160–183. <https://doi.org/10.5553/PLC/.000051>
- Belgian Chamber of Representatives. (2024a). Genesis: Legislative tracking database [Accessed: May 2024]. <https://www.dekamer.be>
- Belgian Chamber of Representatives. (2024b). Historisch ledenbestand – membres historiques [Accessed: May 2024]. <https://www.dekamer.be/kvvcr/showpage.cfm?section=hist>
- Belgian Senate. (2024a). Composition historique du Sénat – historische samenstelling van de Senaat [Accessed: May 2024]. <https://www.senate.be/>
- Belgian Senate. (2024b). Historical parliamentary proceedings of the Belgian Senate (Annales parlementaires / Parlementaire Handelingen), 1880–1912 [Accessed: May 2024]. <https://www.senate.be>
- Berlinski, S., & Dewan, T. (2011). The political consequences of franchise extension: Evidence from the Second Reform Act. *Quarterly Journal of Political Science*, 6(3–4), 329–376. <https://doi.org/10.1561/100.00011013>
- Bertrand, L. (1907). *Histoire de la démocratie et du socialisme en Belgique depuis 1830* (Vol. 2). Dechenne.

- Bertrand, M., Duflo, E., & Mullainathan, S. (2004). How much should we trust differences-in-differences estimates? *The Quarterly Journal of Economics*, 119(1), 249–275. <https://doi.org/10.1162/003355304772839588>
- Besley, T. (2005). Political selection. *Journal of Economic Perspectives*, 19(3), 43–60. <https://doi.org/10.1257/089533005774357761>
- Besley, T., & Coate, S. (1997). An economic model of representative democracy. *The Quarterly Journal of Economics*, 112(1), 85–114. <https://doi.org/10.1162/003355397555136>
- Bevan, S., & Jennings, W. (2014). Representation, agendas and institutions. *European Journal of Political Research*, 53(1), 37–56. <https://doi.org/10.1111/1475-6765.12023>
- Boix, C. (1999). Setting the rules of the game: The choice of electoral systems in advanced democracies. *American Political Science Review*, 93(3), 609–624. <https://doi.org/10.2307/2585577>
- Boix, C. (2003). *Democracy and redistribution*. Cambridge University Press. <https://doi.org/10.1017/cbo9780511804960>
- Bouhon, F. (2018). Le vote obligatoire en Belgique: Une expérience de 125 ans (1893–2018). [https://orbi.uliege.be/bitstream/2268/228484/1/BOUHON%20\(2018\)%20-%20Le%20vote%20obligatoire%20en%20Belgique.%20Une%20exp%C3%A9rience%20de%20125%20ans.pdf](https://orbi.uliege.be/bitstream/2268/228484/1/BOUHON%20(2018)%20-%20Le%20vote%20obligatoire%20en%20Belgique.%20Une%20exp%C3%A9rience%20de%20125%20ans.pdf)
- Brepoels, J. (2015). *Wat zoudt gij zonder 't werkvolk zijn? De geschiedenis van de Belgische arbeidersbeweging 1830–2015*. Van Halewyck.
- Brepoels, J., & Hoekx, R. (2019). *De bloednacht van 18 april 1902: Leuven op de bres voor het algemeen stemrecht* [96 pp.]. Salsa! – Vriendenkring van het Stadsarchief Leuven.
- Budge, I., & Farlie, D. J. (1983). *Explaining and predicting elections: Issue effects and party strategies in twenty-three democracies*. Allen & Unwin.
- Calvert, R. L. (1985). Robustness of the multidimensional voting model: Candidate motivations, uncertainty, and convergence. *American Journal of Political Science*, 29(1), 69–95. <https://doi.org/10.2307/2111212>
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3), 414–427. <https://doi.org/10.1162/rest.90.3.414>
- Carter, A. V., Schnepel, K. T., & Steigerwald, D. G. (2017). Asymptotic behavior of a  $t$ -test robust to cluster heterogeneity. *Review of Economics and Statistics*, 99(4), 698–709. [https://doi.org/10.1162/REST\\_a\\_00639](https://doi.org/10.1162/REST_a_00639)
- Conley, T. G., & Taber, C. R. (2011). Inference with “difference in differences” with a small number of policy changes. *The Review of Economics and Statistics*, 93(1), 113–125. [https://doi.org/10.1162/REST\\_a\\_00049](https://doi.org/10.1162/REST_a_00049)

- Cox, G. W. (1997). *Making votes count: Strategic coordination in the world's electoral systems*. Cambridge University Press. <https://doi.org/10.1017/cbo9781139174954>
- Cusack, T. R., Iversen, T., & Soskice, D. (2007). Economic interests and the origins of electoral systems. *American Political Science Review*, 101(3), 373–391. <https://doi.org/10.1017/S0003055407070384>
- Dahl, R. A. (1971). *Polyarchy: Participation and opposition*. Yale University Press.
- Delwit, P. (2017). Les libéraux belges et les élections. In P. Delwit (Ed.), *Du parti libéral au MR: 170 ans de libéralisme en Belgique* (pp. 159–197). Éditions de l'Université de Bruxelles.
- Deneckere, G. (1993). The transforming impact of collective action: Belgium, 1886. *International Review of Social History*, 38(3), 345–367. <https://doi.org/10.1017/s002085900011212x>
- Dhondt, F. (2020). Justice and equality for all? proportional representation in Belgium and France (1883–1921). *Sequência*, 41(86), 28–62. <https://doi.org/10.5007/2177-7055.2020v41n86p28>
- Donald, S. G., & Lang, K. (2007). Inference with difference-in-differences and other panel data. *The Review of Economics and Statistics*, 89(2), 221–233. <https://doi.org/10.1162/rest.89.2.221>
- Dumoulin, M. (2024). Belges et Italiens dans la “guerre à la guerre” à la veille de 1914. *Les Études sociales*, (179), 271–298. <https://doi.org/10.3917/etsoc.179.0271>
- Elsässer, L., Hense, S., & Schäfer, A. (2021). Not just money: Unequal responsiveness in egalitarian democracies. *Journal of European Public Policy*, 28(12), 1890–1908. <https://doi.org/10.1080/13501763.2020.1801804>
- Emmenegger, P., & Walter, A. (2019). When dominant parties adopt proportional representation: The mysterious case of Belgium. *European Political Science Review*, 11(4), 433–450. <https://doi.org/10.1017/S1755773919000225>
- Ferman, B., & Pinto, C. (2019). Inference in differences-in-differences with few treated groups and heteroskedasticity. *The Review of Economics and Statistics*, 101(3), 452–467. [https://doi.org/10.1162/rest\\_a.00759](https://doi.org/10.1162/rest_a.00759)
- Fisher, R. A. (1935). *The design of experiments*. Oliver & Boyd.
- Flora, P., & Heidenheimer, A. J. (1981). The historical core and changing boundaries of the welfare state. In P. Flora & A. J. Heidenheimer (Eds.), *The development of welfare states in Europe and America* (pp. 17–34). Transaction Books.
- Gagliarducci, S., & Nannicini, T. (2013). Do better paid politicians perform better? disentangling incentives from selection. *Journal of the European Economic Association*, 11(2), 369–398. <https://doi.org/10.1111/jeea.12002>
- Gagliarducci, S., Nannicini, T., & Naticchioni, P. (2011). Electoral rules and politicians' behavior: A micro test. *American Economic Journal: Economic Policy*, 3(3), 144–174. <https://doi.org/10.1257/pol.3.3.144>

- Gennaro, G., & Ash, E. (2022). Emotion and reason in political language. *The Economic Journal*, 132(643), 1037–1059. <https://doi.org/10.1093/ej/ueab104>
- Gentzkow, M., Kelly, B., & Taddy, M. (2019). Text as data. *Journal of Economic Literature*, 57(3), 535–574. <https://doi.org/10.1257/jel.20181020>
- Gilens, M. (2012). *Affluence and influence: Economic inequality and political power in America*. Princeton University Press. <https://doi.org/10.1515/9781400844821>
- Gilens, M., & Page, B. I. (2014). Testing theories of American politics: Elites, interest groups, and average citizens. *Perspectives on Politics*, 12(3), 564–581. <https://doi.org/10.1017/s1537592714001595>
- Gilissen, J. (1958). *Le régime représentatif en Belgique depuis 1790*. La Renaissance du Livre.
- Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3), 267–297. <https://doi.org/10.1093/pan/mps028>
- Hooghe, M., & Stiers, D. (2025). Abolishing compulsory voting: The 2024 experience in the flemish region of belgium. *PS: Political Science & Politics*, 58(4), 606–613. <https://doi.org/10.1017/S1049096525000423>
- Iversen, T., & Soskice, D. (2006). Electoral institutions and the politics of coalitions: Why some democracies redistribute more than others. *American Political Science Review*, 100(2), 165–181. <https://doi.org/10.1017/S0003055406062083>
- Jones, B. D., & Baumgartner, F. R. (2004). Representation and agenda setting. *Policy Studies Journal*, 32(1), 1–24. <https://doi.org/10.1111/j.0190-292X.2004.00050.x>
- Kollman, K., Hicken, A., Caramani, D., Backer, D., & Lublin, D. (2025). Constituency-level elections archive (CLEA). *Center for Political Studies, University of Michigan*. <https://electiondataarchive.org>
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174. <https://doi.org/10.2307/2529310>
- Lee, D. S., Moretti, E., & Butler, M. J. (2004). Do voters affect or elect policies? evidence from the u.s. house. *Quarterly Journal of Economics*, 119(3), 807–859. <https://doi.org/10.1162/0033553041502153>
- Lijphart, A., & Gibberd, R. W. (1977). Thresholds and payoffs in list systems of proportional representation. *European Journal of Political Research*, 5(3), 219–244. <https://doi.org/10.1111/j.1475-6765.1977.tb01289.x>
- Lindbeck, A., & Weibull, J. W. (1987). Balanced-budget redistribution as the outcome of political competition. *Public Choice*, 52(3), 273–297. <https://doi.org/10.1007/BF00116710>
- Lindert, P. H. (2004). *Growing public: Social spending and economic growth since the eighteenth century* (Vol. 1). Cambridge University Press. <https://doi.org/10.1017/cbo9780511510717>

- Lipset, S. M., & Rokkan, S. (1967). Cleavage structures, party systems, and voter alignments: An introduction. In S. M. Lipset & S. Rokkan (Eds.), *Party systems and voter alignments: Cross-national perspectives* (pp. 1–64). Free Press.
- Lizzeri, A., & Persico, N. (2001). The provision of public goods under alternative electoral incentives. *American Economic Review*, 91(1), 225–239. <https://doi.org/10.1257/aer.91.1.225>
- Lizzeri, A., & Persico, N. (2004). Why did the elites extend the suffrage? democracy and the scope of government, with an application to Britain’s “age of reform”. *The Quarterly Journal of Economics*, 119(2), 707–765. <https://doi.org/10.1162/0033553041382175>
- Luykx, T., & Platel, M. (1985). *Politieke geschiedenis van België* (5th ed., Vol. 1). Kluwer.
- MacKinnon, J. G., Nielsen, M. Ø., & Webb, M. D. (2023). Leverage, influence, and the jackknife in clustered regression models: Reliable inference using `summclust`. *The Stata Journal*, 23(4), 942–982. <https://doi.org/10.1177/1536867x231212433>
- MacKinnon, J. G., & Webb, M. D. (2018). The wild bootstrap for few (treated) clusters. *The Econometrics Journal*, 21(2), 114–135. <https://doi.org/10.1111/ectj.12107>
- MacKinnon, J. G., & Webb, M. D. (2020). Randomization inference for difference-in-differences with few treated clusters. *Journal of Econometrics*, 218(2), 435–450. <https://doi.org/10.1016/j.jeconom.2020.04.024>
- Martin, L., Muller, B., Ortiz Suárez, P. J., Dupont, Y., Romary, L., de la Clergerie, É. V., Seddah, D., & Sagot, B. (2020). CamemBERT: A tasty french language model. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 7203–7219. <https://doi.org/10.18653/v1/2020.acl-main.645>
- Meguid, B. M. (2005). Competition between unequals: The role of mainstream party strategy in niche party success. *American Political Science Review*, 99(3), 347–359.
- Meltzer, A. H., & Richard, S. F. (1981). A rational theory of the size of government. *Journal of Political Economy*, 89(5), 914–927. <https://doi.org/10.1086/261013>
- Miller, W. E., & Stokes, D. E. (1963). Constituency influence in Congress. *American Political Science Review*, 57(1), 45–56. <https://doi.org/10.2307/1952717>
- Office du travail de Belgique. (1900–1902). Recensement général des industries et des métiers (31 octobre 1896) [Accessed: 2026]. <https://bib.kuleuven.be/ebib/project-belgische-historische-tellingen/recensement-general-des-industries-des-metiers-31-octobre-1896>
- Osborne, M. J., & Slivinski, A. (1996). A model of political competition with citizen-candidates. *The Quarterly Journal of Economics*, 111(1), 65–96. <https://doi.org/10.2307/2946658>

- Osnabrügge, M., Ash, E., & Morelli, M. (2023). Cross-domain topic classification for political texts. *Political Analysis*, 31(1), 59–80. <https://doi.org/10.1017/pan.2021.37>
- Persson, T., & Tabellini, G. (2000). *Political economics: Explaining economic policy*. MIT Press.
- Persson, T., & Tabellini, G. (2003). *The economic effects of constitutions*. MIT Press. <https://doi.org/10.7551/mitpress/2591.001.0001>
- Petrocik, J. R. (1996). Issue ownership in presidential elections, with a 1980 case study. *American Journal of Political Science*, 40(3), 825–850. <https://doi.org/10.2307/2111797>
- Pitkin, H. F. (1967). *The concept of representation*. University of California Press. <https://doi.org/10.1525/9780520340503>
- Plenum.be. (2024). Digitised plenary proceedings of the belgian chamber of representatives [Accessed: May 2024]. <https://plenum.be>
- Polasky, J. L. (1992). A revolution for socialist reforms: The Belgian general strike for universal suffrage. *Journal of Contemporary History*, 27(3), 449–466. <https://doi.org/10.1177/002200949202700304>
- Popelier, P. (2018). Bicameralism in belgium: The dismantlement of the senate for the sake of multinational confederalism. *Perspectives on Federalism*, 10(2), E-215–E-237. <https://doi.org/10.2478/pof-2018-0024>
- Rae, D. W., Hanby, V., & Loosemore, J. (1971). Thresholds of representation and thresholds of exclusion: An analytic note on electoral systems. *Comparative Political Studies*, 3(4), 479–488. <https://doi.org/10.1177/001041407100300406>
- Rodden, J. A. (2019). *Why cities lose: The deep roots of the urban–rural political divide*. Basic Books.
- Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., & Webb, M. D. (2019). Fast and wild: Bootstrap inference in Stata using `boottest`. *Stata Journal*, 19(1), 4–60. <https://doi.org/10.1177/1536867X19830877>
- Royaume de Belgique. (1899). Loi du 29 décembre 1899 relative à l’application de la représentation proportionnelle aux élections législatives.
- Schakel, W., & Burgoon, B. (2022). The party road to representation: Unequal responsiveness in party platforms. *European Journal of Political Research*, 61(2), 304–325. <https://doi.org/10.1111/1475-6765.12489>
- Sénat de Belgique. (2019). *Suffrage universel pur et simple: 100 ans pour les hommes, 70 ans pour les femmes. Partie III, 1893 à 1918: Quand certains sont plus égaux que d’autres*. Retrieved June 5, 2026, from [https://www.senate.be/home/sections/geschiedenis\\_en\\_erfgoed/AES-SU/art-3\\_fr.pdf](https://www.senate.be/home/sections/geschiedenis_en_erfgoed/AES-SU/art-3_fr.pdf)

- Spirling, A. (2016). Democratization and linguistic complexity: The effect of franchise extension on parliamentary discourse, 1832–1915. *The Journal of Politics*, 78(1), 120–136. <https://doi.org/10.1086/683612>
- Stengers, J. (2004). Histoire de la législation électorale en Belgique. *Revue belge de philologie et d'histoire*, 82(1–2), 247–270. <https://doi.org/10.3406/rbph.2004.4826>
- Strikwerda, C. (1988). The divided class: Catholics vs. Socialists in Belgium, 1880–1914. *Comparative Studies in Society and History*, 30(2), 333–359. <https://doi.org/10.1017/s001041750001522x>
- Strikwerda, C. (1997). *A house divided: Catholics, Socialists, and Flemish nationalists in nineteenth-century Belgium*. Rowman & Littlefield.
- Strömberg, D. (2004). Radio's impact on public spending. *Quarterly Journal of Economics*, 119(1), 189–221. <https://doi.org/10.1162/003355304772839560>
- Van Molle, P. (1972). *Het Belgisch parlement 1894–1972*. Standaard Wetenschappelijke Uitgeverij.
- Vanthemsche, G., & De Peuter, R. (2023). *A concise history of Belgium*. Cambridge University Press. <https://doi.org/10.1017/9781139018005>
- Wang, Y. (2023). Topic classification for political texts with pretrained language models. *Political Analysis*, 31(4), 662–668. <https://doi.org/10.1017/pan.2023.3>
- Wikipedia contributors. (n.d.-a). Auguste Houzeau de Lehaie. Retrieved September 11, 2026, from [https://nl.wikipedia.org/wiki/Auguste\\_Houzeau\\_de\\_Lehaie](https://nl.wikipedia.org/wiki/Auguste_Houzeau_de_Lehaie)
- Wikipedia contributors. (n.d.-b). Liste des sénateurs belges (législature 1900–1902). Retrieved September 11, 2026, from [https://fr.wikipedia.org/wiki/Liste\\_des\\_s%C3%A9nateurs\\_belges\\_\(l%C3%A9gislation\\_1900-1902\)](https://fr.wikipedia.org/wiki/Liste_des_s%C3%A9nateurs_belges_(l%C3%A9gislation_1900-1902))
- Wikipedia contributors. (2024). Kamer van volksvertegenwoordigers en Senaat: Samenstelling per zittingsperiode, 1878–1912. [https://nl.wikipedia.org/wiki/Kamer\\_van\\_volksvertegenwoordigers](https://nl.wikipedia.org/wiki/Kamer_van_volksvertegenwoordigers)
- Witte, E., Craeybeckx, J., & Meynen, A. (2009). *Political history of Belgium: From 1830 onwards*. ASP – Academic & Scientific Publishers.
- Witte, E., Meynen, A., & Luyten, D. (2016). *Politieke geschiedenis van België van 1830 tot heden* (10th ed.). Uitgeverij Manteau/WPG Uitgevers België.
- Wittman, D. (1983). Candidate motivation: A synthesis of alternative theories. *American Political Science Review*, 77(1), 142–157. <https://doi.org/10.2307/1956016>
- Young, A. (2019). Channeling Fisher: Randomization tests and the statistical insignificance of seemingly significant experimental results. *The Quarterly Journal of Economics*, 134(2), 557–598. <https://doi.org/10.1093/qje/qjy029>

# Appendix

## A1 Coverage, denominators and alternative outcome constructions

Table A1 reports the share of interventions the classifier assigns to any substantive domain, by chamber and reform window, on the scored corpus. An intervention counts as classified when the procedural residual is not its modal class, which is the binary counterpart of the continuous classified share used in the estimation sample. The difference in differences across the 1894 reform is  $-0.052$  on raw shares; on the estimation sample, restricted to Catholic and Liberal members and with fixed effects, it attenuates to  $-0.023$  (s.e. 0.010; Table A3). The 1900 margin runs the other way and is smaller still:  $+0.025$  on raw shares and  $-0.002$  (s.e. 0.013) on the estimation sample, where the triple difference is  $-0.007$  (s.e. 0.008). The selection the design already imposes removes most of the problem, and retaining the procedural residual as a seventh bucket removes the rest by construction (Section 5.4).

Table A1: Classification coverage by chamber and reform window

| Chamber      | Window    | Interventions | Classified | Share |
|--------------|-----------|---------------|------------|-------|
| Chamber (DK) | pre-1894  | 157,354       | 71,296     | 0.453 |
|              | 1894–1899 | 81,847        | 33,793     | 0.413 |
|              | post-1900 | 157,204       | 60,692     | 0.386 |
| Senate (SK)  | pre-1894  | 17,080        | 9,991      | 0.585 |
|              | 1894–1899 | 13,705        | 8,184      | 0.597 |
|              | post-1900 | 33,042        | 18,006     | 0.545 |

All scored interventions dated 1880–1912 with a chamber assignment, before the estimation-sample restrictions of Section 4. An intervention counts as classified when the procedural residual is not its modal class.

Table A2 extends the two-denominator comparison of Section 5.4 to two further constructions. Dropping the length weight entirely—so that the estimand is the share of interventions rather than of words—leaves the two mining-interacted objects intact, 4.82 for the triple difference and 5.35 for the entrant gradient, but takes the 1894 difference in differences to 1.76 (s.e. 1.28). That margin is carried by long speeches: the same deputies, speaking at greater length on the social question rather than raising it in more interventions. Aggregating instead to one observation per member per legislature, which removes the possibility that a few long interventions dominate, returns 4.21 (s.e. 1.97) for 1894 and 9.45 for the triple difference, close to the baseline. We report the length-weighted

estimates as the main specification because the estimand is the share of parliamentary language, and we report the unweighted figures because the 1894 margin depends on that choice.

Table A2: The headline estimates under alternative outcome constructions

|                                                                         | Estimate  | Std. err. | Obs.    |
|-------------------------------------------------------------------------|-----------|-----------|---------|
| <i>Chamber <math>\times</math> Post-1894</i>                            |           |           |         |
| Baseline: six buckets, word weight                                      | 4.455**   | (2.100)   | 208,325 |
| All seven buckets                                                       | 3.937*    | (2.030)   | 208,325 |
| Unweighted                                                              | 1.764     | (1.282)   | 208,904 |
| Member $\times$ legislature panel                                       | 4.214**   | (1.973)   | 2,522   |
| <i>Mining <math>\times</math> Post-1900 <math>\times</math> Chamber</i> |           |           |         |
| Baseline: six buckets, word weight                                      | 9.253***  | (2.351)   | 119,448 |
| All seven buckets                                                       | 9.013***  | (2.395)   | 119,448 |
| Unweighted                                                              | 4.823***  | (1.232)   | 119,761 |
| Member $\times$ legislature panel                                       | 9.453***  | (2.349)   | 1,757   |
| <i>Entrant mining gradient</i>                                          |           |           |         |
| Baseline: six buckets, word weight                                      | 11.073*** | (1.833)   | 14,182  |
| All seven buckets                                                       | 10.701*** | (1.858)   | 14,182  |
| Unweighted                                                              | 5.354**   | (2.126)   | 14,218  |
| Member $\times$ legislature panel                                       | 10.803*** | (1.919)   | 349     |

The outcome in the baseline is the labour bucket's share of the six substantive buckets, weighted by intervention length. The all-seven row divides by every bucket including the procedural residual; the unweighted row is the share of interventions; the panel row is one observation per member per legislature. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A3: The classified share as an outcome, estimation sample

|                                                       | Estimate | Std. err. | Obs.    |
|-------------------------------------------------------|----------|-----------|---------|
| Chamber $\times$ Post-1894, 1880–1899                 | -0.023** | (0.010)   | 208,325 |
| Chamber $\times$ Post-1900, 1894–1906                 | -0.002   | (0.013)   | 119,448 |
| Mining $\times$ Post-1900 $\times$ Chamber, 1894–1906 | -0.007   | (0.008)   | 119,448 |

The outcome is the share of an intervention's probability mass that the classifier assigns to the six substantive domains, between zero and one. Estimation sample, Catholic and Liberal members, with the fixed effects, weights and clustering of the corresponding rows of Tables 8 and 11. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A2 Alternative measures of exposure

Table A4 reports the post-1900 triple difference under alternative pressure measures. A binary high-mining indicator gives 24.03 (s.e. 4.35); a binary heavy-industry indicator 26.96 (s.e. 3.01); a continuous heavy-industry measure 4.37 (s.e. 2.19). The Panel A counterparts are negative under every measure that identifies the coalfields:  $-8.34$  for binary mining,  $-7.90$  for binary heavy industry. The sign reversal across regimes is not specific to the continuous proxy.

Table A4: Alternative measures of exposure

| Exposure measure                                 | Panel A: 1894 cutoff |           | Panel B: 1900 cutoff |           |
|--------------------------------------------------|----------------------|-----------|----------------------|-----------|
|                                                  | Estimate             | Std. err. | Estimate             | Std. err. |
| Mining share, continuous                         | -3.238***            | (1.031)   | 9.253***             | (2.351)   |
| Mining, high-exposure indicator ( $>0.25$ share) | -8.339***            | (2.762)   | 24.030***            | (4.351)   |
| Heavy industry, high-exposure indicator          | -7.899***            | (2.876)   | 26.960***            | (3.013)   |
| Heavy industry, continuous                       | -0.239               | (1.222)   | 4.368*               | (2.188)   |

Binary measures give estimates roughly two and a half times the continuous ones, as expected when a threshold discards within-group variation; the continuous measure is the conservative baseline throughout. Two measures built from electoral returns are omitted: their construction codes arrondissements failing to match the electoral source as having zero socialist pressure, which places Liège at the bottom of the distribution. They are not interpretable until that is repaired. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

The mining proxy conflates two things a reader may want separated: the mechanical bite of the 1899 allocation rule, and the geography of socialist pressure that made the rule bind. Under D'Hondt in a district of magnitude  $M_d$ , the vote share at which a list can be entirely shut out falls from one half to  $1/(M_d + 1)$ ; the threshold shift  $0.5 - 1/(M_d + 1)$ , computed from the magnitudes the 1899 law fixed on census population, is predetermined and independent of who had been accumulating votes in the district. It carries nothing on its own: 1.21 (s.e. 1.90) in the post-1900 triple difference, with a clean placebo at the 1894 cutoff ( $-0.52$ ; Table A6). Entered alongside mining intensity it stays at 0.34 while the mining coefficient rises slightly, to 10.02 (s.e. 2.69; same table). District magnitude, which is largest in Brussels, predicts nothing; the rule change mattered where pre-existing pressure gave it something to bind on, and we read the 1900 margin as rule times pressure rather than as either alone. Online Appendix OA6 reports this measure with four further constructions built from the 1894–1898 returns and the 1899 allocation, none of which displaces the mining proxy, and diagnoses one apparent exception.

Table A5 reports the distribution of the standardised mining share across the ten most exposed arrondissements, so that a coefficient per standard deviation can be read against where Liège sits. It also shows where the binary indicator cuts. The four coalfield districts are those whose mining employment share exceeds 0.25, and on the standardised scale

the rule separates Liège at 1.87 from Thuin at 1.51—a gap of 0.36 standard deviations, wider than the gap between Soignies and Liège but not by much. That is the sense in which the binary measure discards information, and it is why the continuous measure is the baseline.

Table A5: Standardised mining share, ten most exposed arrondissements

| Arrondissement | Mining share (s.d.) | Coalfield |
|----------------|---------------------|-----------|
| Mons           | 3.89                | Yes       |
| Charleroi      | 3.15                | Yes       |
| Soignies       | 1.88                | Yes       |
| Liège          | 1.87                | Yes       |
| Thuin          | 1.51                | —         |
| Namur          | 0.45                | —         |
| Ath            | 0.34                | —         |
| Bastogne       | −0.02               | —         |
| Hasselt        | −0.13               | —         |
| Turnhout       | −0.20               | —         |

The ten most mining-intensive arrondissements on the 1896 census measure, standardised to mean zero and unit variance across the 41 districts. A coefficient per standard deviation is read against this scale. The coalfield column marks the four districts whose mining employment share exceeds 0.25; the blank line marks where the rule falls.

Table A6: The threshold shift under D'Hondt

|                                                | Estimate  | Std. err. | Obs.    |
|------------------------------------------------|-----------|-----------|---------|
| <i>Panel A. The rule change alone</i>          |           |           |         |
| Threshold shift $\times$ Post $\times$ Chamber | 1.207     | (1.898)   | 119,448 |
| Entrant gradient                               | 0.152     | (2.345)   | 14,182  |
| Placebo, 1894 cutoff                           | −0.517    | (1.005)   | 208,325 |
| <i>Panel B. Against the mining proxy</i>       |           |           |         |
| Threshold shift                                | 0.340     | (1.605)   | 119,448 |
| Mining intensity                               | 10.023*** | (2.686)   | 119,448 |

The threshold shift is  $0.5 - 1/(M_d + 1)$ , standardised, where  $M_d$  is the district's Chamber magnitude under the 1899 law. It is predetermined: magnitudes were set on census population. The placebo row uses the same variable with the 1894 cutoff, when the rule was not yet in force. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### A3 The estimate under every closing legislature

Table A7 gives the post-1900 triple difference under every closing legislature from 1902 to 1912, and a donut specification dropping the 1900 legislature itself, which Table 19 reports at 9.49. The estimate falls smoothly from 11.61 to 7.44 as legislatures are added past 1906, which is what a control group that is progressively contaminated should produce; all six are significant at the 5 per cent level or better.

Table A7: The estimate under every closing legislature

| Window               | Estimate  | Std. err. | Obs.    |
|----------------------|-----------|-----------|---------|
| 1894–1902            | 11.609*** | (3.322)   | 85,681  |
| 1894–1904            | 11.156*** | (2.469)   | 104,756 |
| 1894–1906 (reported) | 9.253***  | (2.351)   | 119,448 |
| 1894–1908            | 8.665***  | (2.738)   | 138,077 |
| 1894–1910            | 8.806***  | (2.460)   | 155,159 |
| 1894–1912            | 7.437***  | (2.812)   | 176,985 |

The post-1900 triple difference under every closing legislature. The estimate attenuates as the window extends, because directly elected socialist senators enter the control group from the middle of the decade, so the longer windows are conservative rather than contradictory. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

### A4 The full interaction structure

Tables A8, A9 and A10 report every term of the specifications behind Tables 8 and 11. Year effects are absorbed in all of them, so the coefficient on the reform indicator is identified only inside the calendar year in which the rule changes and has no interpretation of its own; the interactions are identified from every year.

At the 1894 margin the mining interactions leave the baseline in place. Evaluated at mean exposure, the Chamber  $\times$  Post-1894 term is 4.76 (s.e. 1.93) without politician effects and 4.14 (s.e. 2.27) with them (Table A8, columns 4 and 6).

At the 1900 margin the lower-order terms show what the triple difference is a difference from. In 1894–99 the Chamber’s mining gradient lies 4.53 points below the Senate’s (s.e. 1.09; Mining  $\times$  Chamber in Table A9, column 4); the triple difference of 9.25 is the change in that gap, which therefore ends 4.72 points above zero. Part of the change is the Senate’s own gradient falling by 2.31 after 1900 (s.e. 0.92; Mining  $\times$  Post-1900, same column). The binary indicator gives the same reading: before 1900 the Chamber–Senate difference is 10.60 points lower inside the coalfields than outside (s.e. 2.45), coalfield senators fall by 5.03 after the reform relative to the others (s.e. 2.33), and the triple term of 24.03 is measured against both (Table A10, column 3).

Table A8: Plural suffrage and labour salience, 1880–1899: every term

|                                            | Difference in differences |                    |                   | Triple difference    |                      |                     |
|--------------------------------------------|---------------------------|--------------------|-------------------|----------------------|----------------------|---------------------|
|                                            | (1)                       | (2)                | (3)               | (4)                  | (5)                  | (6)                 |
| Chamber                                    | 0.279<br>(0.871)          | 0.280<br>(0.881)   | -0.606<br>(0.862) | 0.272<br>(0.814)     | 0.280<br>(0.821)     | -0.679<br>(0.639)   |
| Post-1894                                  | 3.963*<br>(2.340)         | 3.945*<br>(2.291)  | 3.571<br>(2.131)  | 3.406<br>(2.391)     | 3.307<br>(2.321)     | 2.885<br>(1.988)    |
| Chamber $\times$ Post-1894                 | 4.455**<br>(2.100)        | 4.455**<br>(2.108) | 2.917<br>(2.507)  | 4.760**<br>(1.930)   | 4.767**<br>(1.956)   | 4.137*<br>(2.266)   |
| Mining                                     |                           |                    |                   | <i>absorbed</i>      | <i>absorbed</i>      | <i>absorbed</i>     |
| Mining $\times$ Post-1894                  |                           |                    |                   | 1.623**<br>(0.612)   | 1.665***<br>(0.568)  | 1.604*<br>(0.910)   |
| Mining $\times$ Chamber                    |                           |                    |                   | 0.096<br>(0.903)     | 0.095<br>(0.902)     | 2.068***<br>(0.525) |
| Mining $\times$ Post-1894 $\times$ Chamber |                           |                    |                   | -3.238***<br>(1.031) | -3.333***<br>(0.970) | 0.189<br>(1.360)    |
| Incumbent                                  |                           | -0.011<br>(0.533)  |                   |                      | -0.011<br>(0.529)    |                     |
| Entrant                                    |                           | 0.122<br>(1.646)   |                   |                      | 0.597<br>(1.420)     |                     |
| Arrondissement, party, year FE             | Yes                       | Yes                | Yes               | Yes                  | Yes                  | Yes                 |
| Politician FE                              | No                        | No                 | Yes               | No                   | No                   | Yes                 |
| Observations                               | 208,325                   | 208,325            | 208,190           | 208,325              | 208,325              | 208,190             |
| Clusters                                   | 41                        | 41                 | 41                | 41                   | 41                   | 41                  |

Every term of the six specifications of Table 8, in the same order of columns. A blank cell is a term that is not in that specification; *absorbed* is a term that is in it and that the fixed effects absorb. Mining is the standardised mining share of the arrondissement in the 1896 industrial census; it does not vary within arrondissement, so its main effect is absorbed. Incumbent marks members who sit on both sides of 1894 and Entrant members first observed in 1894 or later; the omitted group left before 1894. Sample 1880–1899, observations weighted by intervention length, standard errors clustered by arrondissement. Politician fixed effects are defined on the parliamentary archive’s person identifier. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A9: Proportional representation and labour salience, 1894–1906: every term

|                                            | Difference in differences |                     |                      | Triple difference    |                      |                     |
|--------------------------------------------|---------------------------|---------------------|----------------------|----------------------|----------------------|---------------------|
|                                            | (1)                       | (2)                 | (3)                  | (4)                  | (5)                  | (6)                 |
| Chamber                                    | 5.932**<br>(2.431)        | 6.144**<br>(2.574)  | -0.044<br>(2.171)    | 5.074**<br>(2.157)   | 5.680**<br>(2.156)   | -0.866<br>(2.703)   |
| Post-1900                                  | -4.380<br>(3.042)         | -4.524<br>(3.167)   | -7.125***<br>(2.603) | -3.838<br>(2.678)    | -3.738<br>(2.730)    | -8.244**<br>(3.286) |
| Chamber $\times$ Post-1900                 | 1.475<br>(3.058)          | 0.935<br>(2.792)    | 2.788<br>(2.386)     | 2.648<br>(2.307)     | 1.707<br>(2.131)     | 3.350<br>(2.821)    |
| Mining                                     |                           |                     |                      | <i>absorbed</i>      | <i>absorbed</i>      | <i>absorbed</i>     |
| Mining $\times$ Post-1900                  |                           |                     |                      | -2.311**<br>(0.915)  | -1.989**<br>(0.802)  | 1.731<br>(1.170)    |
| Mining $\times$ Chamber                    |                           |                     |                      | -4.531***<br>(1.094) | -4.425***<br>(1.176) | 2.109<br>(2.176)    |
| Mining $\times$ Post-1900 $\times$ Chamber |                           |                     |                      | 9.253***<br>(2.351)  | 7.503***<br>(1.832)  | -3.474<br>(2.136)   |
| Incumbent                                  |                           | -7.080**<br>(2.678) |                      |                      | -7.316***<br>(2.666) |                     |
| Entrant                                    |                           | 0.906<br>(4.724)    |                      |                      | -0.671<br>(4.137)    |                     |
| Arrondissement, party, year FE             | Yes                       | Yes                 | Yes                  | Yes                  | Yes                  | Yes                 |
| Politician FE                              | No                        | No                  | Yes                  | No                   | No                   | Yes                 |
| Observations                               | 119,448                   | 119,448             | 119,341              | 119,448              | 119,448              | 119,341             |
| Clusters                                   | 41                        | 41                  | 41                   | 41                   | 41                   | 41                  |

Every term of the six specifications in the upper panel of Table 11, in the same order of columns. A blank cell is a term that is not in that specification; *absorbed* is a term that is in it and that the fixed effects absorb. Mining is the standardised mining share of the arrondissement in the 1896 industrial census; it does not vary within arrondissement, so its main effect is absorbed. Incumbent marks members who sit on both sides of 1900 and Entrant members first observed in 1900 or later; the omitted group left before 1900. Sample 1894–1906, observations weighted by intervention length, standard errors clustered by arrondissement. Politician fixed effects are defined on the parliamentary archive’s person identifier.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table A10: Proportional representation by geography and by party: every term

|                                               | By geography        |                       |                       | By party           |                      |
|-----------------------------------------------|---------------------|-----------------------|-----------------------|--------------------|----------------------|
|                                               | Outside             | Inside                | Joint model           | Catholic           | Liberal              |
| Chamber                                       | 7.155***<br>(2.587) | -3.110***<br>(0.502)  | 6.615**<br>(2.589)    | 3.537<br>(5.385)   | 3.899*<br>(2.266)    |
| Post-1900                                     | -2.535<br>(2.964)   | -16.489***<br>(2.725) | -3.124<br>(2.837)     | -6.807<br>(5.615)  | -6.226<br>(4.955)    |
| Chamber $\times$ Post-1900                    | -0.817<br>(2.675)   | 13.441***<br>(2.071)  | -0.958<br>(2.661)     | 2.965<br>(5.094)   | 2.378<br>(2.213)     |
| Coalfield                                     |                     |                       | <i>absorbed</i>       |                    |                      |
| Coalfield $\times$ Chamber                    |                     |                       | -10.601***<br>(2.452) |                    |                      |
| Coalfield $\times$ Post-1900                  |                     |                       | -5.027**<br>(2.331)   |                    |                      |
| Coalfield $\times$ Post-1900 $\times$ Chamber |                     |                       | 24.030***<br>(4.351)  |                    |                      |
| implied Chamber $\times$ Post-1900 inside     |                     |                       | 23.072***<br>(3.450)  |                    |                      |
| Mining                                        |                     |                       |                       | <i>absorbed</i>    | <i>absorbed</i>      |
| Mining $\times$ Post-1900                     |                     |                       |                       | -4.128<br>(13.181) | -1.511<br>(1.160)    |
| Mining $\times$ Chamber                       |                     |                       |                       | -6.658<br>(13.209) | -3.721***<br>(1.301) |
| Mining $\times$ Post-1900 $\times$ Chamber    |                     |                       |                       | 7.436<br>(12.344)  | 5.850***<br>(1.675)  |
| Arrondissement and year FE                    | Yes                 | Yes                   | Yes                   | Yes                | Yes                  |
| Party FE                                      | Yes                 | Yes                   | Yes                   | No                 | No                   |
| Observations                                  | 106,673             | 12,775                | 119,448               | 86,578             | 32,867               |
| Clusters                                      | 37                  | 4                     | 41                    | 40                 | 32                   |

Every term of the regressions behind the lower panels of Table 11. Outside and Inside are the two split samples; Joint model interacts the coalfield indicator with the full design on the whole sample, and the implied effect inside the coalfields is the sum of the two Chamber  $\times$  Post-1900 terms. Catholic and Liberal re-estimate column 4 on one party each. A blank cell is a term that is not in that specification; *absorbed* is a term that is in it and that the fixed effects absorb. The coalfields are an indicator for four arrondissements, Mining the continuous mining share: neither varies within arrondissement, so both main effects are absorbed. In the Inside column the standard errors rest on the four coalfield clusters. Sample 1894–1906, observations weighted by intervention length, standard errors clustered by arrondissement.

\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A5 Party-by-year fixed effects and the pre-1894 mining gradient

The estimation samples change party composition sharply across both reforms—the Liberal collapse of 1894–96 most of all—so a party-specific national trend is the natural remaining confound. Table A11 absorbs party-by-year fixed effects, identifying every coefficient within party-year cells. The two mining-interacted objects are unmoved: 9.11 against a baseline of 9.25 for the post-1900 triple difference, and  $-2.79$  against  $-3.24$  for the 1894 reversal. The 1894 difference in differences falls to 3.89 with a  $t$ -statistic of 1.72. The censitary-sample rows report the pre-trend and placebo cutoffs for the coalfield reversal discussed in Section 8.

Table A11: Party-by-year fixed effects and the pre-1894 mining gradient

|                                                                                    | Estimate  | Std. err. | Obs.    |
|------------------------------------------------------------------------------------|-----------|-----------|---------|
| <i>Chamber <math>\times</math> Post-1894, 1880–1899</i>                            |           |           |         |
| Baseline                                                                           | 4.455**   | (2.100)   | 208,325 |
| Party-by-year fixed effects                                                        | 3.889*    | (2.263)   | 208,325 |
| <i>Mining <math>\times</math> Post-1900 <math>\times</math> Chamber, 1894–1906</i> |           |           |         |
| Baseline                                                                           | 9.253***  | (2.351)   | 119,448 |
| Party-by-year fixed effects                                                        | 9.110***  | (2.355)   | 119,448 |
| <i>Mining <math>\times</math> Post-1894 <math>\times</math> Chamber, 1880–1899</i> |           |           |         |
| Baseline                                                                           | -3.238*** | (1.031)   | 208,325 |
| Party-by-year fixed effects                                                        | -2.792**  | (1.236)   | 208,325 |
| <i>Censitary sample only, 1880–1892</i>                                            |           |           |         |
| Mining $\times$ Chamber $\times$ linear trend                                      | 0.006     | (0.053)   | 156,523 |
| Mining placebo cutoff, 1886                                                        | -0.084    | (1.023)   | 156,523 |
| Mining placebo cutoff, 1888                                                        | -0.267    | (0.759)   | 156,523 |
| Mining placebo cutoff, 1890                                                        | 0.260     | (0.499)   | 156,523 |

Party-by-year fixed effects absorb any party-specific time path, so the coefficients are identified within party-year cells; a Catholic-specific national trend cannot carry them. The censitary-sample rows estimate the mining-interacted object on 1880–1892 only, with both sides of each placebo cutoff before the reform. DiD rows use the raw constituency coding, triple-difference rows the harmonised coding, as in the corresponding main tables. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A6 Domain specificity

If the design were picking up a general change in parliamentary language rather than a shift towards the social question, the same triple difference estimated on other policy domains would be comparably large. Table A12 estimates it on each of the six substantive buckets in turn. Labour is the only bucket that rises: 9.25 ( $t = 3.94$ ). The largest other positive coefficient is 0.20 on territory and infrastructure, with a  $t$ -statistic of 0.10. Confessional and state governance both fall, by 2.10 and 5.03, which is the arithmetic consequence of the buckets summing to a constant within the classified share. The test is more demanding under six categories than under a finer taxonomy, because each absorbs a large share of the corpus and there is little room for a placebo domain to look inert by construction. Online Appendix OA4 repeats the exercise under the all-class denominator.

Table A12: Domain specificity: the same triple difference on each bucket

| Policy bucket    | Estimate | Std. err. | $t$   |
|------------------|----------|-----------|-------|
| LAB              | 9.253*** | (2.351)   | 3.94  |
| confessional     | -2.099*  | (1.107)   | -1.90 |
| economic fiscal  | -0.596   | (3.310)   | -0.18 |
| state governance | -5.034   | (3.362)   | -1.50 |
| foreign defence  | -1.723   | (1.811)   | -0.95 |
| territory infra  | 0.199    | (2.035)   | 0.10  |

The post-1900 triple difference estimated on each substantive bucket in turn. The buckets sum to 100 within the classified share, so a rise in one implies a fall elsewhere; what matters is that no other bucket carries a comparable positive coefficient. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A7 Identification checks at the 1894 margin

Table A13 collects the checks on the 1894 difference in differences cited in Section 8: the pre-reform trend, the placebo cutoffs inside the censitary period, the two timing variants, and the exclusion of one arrondissement at a time.

Table A13: Identification checks at the 1894 margin

|                                                          | Estimate | Std. err.    | Obs.    |
|----------------------------------------------------------|----------|--------------|---------|
| Baseline, Chamber $\times$ Post-1894, 1880–1899          | 4.455**  | (2.100)      | 208,325 |
| <i>Panel A. Pre-trend, 1880–1892</i>                     |          |              |         |
| Chamber $\times$ linear trend                            | -0.022   | (0.079)      | 156,523 |
| <i>Panel B. Placebo cutoffs, 1880–1892 only</i>          |          |              |         |
| Chamber $\times$ Post-1886                               | -1.160   | (0.901)      | 156,523 |
| Chamber $\times$ Post-1888                               | -0.973   | (0.950)      | 156,523 |
| Chamber $\times$ Post-1890                               | 1.961*   | (1.143)      | 156,523 |
| <i>Panel C. Timing</i>                                   |          |              |         |
| Cutoff moved to 1896                                     | 6.364*** | (1.854)      | 208,325 |
| 1894 legislature excluded                                | 6.578*** | (2.058)      | 189,385 |
| <i>Panel D. Leaving out one arrondissement at a time</i> |          |              |         |
| Exclusions with a positive estimate                      |          | 41 of 41     |         |
| Exclusions significant at 5 per cent                     |          | 38 of 41     |         |
| Range across exclusions                                  |          | [2.71, 5.58] |         |
| Standard deviation across exclusions                     |          | 0.37         |         |
| Excluding Alost ( $t = 1.77$ )                           | 3.681    | (2.080)      | 191,742 |
| Excluding Brussels ( $t = 1.89$ )                        | 4.463    | (2.358)      | 182,792 |
| Excluding Leuven ( $t = 1.91$ )                          | 2.710    | (1.415)      | 201,704 |

Every row is a difference in differences between the Chamber and the Senate with arrondissement, party and year fixed effects, weighted by intervention length, standard errors clustered by arrondissement. Panel A interacts the Chamber indicator with a linear trend over the censitary legislatures. Panel B places both sides of each placebo cutoff before the reform. Panel C moves the cutoff to the first legislature that sat a full session under plural suffrage, and drops the 1894 legislature. Panel D re-estimates the baseline 41 times, each time without one arrondissement; an exclusion counts as significant when its cluster-robust  $t$  exceeds 1.96, and the exclusions that fall below that value are listed. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A8 The wild cluster bootstrap, and where it applies

With few effective clusters the standard remedy is the wild cluster bootstrap with the null imposed (Cameron et al., 2008; Roodman et al., 2019). For the 1894 margin the treatment is Chamber  $\times$  Post, which varies within every arrondissement; the bootstrap is informative there and it passes: 9,999 replications with Webb weights give  $p = 0.036$ , with a confidence set of [0.22, 10.01]; without the null imposed the  $p$ -value is 0.072 (Table A14).

For the 1900 margin it is not informative. Mining exposure is constant within arrondissement, and the interaction that identifies the effect draws almost all of its variation from the four coalfield districts, so the design is the few-treated-clusters case MacKinnon and Webb (2018) analyse. They show that in that case the wild cluster bootstrap with the null imposed cannot reproduce the variation in treatment across treated clusters that the test requires, and that it under-rejects severely: the  $p$ -values it returns are far too large. For the triple difference the bootstrap returns  $p = 0.184$ , with a 95 per cent confidence set of  $[-3.76, 16.79]$ ; for the entrant gradient,  $p = 0.156$  and a confidence set of  $[-9.85, 43.26]$  (Table A14). Without the null imposed the same two tests return  $p = 0.005$  and  $p = 0.002$ . A gap of that size between the two versions is the symptom of the few-treated-clusters case, in which the restricted bootstrap under-rejects and the unrestricted one over-rejects, and neither is informative. Randomization inference is the appropriate test for the 1900 margin: it permutes the exposure map across districts and requires no within-cluster variation in the treatment. The bootstrap is implementable in this fixed-effects setup—`boottest` runs after `areg` with explicit dummies, and the explicit and absorbed estimates agree to four decimal places—so what limits it is a property of the regressor, not of the software.

Table A14: Wild cluster bootstrap

|                                            | Estimate | Cluster $t$ | Null imposed |                    | Null not imposed |
|--------------------------------------------|----------|-------------|--------------|--------------------|------------------|
|                                            |          |             | $p$          | 95% confidence set | $p$              |
| Chamber $\times$ Post-1894                 | 4.455    | 2.12        | 0.036        | [0.22, 10.01]      | 0.072            |
| Mining $\times$ Post-1900 $\times$ Chamber | 9.253    | 3.93        | 0.184        | $[-3.76, 16.79]$   | 0.005            |
| Entrant mining gradient                    | 11.073   | 6.04        | 0.156        | $[-9.85, 43.26]$   | 0.002            |

Wild cluster bootstrap, 9,999 replications, Webb weights, clusters at the arrondissement. The confidence set is obtained by inverting the test with the null imposed; the last column repeats the test without imposing it. For the two post-1900 rows, the exposure measure is constant within cluster and the identifying variation is concentrated in very few clusters (Table A15), the case in which the first version under-rejects and the second over-rejects (MacKinnon & Webb, 2018); the inference for that margin is the randomization inference of Table 16.

Table A15: Where the identifying variation of the post-1900 triple difference sits

| Arrondissement                                | Mining share<br>(s.d.)              | Partial leverage<br>(per cent) | Treated-cell words<br>(per cent) |
|-----------------------------------------------|-------------------------------------|--------------------------------|----------------------------------|
| Mons                                          | 3.89                                | 5.6                            | 16.0                             |
| Charleroi                                     | 3.15                                | 13.2                           | 10.8                             |
| Soignies                                      | 1.88                                | 3.1                            | 8.0                              |
| Liège                                         | 1.87                                | 56.0                           | 65.2                             |
| The other 37                                  |                                     | 22.1                           | 0.0                              |
| Clusters                                      |                                     | 41                             |                                  |
| Effective number of clusters                  |                                     | 2.95                           |                                  |
| Triple difference without the four coalfields | 15.97 (11.59), 106,673 observations |                                |                                  |

Post-1900 triple difference, 1894–1906. Partial leverage is a district’s share of the weighted sum of squares of the triple-interaction regressor once every other regressor and the fixed effects are partialled out. The effective number of clusters is the inverse of the sum of the squared leverages, the measure of Carter et al. (2017) for a single coefficient with uncorrelated errors within cluster. Treated-cell words are the words of Catholic and Liberal deputies for the four coalfield districts after 1900. Mining share is the standardised 1896 census measure.

## A9 The long difference, skipping the majoritarian legislatures

Table A16 compares the censitary period (1880–1892) directly with the proportional one (1900–1906), so that the base is the level the coalfield districts had before any displacement occurred. On the continuous exposure measure the long difference is 3.12 (s.e. 3.01); on the coalfield indicator it is 12.75 (s.e. 7.47). Both point estimates are positive and neither is precise. Within politician the long difference is negative (−2.89, s.e. 1.00), which is what the recomposition mechanism implies: the members who span both periods are precisely the incumbents who did not drive the post-1900 rise, and the rise is carried by entrants who by construction have no censitary observations. Proportional representation certainly reversed the displacement—the coalfield Chamber cell rises from 12.71 to 34.88, against a censitary level of 9.94 in the same districts (Table A16)—but whether it carried those districts *above* where they had been before the franchise reform, relative to the Senate and to the rest of the country, is not something these data settle.

Table A16: The long difference, skipping the majoritarian legislatures

|                                                                | Estimate  | Std. err. | Obs.    |
|----------------------------------------------------------------|-----------|-----------|---------|
| Baseline, 1894–1906                                            | 9.253***  | (2.351)   | 119,448 |
| <i>Long difference, 1880–1892 against 1900–1906</i>            |           |           |         |
| Continuous mining exposure                                     | 3.119     | (3.013)   | 224,169 |
| Coalfield indicator                                            | 12.751*   | (7.469)   | 224,169 |
| Continuous, politician fixed effects                           | -2.893*** | (1.001)   | 224,028 |
| <i>The first reform on the same design</i>                     |           |           |         |
| Censitary against majoritarian                                 | -3.238*** | (1.031)   | 208,325 |
| <i>Level of the coalfield Chamber cell, word-weighted mean</i> |           |           |         |
| Censitary legislatures, 1880–1892                              | 9.94      |           | 23,105  |
| Majoritarian legislatures, 1894–1898                           | 12.71     |           | 1,525   |
| Proportional legislatures, 1900–1906                           | 34.88     |           | 7,273   |

The long difference drops the plural-suffrage majoritarian legislatures and compares the censitary period directly with the proportional one. It answers whether the post-1900 coalfield rise clears the level those districts had before the 1894 displacement, or merely returns to it. The row under it estimates the displacement itself on the same design, and the last three rows give the level of the treated cell in each regime. Politician fixed effects are defined on the parliamentary archive’s person identifier. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A10 Leaving out one pre-reform legislature

The pre-period of the 1894 design is not perfectly flat: in the 1888 legislature the senators observed on both sides of the reform sit above the deputies, and that legislature carries the Senate’s debates on the labour laws of 1889. Table A17 re-estimates the pooled and the within-person difference in differences dropping one censitary legislature at a time. The pooled estimate runs from 3.48 to 4.86 against a baseline of 4.46; the within-person estimate from 2.00 to 4.31 against a baseline of 3.24. Both are lowest without the 1888 legislature; the highest value comes from dropping 1882 in the pooled column and 1892 in the within-person one.

Because a legislature is a coarse unit, the table also reports the estimate with only the three sittings that carry 30 per cent of the labour content of the 1888 Senate cell removed: 3.86 pooled and 2.56 within person, on 85 of 208,325 rows (Table A17). Most of what the legislature contributes is contributed by those three sittings.

Table A17: Leave-one-out over the censitary legislatures

| Dropped                           | Pooled   |           | Within person |           |
|-----------------------------------|----------|-----------|---------------|-----------|
|                                   | Estimate | Std. err. | Estimate      | Std. err. |
| Nothing (baseline)                | 4.455**  | (2.100)   | 3.236         | (2.524)   |
| Legislature of 1880               | 4.776**  | (2.048)   | 3.288         | (2.500)   |
| Legislature of 1882               | 4.855**  | (2.081)   | 3.252         | (2.462)   |
| Legislature of 1884               | 4.477**  | (2.104)   | 3.434         | (2.453)   |
| Legislature of 1886               | 4.315**  | (2.093)   | 3.133         | (2.405)   |
| Legislature of 1888               | 3.477    | (2.336)   | 1.998         | (2.813)   |
| Legislature of 1890               | 4.569**  | (2.116)   | 3.999         | (2.747)   |
| Legislature of 1892               | 4.728**  | (2.081)   | 4.306*        | (2.532)   |
| Three sittings of the 1888 Senate | 3.860*   | (2.177)   | 2.563         | (2.559)   |

Chamber  $\times$  Post-1894 on 1880–1899, dropping one censitary legislature at a time. The pooled column is the specification of Table 8, column 1. The within-person column restricts the sample to members observed on both sides of the reform and absorbs politician fixed effects on the parliamentary archive’s person identifier, as in Table 9, Panel B. The last row drops only the three sitting days that carry most of the labour content of the 1888 Senate cell (1889-11-19, 1889-11-13 and 1889-07-26): 85 interventions, which hold 30 per cent of that cell’s labour content. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## A11 The series by legislature

Table A18 gives the numbers plotted in Figures 1, 3 and 4.

Table A18: Levels, Chamber–Senate difference and mining gradients, by legislature

| Legislature | Level   |        | Chamber minus Senate |        | Mining gradient, Chamber |        | Mining gradient, Senate |        | Chamber words |
|-------------|---------|--------|----------------------|--------|--------------------------|--------|-------------------------|--------|---------------|
|             | Chamber | Senate | Coef.                | (s.e.) | Coef.                    | (s.e.) | Coef.                   | (s.e.) |               |
| 1880        | 4.88    | 4.51   | 0.44                 | (0.57) | 0.62                     | (0.40) | -0.44                   | (0.36) | 1,288,938     |
| 1882        | 7.40    | 5.49   | 3.09                 | (1.02) | 0.01                     | (0.47) | 0.46                    | (0.71) | 1,370,867     |
| 1884        | 5.17    | 5.13   | 0.30                 | (0.86) | 0.20                     | (0.15) | 0.51                    | (0.31) | 1,519,023     |
| 1886        | 8.85    | 8.53   | -1.22                | (1.15) | 0.88                     | (0.70) | 0.86                    | (0.73) | 1,220,835     |
| 1888        | 11.80   | 15.25  | -3.34                | (3.13) | 2.57                     | (0.83) | 3.02                    | (1.75) | 1,164,358     |
| 1890        | 11.01   | 9.08   | 0.34                 | (1.47) | 0.60                     | (0.84) | 0.34                    | (0.65) | 1,018,741     |
| 1892        | 7.73    | 5.93   | 1.82                 | (0.99) | 0.60                     | (0.25) | -0.41                   | (0.42) | 1,440,462     |
| 1894        | 12.94   | 13.25  | -0.10                | (2.85) | 0.97                     | (2.03) | 4.73                    | (1.44) | 963,271       |
| 1896        | 21.37   | 12.19  | 9.59                 | (4.53) | 5.12                     | (0.97) | 3.13                    | (1.82) | 800,057       |
| 1898        | 15.16   | 9.85   | 3.59                 | (1.64) | 0.95                     | (0.40) | -0.90                   | (0.70) | 1,000,201     |
| 1900        | 11.56   | 6.65   | 3.28                 | (1.23) | 1.52                     | (0.62) | 0.23                    | (0.36) | 783,815       |
| 1902        | 23.52   | 10.73  | 13.60                | (4.40) | 9.43                     | (5.08) | -1.12                   | (0.76) | 683,302       |
| 1904        | 20.22   | 8.85   | 11.33                | (2.50) | 7.06                     | (1.64) | 2.07                    | (0.72) | 725,389       |
| 1906        | 18.26   | 15.38  | 0.03                 | (2.08) | 4.16                     | (0.68) | 6.04                    | (2.56) | 632,981       |
| 1908        | 17.01   | 15.24  | 0.34                 | (2.56) | 1.83                     | (1.97) | 4.89                    | (2.70) | 761,904       |
| 1910        | 18.25   | 13.72  | 1.19                 | (3.17) | 4.15                     | (1.98) | 3.09                    | (2.84) | 604,030       |
| 1912        | 12.70   | 10.80  | 2.71                 | (2.13) | -0.74                    | (0.71) | 3.03                    | (2.53) | 857,990       |

Catholic and Liberal members, by legislature. Level is the word-weighted mean of the labour salience measure (Figure 1). Chamber minus Senate is the coefficient on the Chamber indicator estimated within the legislature, with arrondissement and party fixed effects (Figure 3). The mining gradient is the coefficient on the standardised mining share estimated within a chamber and legislature, with party fixed effects (Figure 4). All estimates are weighted by intervention length, with standard errors clustered by arrondissement. The final column reports the number of words recorded for Catholic and Liberal deputies. The blank lines mark 1894 and 1900.

## A12 The aggregate difference under every closing legislature

Section 9 reports the aggregate Chamber–Senate difference after 1900 on the 1894–1906 window, and Section 11 reports the triple difference under every closing legislature. The aggregate is given the same treatment here, because the estimation window is chosen on the Senate’s behaviour and a null that depended on that choice would not be informative. Table A19 reports it, for all districts and by geography. The aggregate is indistinguishable from zero under every closing legislature from 1902 to 1912.

Outside the coalfields the difference is likewise null over the window,  $-0.82$  (s.e. 2.68), but it is not flat legislature by legislature: 2.10 in 1894, 12.15 in 1896, 4.17 in 1898, 1.80 in 1900, 8.58 in 1902, 9.94 in 1904 and 0.88 in 1906. The estimate over the window is a cancellation between a pre-period averaging 6.14 and a post-period averaging 5.30 (Table A20), and the movement after 1900 is of the same size as the movement the

difference already had under the majoritarian rule. Inside the coalfields the same series is  $-7.84$  in 1894 and  $27.96$  in 1902; it is estimated on four arrondissements, so we report the coefficients and not their standard errors, which the cluster structure does not support.

Table A19: Chamber  $\times$  Post-1900 under every closing legislature

| Sample                 | Closing legislature |                  |                  |                  |                  |                  |
|------------------------|---------------------|------------------|------------------|------------------|------------------|------------------|
|                        | 1902                | 1904             | 1906             | 1908             | 1910             | 1912             |
| All districts          | 3.68<br>( 3.60)     | 3.53<br>( 3.13)  | 1.48<br>( 3.06)  | -1.40<br>( 3.28) | -2.00<br>( 3.38) | -1.86<br>( 3.48) |
| Outside the coalfields | 0.26<br>( 2.94)     | 0.80<br>( 2.75)  | -0.82<br>( 2.67) | -4.05<br>( 3.05) | -5.10<br>( 3.14) | -4.96<br>( 3.20) |
| Inside the coalfields  | 24.83<br>( 3.71)    | 20.19<br>( 3.17) | 13.44<br>( 2.07) | 12.38<br>( 4.13) | 12.40<br>( 3.75) | 9.85<br>( 4.53)  |

Table A20: The Chamber–Senate difference by legislature, outside and inside the coalfields

| Legislature | Outside the coalfields |        | Inside the coalfields |
|-------------|------------------------|--------|-----------------------|
|             | Coef.                  | (s.e.) | Coef.                 |
| 1880        | 0.22                   | (0.64) | 1.64                  |
| 1882        | 3.34                   | (1.13) | 1.87                  |
| 1884        | 0.55                   | (1.00) | -0.75                 |
| 1886        | -1.48                  | (1.24) | 0.87                  |
| 1888        | -2.69                  | (2.49) | -5.11                 |
| 1890        | 1.15                   | (1.10) | -1.68                 |
| 1892        | 1.65                   | (1.20) | 2.42                  |
| 1894        | 2.10                   | (3.04) | -7.84                 |
| 1896        | 12.15                  | (4.57) | -1.58                 |
| 1898        | 4.17                   | (2.06) | 1.75                  |
| 1900        | 1.80                   | (0.99) | 11.29                 |
| 1902        | 8.58                   | (2.59) | 27.96                 |
| 1904        | 9.94                   | (2.22) | 9.00                  |
| 1906        | 0.88                   | (1.67) | -5.73                 |
| 1908        | -0.22                  | (3.09) | 2.69                  |
| 1910        | 0.24                   | (3.57) | 6.49                  |
| 1912        | 4.78                   | (2.16) | -4.57                 |

Coefficient on the Chamber indicator estimated within a legislature, separately for the 37 arrondissements outside the coalfields and for the four inside them, with arrondissement and party fixed effects and observations weighted by intervention length. Outside the coalfields the unweighted mean of the coefficients is 6.14 over 1894–1898 and 5.30 over 1900–1906. The inside series is estimated on four clusters, so its standard errors are not reported; a dash marks a legislature with fewer than 100 interventions in one of the two houses. The blank lines mark 1894 and 1900.

## A13 The agenda-conditional sample

The estimate of 3.02 in Section 11.4 is computed on sitting days that members outside the estimation sample did not make about labour, where "about labour" means above the median of the day's outside labour salience within the same chamber and period. Table A21 reports, for each chamber-by-period cell, the sitting days for which that measure is defined, the days for which it is not, the interventions retained and the share of words retained.

A median split retains about half of a cell by construction, and does so in three of the four: 48 per cent of pre-reform Chamber words, 53 per cent of post-reform Chamber

words, 45 per cent of post-reform Senate words (Table A21). The censitary Senate is the exception at 14 per cent, because on 363 of its 492 sitting days no member outside the estimation sample spoke at all—before 1894 the upper house was almost wholly Catholic and Liberal, so for most of its sittings there is no outside speech against which to classify the day’s business. That is a limitation of the test rather than of the reform: the agenda channel can only be measured where somebody outside the sample sets an agenda. It is the reason the conditional estimate is reported alongside the unconditional one rather than in place of it.

Table A21: Observations behind the agenda-conditional estimate

|                                                | Chamber  |         | Senate   |         |
|------------------------------------------------|----------|---------|----------|---------|
|                                                | pre-1894 | 1894–99 | pre-1894 | 1894–99 |
| Sitting days, all                              | 1,389    | 672     | 129      | 177     |
| on the non-labour agenda                       | 692      | 324     | 70       | 102     |
| no outside speech that day                     | 164      | 71      | 363      | 74      |
| Interventions, all                             | 142,773  | 47,166  | 14,181   | 4,784   |
| retained                                       | 69,464   | 24,477  | 2,142    | 2,042   |
| Share of interventions retained                | 0.487    | 0.519   | 0.151    | 0.427   |
| Share of words retained                        | 0.483    | 0.528   | 0.143    | 0.453   |
| Share of sitting days with a socialist speaker | 0.071    | 0.853   | 0.010    | 0.622   |
| Socialist share of the words spoken, per cent  | 0.2      | 19.9    | 0.0      | 8.6     |

Panel A estimation sample. A chamber-day is on the labour agenda if the word-weighted labour salience of speech delivered that day by members outside the estimation sample exceeds the median of its own chamber-by-period cell; days on which no such member spoke enter neither arm, and the first row counts the days on which one did. The last two rows are computed over every sitting day of the cell. Party labels are fixed over a member’s career, so their entries before 1894 are the interventions of a member who sat as a Liberal at the time and joined the socialists later. The within-politician estimate on the retained days is 3.02 ( $t = 2.30$ ), absorbing politician, party, arrondissement and year, with the politician defined on the parliamentary archive’s person identifier.

## A14 The incumbents’ post-1900 cross-section gradient

The entrant gradient of Table 12 is a post-1900 cross-section, because entrants have no pre-reform observations. The comparable object for incumbents is their own Mining  $\times$  Chamber gradient on the same post-1900 window, without the pre-period, which Table A22 reports alongside the entrant figure and the incumbents’ difference in differences.

Table A22: Mining gradient after 1900: incumbents and entrants on the same window

|                                                  | Coefficient | (s.e.)  | N       |
|--------------------------------------------------|-------------|---------|---------|
| Entrants, post-1900 cross-section                | 11.07       | ( 1.83) | 14,182  |
| Incumbents, post-1900 cross-section              | 0.63        | ( 0.73) | 53,460  |
| Entrants minus incumbents, cross-section         | 8.59        | ( 1.61) |         |
| Incumbents, difference in differences (Table 12) | 4.49        | ( 1.74) | 100,174 |

Coefficient on Mining  $\times$  Chamber over the post-1900 legislatures of the Panel B window, with arrondissement, party and year fixed effects, length weights and clustering by arrondissement. Incumbents sat before 1900 and after; entrants first appear in 1900 or later. The last row is the incumbents' triple difference over the whole Panel B window.

## A15 Stage reached: labour bills against other bills

Table 15 follows the members' labour bills to the furthest stage the archive records. Table A23 applies the same classification to members' bills on every other subject, by period, so that the share never reaching a Chamber vote can be compared across subjects.

Table A23: Stage reached by members' bills, labour against other subjects

|           | No Chamber vote |       | Promulgated |       | Promulgated if voted |       | Bills  |       |
|-----------|-----------------|-------|-------------|-------|----------------------|-------|--------|-------|
|           | Labour          | Other | Labour      | Other | Labour               | Other | Labour | Other |
| 1880–1893 | 0.66            | 0.59  | 0.29        | 0.30  | 0.84                 | 0.73  | 73     | 508   |
| 1894–1899 | 0.74            | 0.67  | 0.26        | 0.30  | 1.00                 | 0.90  | 87     | 325   |
| 1900–1912 | 0.70            | 0.77  | 0.27        | 0.14  | 0.89                 | 0.63  | 215    | 833   |
| All       | 0.70            | 0.70  | 0.27        | 0.22  | 0.90                 | 0.73  | 375    | 1666  |

Bills deposited by Catholic and Liberal members, dated by deposit, classified by the furthest stage the Genesis archive records, on the definition of Table 15. Labour bills carry the tags *Labor & Work*, *Welfare & Social Protection* or *Health*. With period fixed effects, the labour-minus-other difference is -0.009 (s.e. 0.026) in the share never reaching a vote and 0.058 (s.e. 0.025) in the share promulgated; in 1900–1912 alone the second is 0.121 (s.e. 0.032). The third pair of columns is the share promulgated among the bills that reached a vote in the Chamber.

# Online Appendix

Not intended for print publication

## OA1 Historical background

**Industrial Belgium and the identification problem.** By the late nineteenth century, Belgium was one of the most industrialised economies in Europe. Coal, steel, and textiles had created a large working class in Wallonia and in the main urban-industrial districts (Deneckere, 1993; Strikwerda, 1997; Witte et al., 2009). This produced recurrent political mobilisation. The main episodes were the unrest of 1886 and the general strike of 1893 (Deneckere, 1993; Witte et al., 2009).

These were not only wage conflicts. The Belgian Workers' Party linked them to demands for political inclusion. Political rights were presented as a necessary step toward addressing the social question in parliament (Deneckere, 1993; Polasky, 1992).

This connection between mobilisation and reform creates the identification problem. Mobilisation helped produce institutional change. But it may also have changed parliamentary behaviour directly by increasing the public salience of labour and social policy (Deneckere, 1993; Polasky, 1992; Witte et al., 2009). A simple before-and-after comparison would therefore mix the effect of the reforms with the effect of the events that produced them. We instead need variation in exposure to reform that does not simply reproduce this common shock. The Chamber–Senate difference-in-exposure design of Section 7 provides this variation.

This political pressure also had a clear geography. Working-class mobilisation and socialist electoral support were concentrated in the coal and heavy-industry districts of the Walloon *sillon industriel*. These included the Borinage, the Centre, Charleroi, and Liège (Deneckere, 1993; Strikwerda, 1988, 1997; Witte et al., 2009). Miners and metalworkers formed the organisational and electoral base of Belgian socialism. The Workers' Party built its strongest cooperatives, unions, and turnout organisations in these districts. Its vote also became concentrated there once the franchise expanded (Brepoels, 2015; Strikwerda, 1988, 1997).

The location of this workforce was largely determined by the geology of the coal seams. This endowment was fixed long before the electoral reforms. We therefore measure socialist electoral pressure using a predetermined proxy: the share of mining workers in arrondissement employment in the 1896 industrial census. This avoids using realised socialist vote shares, which were themselves affected by the reforms. Section 6.2 documents the relationship.

**Restricted suffrage and elite insulation.** Before 1893, national politics was dominated by Catholic and Liberal elites. Voting rights depended on tax and property qual-

ifications, and only a small share of adult men could vote (Witte et al., 2009). Industrialisation created social pressure, but the restricted franchise gave workers little direct influence in parliament (Deneckere, 1993; Witte et al., 2009).

The Senate added another layer of insulation. It inherited the conservative logic of the Dutch constitutional framework. Candidates faced strict eligibility rules, including a high tax qualification and a minimum age of forty. Senators came almost entirely from property-owning and professional groups (Vanthemsche & De Peuter, 2023; Witte et al., 2009).

The pool of men eligible to sit in the Senate remained very small. There were 996 eligible men in 1890, 1,527 in 1900, and 1,637 in 1910, in a country of more than six million people (Luykx & Platel, 1985, p. 196). The Senate’s own historical dossier reports the same series (Sénat de Belgique, 2019).

The two chambers therefore differed sharply in socialist representation. The Workers’ Party entered the Chamber of Representatives in 1894. It won 28 seats in the first election held under plural suffrage (Witte et al., 2009, 2016). By 1910, it held 35 of 166 seats and was the third-largest group after the Catholics and Liberals (Witte et al., 2016). Table OA1 reports the seat counts by legislature and chamber.

Socialist representation developed more slowly in the Senate. Socialist senators were present from 1894, but the first entered through the provincial route. They were chosen by provincial councils rather than directly elected by the party’s industrial electorate.

The first socialist returned for a directly contested Senate seat was Auguste Houzeau de Lehaie. He was elected in 1900 for Charleroi–Thuin. He had previously been a Liberal deputy and had sat as a Liberal senator for Mons since 1897 (Dumoulin, 2024).<sup>29</sup> He is the single directly elected socialist senator recorded in Table OA1 for 1900 and 1902.

The gap between the chambers remained large. In 1910, socialists held 11 of 110 Senate seats, compared with 35 of 166 seats in the Chamber. Only 7 of the 11 socialist senators were directly elected (Vanthemsche & De Peuter, 2023; Witte et al., 2016).

The Senate was therefore more insulated from the new electorate than the Chamber. This mattered because every law required approval from both chambers. The Senate could therefore limit the policy consequences of democratisation in the lower house (Vanthemsche & De Peuter, 2023; Witte et al., 2009).

**Plural suffrage as elite management.** The first major reform came in 1893. Under pressure from socialist mobilisation and the general strike, Belgium adopted plural male suffrage. The reform expanded political participation, but it was designed around the priorities of the incumbent parties rather than as a full concession to working-class demands (Deneckere, 1993; Witte et al., 2009).

---

<sup>29</sup>The Senate’s historical dossier dates the party’s first directly elected senator to 1904 (Sénat de Belgique, 2019).

Three features matter for our analysis. First, plural voting kept electoral weight unequal. The system created three routes to supplementary votes (Emmenegger & Walter, 2019; Gilissen, 1958; Witte et al., 2016).

A head of household aged at least thirty-five who paid five francs or more in direct property tax received an additional vote. A man meeting a property or income threshold also received an additional vote. Holders of a higher-education diploma, and members of specified offices and professions, the *capacitaires*, could receive up to two additional votes (Luykx & Platel, 1985, p. 195). No man could cast more than three votes.

The electorate expanded dramatically. It grew from 136,707 registered voters under the censitary system (Kollman et al., 2025) to about 1.37 million (Witte et al., 2016). But electoral weight remained tilted toward property owners and the educated. The reform therefore expanded the electorate while limiting the relative weight of newly enfranchised workers (Deneckere, 1993; Gilissen, 1958; Witte et al., 2009).

Second, compulsory voting was introduced at the same time. This was not a politically neutral technical change. The authors of the reform feared that traditional Catholic and Liberal voters might abstain after the arrival of large numbers of working-class voters. This could have benefited the Workers' Party.

The Beernaert government stated this concern in the Chamber and presented compulsory voting as a conservative measure. It also reduced the cost to parties of mobilising their supporters. The socialists nevertheless supported compulsory voting as a civic duty (Bouhon, 2018). Turnout increased from 48 per cent in 1892 to 94.6 per cent in 1894 (Bouhon, 2018).

Plural suffrage and compulsory voting therefore widened political participation while limiting some of its consequences. The 1894 election involved a much larger electorate, but under rules designed to contain the gains of the Workers' Party (Gilissen, 1958; Witte et al., 2009, 2016). Mass politics entered parliament without eliminating elite influence. Democratisation was managed from above (Barzachka, 2025; Deneckere, 1993; Emmenegger & Walter, 2019).

**Proportional representation as a political deal.** The second major reform was proportional representation. It was adopted in 1899 and first used in the election of 1900. Belgium was among the first countries to use proportional representation in national legislative elections (Ahmed, 2010; Emmenegger & Walter, 2019).

The choice was politically strategic. Socialist and progressive movements demanded the single male vote. The incumbent parties instead chose proportional representation (Barzachka, 2025; Boix, 1999; Emmenegger & Walter, 2019).

Before 1900, seats were allocated under a majoritarian rule in multi-member constituencies. A list that won the district could take all of its seats (Barzachka, 2025; Boix, 1999; Emmenegger & Walter, 2019). As socialist support grew in industrial con-

stituencies, this system became especially dangerous for the Liberals. They were squeezed between Catholic strength in the countryside and socialist growth in urban and industrial districts (Boix, 1999; Emmenegger & Walter, 2019; Witte et al., 2009).

Under the majoritarian rule, Liberals could lose all representation even when they finished second. Where socialists became the largest party, Liberal votes could therefore produce no seats (Boix, 1999; Emmenegger & Walter, 2019).

Proportional representation offered a solution. Catholics and Liberals reached a political compromise around the new system (Barzachka, 2025; Boix, 1999; Emmenegger & Walter, 2019; Witte et al., 2009).

This fits a broader argument in comparative politics. Moderate incumbent parties may adopt proportional representation when a new party threatens to eliminate them under majoritarian rules (Boix, 1999). Belgium is often presented as a leading example. Proportional representation allowed Liberal votes to translate into seats instead of disappearing in winner-take-all contests (Barzachka, 2025; Boix, 1999; Emmenegger & Walter, 2019).

In return, the Liberals supported proportional representation rather than the socialist demand for the single male vote (Barzachka, 2025; Emmenegger & Walter, 2019).

The reform also served Catholic interests. Under the majoritarian rule, concentrated socialist support could produce very large seat shares in industrial districts. Proportional representation limited this advantage and therefore contained socialist growth in parliament (Barzachka, 2025; Boix, 1999; Emmenegger & Walter, 2019).

The socialists had preferred the single male vote, which would have given working-class voters equal electoral weight. Proportional representation still increased their representation relative to the majoritarian rule, but by less than the single male vote would have (Barzachka, 2025; Emmenegger & Walter, 2019).

**Did elite management work?** Belgian democratisation did not simply transfer power from political elites to the working class. It was a sequence of institutional reforms designed by incumbent parties to manage the consequences of mass mobilisation while preserving their own influence (Barzachka, 2025; Deneckere, 1993; Emmenegger & Walter, 2019).

Plural suffrage expanded participation but preserved unequal electoral weight (Gillis, 1958; Witte et al., 2009, 2016). Proportional representation limited socialist dominance in industrial districts and helped preserve Liberal representation (Barzachka, 2025; Boix, 1999; Emmenegger & Walter, 2019). The Senate remained comparatively insulated and continued to act as a constraint on legislation that incumbent elites opposed (Vanthemsche & De Peuter, 2023; Witte et al., 2009).

But preserving political control does not imply that democratisation had no effect on parliamentary behaviour. The question of this paper is whether these reforms changed

the incentives of Catholic and Liberal politicians enough to change what they discussed and what legislation they proposed and supported.

A second question is how such a change occurred. Did politicians already in office adapt their behaviour? Or did parliamentary attention change because different politicians entered office?

A reform designed to limit its own political consequences can still change incentives at the margin. It can force established parties to respond to demands that a restricted franchise had previously allowed them to ignore. Whether this happened in Belgium, and through which channel, is what the paper tests.

## **OA2 Socialist representation in the two chambers**

This appendix records socialist representation in the Chamber of Representatives and the Senate over the period of the analysis: when the Belgian Workers' Party (BWP) entered the legislature, how fast its presence grew after the extension of the franchise, and how the two chambers differed, which is the distinction the difference-in-exposure design rests on.

The Chamber was directly exposed to the electorate created in 1893; the Senate was more insulated. Before the revision senators were directly elected under restrictive eligibility conditions; after it the Senate combined directly elected senators with provincial senators chosen by the provincial councils. The distinction matters because the socialists entered the Senate first by the provincial route, and their directly elected presence there stayed small and came late relative to the Chamber.

Table OA1 reports the standing composition after each election year: for the Chamber, BWP seats and total seats; for the Senate, BWP seats, total seats, the number of directly elected seats and the number of BWP senators who were directly elected, a column that excludes provincial senators and senators by right.

Three patterns in the table bear on the design. There was no socialist representation in either chamber before 1894. The BWP entered the Chamber at the 1894 election with 28 seats of 152, a breakthrough that made the lower house the main channel through which the newly enfranchised reached parliamentary representation.

Socialist representation in the Chamber then grew fast but unevenly: about a fifth of the seats after 1894, more after the introduction of proportional representation, and 39 of 186 by 1912. This is the Chamber as the more exposed arena, the one in which party competition and the pressure of the mass electorate turned most directly into parliamentary presence, and it is why the speech analysis compares the Chamber with the Senate.

The Senate stayed more insulated. Socialist senators sit from 1894, but the first came in as provincial senators, not by direct election, and their number stayed far below the

Table OA1: Socialist representation by election year, Chamber and Senate, 1880–1912

| Year | Chamber   |             | Senate    |             |                        |                            |
|------|-----------|-------------|-----------|-------------|------------------------|----------------------------|
|      | BWP seats | Total seats | BWP seats | Total seats | Directly elected seats | Directly elected BWP seats |
| 1880 | 0         | 132         | 0         | 66          | 66                     | 0                          |
| 1882 | 0         | 138         | 0         | 69          | 69                     | 0                          |
| 1884 | 0         | 138         | 0         | 69          | 69                     | 0                          |
| 1886 | 0         | 138         | 0         | 69          | 69                     | 0                          |
| 1888 | 0         | 138         | 0         | 69          | 69                     | 0                          |
| 1890 | 0         | 138         | 0         | 69          | 69                     | 0                          |
| 1892 | 0         | 152         | 0         | 76          | 76                     | 0                          |
| 1894 | 28        | 152         | 2         | 102         | 76                     | 0                          |
| 1896 | 29        | 152         | 2         | 102         | 76                     | 0                          |
| 1898 | 28        | 152         | 1         | 102         | 76                     | 0                          |
| 1900 | 32        | 152         | 6         | 102         | 76                     | 1 <sup>†</sup>             |
| 1902 | 35        | 166         | 7         | 110         | 83                     | 1 <sup>†</sup>             |
| 1904 | 29        | 166         | 8         | 110         | 83                     | 4 <sup>‡</sup>             |
| 1906 | 31        | 166         | 9         | 111         | 83                     | 4                          |
| 1908 | 35        | 166         | 11        | 111         | 83                     | 7                          |
| 1910 | 35        | 166         | 11        | 110         | 83                     | 7                          |
| 1912 | 39        | 186         | 15        | 120         | 93                     | 10                         |

*Notes:* Values report the full standing composition after the year indicated. For the Senate, “directly elected seats” excludes provincial senators and senators by right. In 1896, 1906, and 1910 there was no regular Senate renewal corresponding to a full new Senate composition; those rows report the composition in force after that year, based on the composition lists and recorded within-legislature changes. Senate totals include provincial senators from 1894 onward and the senator by right in 1906 and 1908. The last column counts senators who sit among the directly elected and whom the composition lists for that legislature carry with a socialist label (Wikipedia contributors, 2024). Both conditions must hold: a socialist senator elected by a provincial council is not counted, and neither is a directly elected senator the lists leave unlabelled.

<sup>†</sup> The 1900 and 1902 entries are the same man, Auguste Houzeau de Lehaie, Liberal senator for Mons from 1897, elected in 1900 for Charleroi–Thuin for the Belgian Workers’ Party (Dumoulin, 2024; Wikipedia contributors, n.d.-a, n.d.-b). These are documented minima rather than exhaustive counts: the lists give a party label for some members only, and a member the lists leave unlabelled is not counted here even if he was a socialist.

<sup>‡</sup> The 1904 entry is qualified in turn by the annulment of Elbers’s election.

The rows from 1904 to 1912 are carried over from the earlier compilation and have not been rechecked under the rule stated above; they may therefore be minima on a different definition from the two rows marked <sup>†</sup>.

Chamber’s throughout: in 1912, with 39 seats in the Chamber, the BWP held 15 in the Senate, 10 of them directly elected. The Senate is therefore not an untreated control. It faced the same national reforms and in time it too held socialist members; but its composition was filtered by the eligibility census, by provincial selection and by its own institutional structure, and it was less exposed to the new electorate than the Chamber.

The comparison between the chambers is accordingly one between two institutions that faced the same national shocks with different degrees of electoral exposure, not between a treated and an untreated institution, and the question the design asks is whether speech and legislative behaviour changed more in the chamber more directly exposed to the new electorate.

The same counts explain why the estimation sample is Catholic and Liberal. Socialist

members entered the Chamber in 1894 and shaped its debates more and more, but the design asks whether the established parties changed what they said and proposed in response to democratisation. The presence of the BWP is part of the environment that generated the competitive pressure; the effect is estimated among the Catholic and Liberal politicians whose incentives changed with the extension of the franchise and the introduction of proportional representation.

## **OA3 Construction and validation of the classifier**

### **OA3.1 Categories and topics**

The categories were built from the corpus in two steps. The proceedings record every debate under a heading that names the item on the agenda, and the heading is a good summary of the subject under discussion. We grouped the headings into topics with an unsupervised clustering procedure and checked the resulting list by hand. A large language model then grouped the topics into six policy domains and a procedural residual, the seven categories of the main text, and we checked the grouping by hand. Table OA2 lists the topics under the category each belongs to. The grouping is complete: every topic is mapped, and there is no leftover category beyond the procedural one.

Table OA2: Topics, by category

| Category                     | Topics                                                                                                                                                                         |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Labour and social            | Labour and worker social policy; general welfare and public assistance; public health and healthcare                                                                           |
| Confessional                 | Religion, secularism and church–state relations; education and science                                                                                                         |
| Economic and fiscal          | Budget and fiscal policy; taxation, customs and property; finance, banking and monetary policy; commerce and trade; industry and economic policy; agriculture and rural policy |
| State and governance         | Governance and administration; electoral and constitutional policy; justice and legal system; public order and policing; citizenship, immigration and minority rights          |
| Foreign and defence          | Colonial, foreign and international relations; defence and military                                                                                                            |
| Territory and infrastructure | Infrastructure, transport and public works; land use, housing and urban policy; environment and natural resources; communication, media and intellectual property              |
| Procedural residual          | Procedural, ceremonial or fragmentary material with no policy content                                                                                                          |

The main text states three boundary decisions: education is grouped with religion rather than with social policy, mining and industry with the economy rather than with labour, and the pay and status of the state’s own employees with governance. Each keeps material that is adjacent to the social question outside the outcome.

### OA3.2 Supervision

No hand-labelled corpus of nineteenth-century Belgian parliamentary speech exists. Training labels are generated by GPT-4o (snapshot `gpt-4o-2024-08-06`, temperature 0, fixed seed, every raw response retained). The prompt, reproduced below, gives the model the topics of Table OA2, a definition of each, and rules for the boundaries most likely to be contested in this corpus. The model returns one primary topic and up to three secondary ones. Only the primary topic is used: it is mapped to its category in Table OA2, and that category is the label the classifier is trained on.

## OA3.3 The annotation prompt

The listing gives the system message and the user message as the model receives them. Line breaks are ours, and the four bracketed placeholders stand for text inserted at run time.

SYSTEM

You are a strict classifier of nineteenth-century Belgian parliamentary speech. Return valid JSON only. Use only the allowed labels. Prefer precision over recall.

USER

Classify the following French parliamentary speech excerpt (Belgium, 1880-1912).

Allowed topics:

[ALLOWED TOPICS]

Topic definitions:

[TOPIC DEFINITIONS]

General rules:

- Return exactly ONE primary topic: the single dominant substantive theme.
- Return up to THREE secondary topics: other themes genuinely present. May be empty.
- Do not classify from isolated words alone.
- The text is OCR output and may contain errors; classify on the readable content.
- If the excerpt is procedural, fragmentary, ceremonial or too vague to classify reliably, set primary to "[RESIDUAL]" and leave secondary empty. Short excerpts are common and are usually "[RESIDUAL]"; do not force a policy label onto them.
- Never put "[RESIDUAL]" in secondary.
- Every label must be exactly one of the allowed topics. Do not invent labels.

Boundary rules that matter for this corpus:

- Labour vs General Welfare: MUTUALLY EXCLUSIVE. Choose Labour only if the beneficiaries are workers as workers (ouvriers, travailleurs, salaries, employes, mineurs, or another occupational group). If they are the poor, the needy, the general population, or unspecified, choose General Welfare.
- Labour vs Industry: a speech may describe factories, mines, firms or output and still not be about labour. Choose Labour only when the employment relationship or worker protection is central, not incidental.
- Education vs Religion: if the speech is about who controls or funds schools, or about religious instruction, the primary label is whichever dominates and the other belongs in secondary. Curricula, examinations or teacher pay with no confessional dimension is Education alone.
- Defense vs Colonial: choose Defense when military organisation, forces, service, fortification or armament is central; Colonial when the subject is the colony, diplomacy or a treaty, even if war is mentioned.
- Public Order vs Defense: police, gendarmerie, riots and repression are Public Order. The regular army is Defense.
- Budget vs the substantive domain: for a budget debate that is

substantively about a policy area, the primary label is that policy area and Budget goes in secondary.

Output format. Return JSON only, no prose, no markdown fences:

```
{"primary": "topic", "secondary": ["topic", "topic"]}
```

Speech:

[SPEECH]

[ALLOWED TOPICS] is the list of topics in Table OA2 together with the label of the procedural residual, which also replaces [RESIDUAL]. [TOPIC DEFINITIONS] is a one-sentence definition of each topic, given in the replication materials, and [SPEECH] is the text of the intervention. The rules on labour and general welfare, on education and religion, and on defence and colonial affairs separate topics that belong to the same category, so they have no bearing on the label the classifier is trained on. The rules that do are labour against industry, policing against the army, and the assignment of a budget debate to the policy area it concerns.

## OA3.4 Sampling

The labelled frame is a probability sample of 7,000 interventions stratified on chamber  $\times$  reform window, with a floor per cell so that the thinner Senate cells are represented. A further 1,500 interventions are drawn from the cells with the fewest interventions and enter training alone: the test and validation splits are drawn only from the probability sample.

Selection uses no topic keywords. A design that oversampled interventions rich in labour vocabulary would have anchored the topic partition to the same lexicon a dictionary uses, and would in addition have made the validation split unrepresentative of the corpus.

## OA3.5 Model

A CamemBERT encoder with a seven-way softmax head over the six policy domains and the residual, trained with class-weighted cross-entropy. An auxiliary head over the topics of Table OA2 is trained alongside it at weight 0.3, which helps the encoder with the rarer topics; its output is used nowhere in the paper. Long interventions are chunked with overlap and chunk probabilities averaged. Decision thresholds do not arise for the seven-way assignment, which is an argmax; the main-text outcome uses the normalised probability rather than the assignment.

### OA3.6 Validation against human coding

Held-out performance measures fidelity to the annotator, not accuracy. To measure accuracy we hand-coded a blind subsample of 250 interventions drawn from the probability sample: the sheet was shuffled, carried no model output, and the chamber and period columns were hidden during coding.

Against that standard the classifier reaches an accuracy of 0.86 with  $\kappa = 0.80$  and macro F1 of 0.823; the GPT-4o annotation on which it was trained reaches 0.84,  $\kappa = 0.77$  and macro F1 of 0.774 (Table 4 in the main text). By the conventional reading (Landis & Koch, 1977) this is substantial agreement, at the threshold content analysis asks of a primary measure. That the classifier agrees with the coder at least as closely as its own annotator does indicates that distillation does not compound the annotator’s error.

The remaining question is whether classification *accuracy*, as opposed to coverage, varies across chamber and period. If the classifier were systematically more accurate in the Chamber after 1900 than in the Senate before 1894, it could generate the estimates mechanically. The hand-coded subsample is drawn from the probability sample rather than re-stratified on the design’s cells, so it does not support a reliable accuracy rate within each of the four chamber-by-period cells: the Senate cells in particular are thin. We report the aggregate figures and do not manufacture a cell-level table the sample cannot carry. What can be said is that the corresponding *coverage* question, which is the one that bears on the estimand, is answered on the full corpus rather than on 250 items (Table A1), and that the two normalisations of the outcome agree, which is the check that differential classification would fail.

### OA3.7 Coverage and the procedural residual

The classifier scores every intervention. 55.8 per cent carry no substantive policy content—procedural exchanges, communications, votes and fragments—which is unremarkable for a plenary record. Retaining them as a seventh bucket rather than dropping them is what keeps the estimation sample free of selection on treatment. The coverage table is reported as Table A1 in Appendix A1, because the differential it documents bears on the choice of denominator rather than being a diagnostic detail.

### OA3.8 What the classifier is and is not used for

The classifier produces the dependent variable and nothing else. It does not enter the construction of the mining exposure measure, the party assignment, the career variables, or the sample definition. The identification comes from the bicameral contrast and from predetermined industrial structure, neither of which passes through the model.

## OA4 Domain specificity

Table A12 in Appendix A6 normalises each bucket over the six substantive categories. Table OA3 repeats the exercise dividing by all seven, so the procedural residual appears both in every denominator and as a row of its own. The two denominators cover identical rows, so the comparison is rescaling and not selection.

The conclusion is unchanged: labour is the only bucket that rises, at 9.01, and the largest other positive is 0.11 (Table OA3). The residual bucket itself is flat—0.74 with a standard error of 0.79—which is what makes the seven-way taxonomy usable: a residual that moved with the treatment would contaminate every bucket through the denominator, and this one does not move.

Table OA3: Domain specificity under the all-class denominator

| Policy bucket       | Estimate | Std. err. | <i>t</i> |
|---------------------|----------|-----------|----------|
| labour and social   | 9.013*** | (2.395)   | 3.76     |
| confessional        | -2.131*  | (1.120)   | -1.90    |
| economic fiscal     | -0.687   | (3.227)   | -0.21    |
| state governance    | -5.260   | (3.276)   | -1.61    |
| foreign defence     | -1.787   | (1.779)   | -1.00    |
| territory infra     | 0.113    | (1.990)   | 0.06     |
| procedural residual | 0.738    | (0.792)   | 0.93     |

The post-1900 triple difference estimated on each bucket in turn, with every bucket divided by the full seven-category denominator rather than by the six substantive ones. The procedural residual therefore appears both in the denominator and as a row. The buckets sum to 100 by construction, so a rise in one implies a fall elsewhere; what matters is whether labour remains the only category rising. Compare Table A12 in the main text. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## OA5 Language composition

Table OA4: Language composition of the corpus

| Legislature | Chamber       |          | Senate        |          |
|-------------|---------------|----------|---------------|----------|
|             | Interventions | Dutch, % | Interventions | Dutch, % |
| 1880        | 16,243        | 0.00     | 1,481         | 0.20     |
| 1882        | 19,747        | 0.00     | 1,473         | 0.20     |
| 1884        | 23,845        | 0.00     | 2,177         | 0.00     |
| 1886        | 18,115        | 0.00     | 2,119         | 0.05     |
| 1888        | 19,233        | 0.01     | 1,990         | 0.05     |
| 1890        | 17,881        | 0.00     | 2,343         | 0.17     |
| 1892        | 27,709        | 0.01     | 2,598         | 0.00     |
| 1894        | 17,290        | 0.01     | 1,719         | 0.17     |
| 1896        | 13,519        | 0.03     | 1,579         | 0.25     |
| 1898        | 16,357        | 0.07     | 1,486         | 0.13     |
| 1900        | 15,250        | 0.01     | 2,287         | 0.17     |
| 1902        | 13,999        | 0.23     | 2,418         | 0.25     |
| 1904        | 15,969        | 0.05     | 3,150         | 0.16     |
| 1906        | 11,583        | 0.00     | 3,155         | 0.22     |
| 1908        | 14,395        | 0.08     | 4,281         | 0.23     |
| 1910        | 13,242        | 0.17     | 3,897         | 0.08     |
| 1912        | 17,464        | 0.05     | 4,418         | 0.27     |

Interventions recorded in Dutch as a percentage of all interventions, estimation sample, by legislature and chamber. Maximum in any cell: 0.27 per cent.

The corpus is 99.95 per cent French, with Dutch never exceeding 0.27 per cent of interventions in any legislature (Table OA4). Parliamentary use of Dutch expanded only after the De Vriendt–Coremans law of 1898, and in plenary debate the change is barely visible before 1912. Differential performance of a French-language model across languages therefore cannot generate the estimates.

Weighted by words rather than by interventions the share is smaller still. In the four coalfield districts, which supply the whole of the 1900 margin, Dutch-language Chamber speech is virtually absent; those constituencies are Walloon and their proceedings are French.

Re-estimating on French-language interventions alone changes nothing: both headline estimates are virtually identical to those on the full sample. The design does not predict the language mix either: the same triple difference with a Dutch-language indicator as the outcome is indistinguishable from zero.

## OA6 Five electoral measures, and why none replaces the proxy

Mining intensity is a proxy for where the majoritarian rule had been distorting outcomes. It is predetermined and it is measured from an administrative source independent of parliamentary activity, but it is a proxy: it also captures industrial structure, constituency composition and the geography of strike activity, and the design cannot separate those from electoral competition by construction.

This appendix reports five attempts to replace it with a direct measure of the rule change, built from the constituency-level returns of 1894–1898 and the seat allocation set by the 1899 law. None of them displaces the mining proxy. We report the exercise in full because the question is one a reader will ask, and because the pattern of failure is itself informative about what the design can and cannot resolve.

Each measure is one implementation of an idea that admits several. The last pre-reform election of a district is taken to be 1898 if it voted in 1898, else 1896, else 1894; the threshold shift uses the magnitude of the post-1900 constituency *containing* the district, so the components of a merged constituency share a value; and a district whose last pre-reform election was uncontested has no counterfactual allocation at all. Three districts are in that position—Eeklo, Turnhout and Hasselt, each with a single list—so the four measures built on the pre-reform returns are computed on 38 arrondissements rather than 41, while the threshold shift is computed on all 41.

### OA6.1 The five measures

*Threshold shift.* Under the majoritarian list rule a district is winner-take-all: a list needs an absolute majority to win in the first round, and the run-off is contested by the two leading lists, so the effective threshold of representation is about one half; under D'Hondt with  $M$  seats it is about  $1/(M+1)$ . The reform therefore moved the threshold by  $0.5 - 1/(M_d + 1)$ . This is monotone in district magnitude, and that is its defect: Brussels, with eighteen seats, has the largest shift in the country and no under-represented industrial minority.

*Counterfactual seat reallocation.* Applying D'Hondt to the votes cast at each district's last pre-reform election and comparing with the allocation the majoritarian rule actually produced gives the number of seats that would have changed hands. This measure has a defect exactly where it matters. At the last pre-reform election in Liège and in Waremme the socialists did not run separately but inside a liberal–socialist cartel—from 1894 in Liège, and from 1898 in Waremme, where in 1894 the Belgian Workers' Party had still run alone on 7.3 per cent (Kollman et al., 2025)—and the cartel *was* the response to the majoritarian rule: two parties coalescing because separately both would lose. Allocating

seats to a bloc that exists only because of the rule being counterfactualled is not a well-posed exercise.

*Unrepresented votes.* One minus the leading list’s share, at the same election: the fraction of the electorate that elected nobody. This measure requires no counterfactual allocation and is therefore immune to the cartel problem. But it measures how *contested* a district was rather than whether it contained a constituency without representation, and those are not the same: the most distorted districts on this measure are Thuin, Verviers, Nivelles and Brussels, not the coalfields.

*Representational concentration.* How many seats a single list held alone under the old rule, weighted by how decisively it won. Proportional representation does not reward closeness; it breaks concentration, and a district electing eight members all from one list is a different object from one electing two.

*Concentration on socialist-held seats.* The same measure restricted to districts whose leading list was socialist or socialist-allied. This is the sharpest version of the idea that proportional representation mattered where an organised left was already winning, and it is also the one that most nearly reproduces the treatment variable: it is positive in seven districts—Liège, Charleroi, Mons, Namur, Soignies, Thuin and Huy—and is highly correlated with the coalfield indicator and with mining intensity. It is reported with the others rather than as a separate case.

## OA6.2 What they show

Table OA5 reports each measure in the paper’s post-1900 specification, alone and alongside mining intensity, with a placebo at the 1894 cutoff when the 1899 law was not yet in force.

The result is uniform on the point that matters. None of the five predicts the post-1900 shift in the direction the mechanism would require, and none displaces mining intensity when both are included: the mining coefficient runs between 7.0 and 10.1 across the with-mining columns, against 9.25 in the baseline (Table OA5).

The failures are not all of the same kind, and two of them are worse than a null. Threshold shift and unrepresented votes are simply uninformative, with  $t$ -statistics below 0.7 in every column. Concentration on socialist-held seats is significant only alongside mining and with the wrong sign,  $-2.72$  ( $t = -4.06$ ), which is what a second, noisier copy of the treatment variable does when the first is already in the regression. And two measures fail their own placebo, or come close to it: seat reallocation returns 2.93 ( $t = 1.97$ ) at the 1894 cutoff, when the 1899 law did not exist, and representational concentration is  $-6.72$  ( $t = -2.52$ ) after the reform against 3.62 ( $t = 1.65$ ) before it (Table OA5). A predetermined variable that moves in the pre-treatment window is not measuring the reform, whatever it does in the post-treatment one.

The threshold-shift row is the measure discussed in Appendix A2.

Table OA5: Five predetermined electoral measures

|                                       | Alone            | With mining      | Placebo 1894     |
|---------------------------------------|------------------|------------------|------------------|
| Threshold shift                       | 1.41<br>( 2.22)  | 0.40<br>( 1.87)  | -0.60<br>( 1.17) |
| Seat reallocation                     | -3.61<br>( 1.86) | -1.95<br>( 1.47) | 2.93<br>( 1.49)  |
| Unrepresented votes                   | -0.45<br>( 2.36) | -1.06<br>( 1.91) | -0.44<br>( 2.05) |
| Representational concentration        | -6.72<br>( 2.67) | -5.11<br>( 2.57) | 3.62<br>( 2.19)  |
| Concentration, socialist-held seats   | -0.93<br>( 0.59) | -2.72<br>( 0.67) | -0.34<br>( 0.49) |
| Mining intensity, with-mining columns | 9.25             | 7.0–10.1         |                  |

Each measure standardised and entered in place of, then alongside, mining intensity in the specification of equation (12). Standard errors clustered by arrondissement. The placebo column applies the same variable with the 1894 cutoff. The mining row reports the range of the mining coefficient across the with-mining columns.

*Notes:* Each measure standardised and entered in place of, then alongside, mining intensity in the specification of equation (12). Standard errors clustered by arrondissement. The placebo column applies the same variable with the 1894 cutoff, when the 1899 law was not yet in force. The mining row reports the range of the mining coefficient across the five with-mining columns. The threshold shift is defined for all 41 arrondissements; the four measures built on the last pre-reform returns are defined for 38, because Eeklo, Turnhout and Hasselt returned a single list at that election and a counterfactual allocation does not exist for them.

### OA6.3 A note on where these numbers come from

An earlier version of this appendix reported a different set of coefficients, and in particular reported the socialist-restricted concentration as displacing the mining proxy with a triple difference of 5.90. Those numbers had no producing code in the replication materials. The stage that now produces them, written for this version and documented line by line, returns the table above. We report what the code returns and note the discrepancy rather than reconciling it after the fact: the construction is one implementation of each idea, several of the choices could reasonably be made otherwise, and a reader who disputes one can change it and rerun.

One of those choices is not innocuous. The counterfactual allocation is computed on the constituency returns as CLEA records them, and CLEA codes a liberal–socialist cartel as a single list. In the districts where the cartel existed the allocation is therefore performed on the bloc rather than on its parts, which is the same objection raised against the seat-reallocation measure above, now stated as a property of the data rather than of the measure.

## OA6.4 What we conclude

The design does not contain the variation needed to separate where the electoral rule bound from where the industrial workforce was. With four districts carrying most of the treated cell and an effective cluster count of 2.95 (Section 11.1, Appendix Table A15), any district-level variable competes to identify the same handful of cells, and the two candidate explanations are not distinguishable within this sample.

We therefore report the mining proxy as what it is: a predetermined measure of where proportional representation had something to change, not a decomposition of the channel through which it changed it. The exercise in this appendix is not a failed instrument. It rules out four specific alternatives—that the post-1900 effect is a mechanical consequence of how the new rule redistributed seats, of how many seats each district elected, of how closely contested it had been, or of where an organised left already held seats—and those are alternatives a reader would otherwise be entitled to suspect.

Separating the channel properly would require either a setting with more districts, or within-district variation in the rule change, and neither is available in Belgium in 1900.

## OA7 Three interventions and their scores

Table OA6 is the measure of Section 5 applied to three interventions, chosen by rule from the Chamber interventions of Catholic and Liberal members running to between 60 and 150 words. The first is the one whose labour score is closest to 0.95 among interventions using the vocabulary of work. The second is the one closest to 0.50 among those, with the economic domain carrying at least a quarter of the remaining mass. The third is the one closest to 0.05 among interventions naming the pit or the coal trade with the economic domain carrying most of the mass, which is the case a keyword dictionary cannot separate from the first: it is about the legal form of mining companies and the majority of shares required to change it, and the word *mines* runs through it.

Table OA6: Three interventions and their scores

|                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Jules Devigne, Chamber, 1905-02-09          | <p><i>Dans tous les cas, en ce qui me concerne et en ce qui concerne un grand nombre de mes amis, je proteste énergiquement contre cette insinuation qui est réellement calomnieuse. [Très bien ! sur les bancs libéraux.] Je n'aime pas parler de moi même, et si je vais me résoudre à le faire, c'est que mon cas est un peu celui de beaucoup de mes amis. J'ai été un des combattants de la première heure pour le repos hebdomadaire. J'ai fait partie, il y a plus dix ans, à Gand, du premier comité constitué dans ce but et dont faisaient partie également [...]</i></p> <p>Class probabilities: labour social 0.93; confessional 0.01; economic fiscal 0.01; state governance 0.01; foreign defence 0.01; territory infra 0.01; other 0.01; labour salience score = 94.2.</p>  |
| (b) Paul de Smet de Naeyer, Chamber, 1897-12-17 | <p><i>Je ne parle pas de l'alcool, sur lequel nous avons relevé les droits malgré toutes les clameurs. Je n'ai pas à tenir compte de la charge qu'assume, pour son malheur et pour le malheur des siens, l'homme qui, au lieu d'employer son salaire à procurer le nécessaire et un peu de bien être à sa famille, dépense son argent sur le comptoir du cabaret. (Très bien! à droite.) Je ne parle ici que de l'ouvrier rangé, qui travaille et qui tâche d'épargner. Quant à celui là, je dis que notre système fiscal et économique tout entier vise, d'une part, [...]</i></p> <p>Class probabilities: labour social 0.49; confessional 0.02; economic fiscal 0.42; state governance 0.02; foreign defence 0.02; territory infra 0.01; other 0.02; labour salience score = 50.2.</p>  |
| (c) Charles Woeste, Chamber, 1886-02-19         | <p><i>Les sociétés qui ont pour objet l'exploitation des carrières n'existent pas? Je vous demande pardon ! Il y en a au contraire en assez bon nombre. Dès lors pourquoi ne pas leur permettre de se transformer ? En second lieu, l'honorable membre propose de dire que la décision qu'il reconnaît aux assemblées d'actionnaires le droit de prendre, en vue de la transformation des sociétés de mines, devra avoir l'adhésion des neuf dixièmes des parts sociales. C'est trop exiger, me semble t il. Il y a des sociétés de mines dont les parts sont répandues [...]</i></p> <p>Class probabilities: labour social 0.05; confessional 0.01; economic fiscal 0.88; state governance 0.01; foreign defence 0.02; territory infra 0.01; other 0.02; labour salience score = 5.1.</p> |

Three interventions chosen by rule from the Chamber interventions of Catholic and Liberal members running to between 60 and 150 words. Each rule selects the intervention whose labour score is closest to a target among those meeting a condition on the text: (a) closest to 0.95 among interventions using the vocabulary of work; (b) closest to 0.50 among those using that vocabulary with the economic domain carrying at least a quarter of the remaining mass; (c) closest to 0.05 among those naming the pit or the coal trade with the economic domain carrying at least 0.45 of the mass, which is the case Section 5 argues a keyword dictionary cannot separate. The excerpt is the opening of the text as transcribed, cut at a word boundary; the only edits are the removal of the soft hyphens the original print left where a word ran across two lines, and of line breaks. The transcription does not carry the hyphens of compound forms, so *semble-t-il* appears as three words; that is the state of the digitised record and not an editorial choice. The score is the labour probability as a share of the six substantive probabilities, multiplied by 100.

## OA8 Decomposing the change in the Chamber

Sections 8 and 9 identify differences between the two chambers. This appendix reports an accounting decomposition of the Chamber’s own movement, which identifies nothing but says how the movement was composed, and whose within term can be compared with the within-person estimates of the main text.

The unit is the member by regime, where the regimes are the censitary legislatures (1880–92), the plural-suffrage majoritarian legislatures (1894–98) and the first proportional legislatures (1900–06). For each member and regime we take the word-weighted mean of his labour salience and then average across members, so the object is the typical member rather than the typical word. For a comparison of two adjacent regimes, let  $S$  be the members present in both,  $L$  those present only in the earlier and  $E$  those present only in the later. The change in the mean then decomposes exactly:

$$\bar{x}_1 - \bar{x}_0 = \underbrace{(\bar{x}_{S1} - \bar{x}_{S0})}_{\text{within}} + \underbrace{\frac{n_E}{N_1}(\bar{x}_E - \bar{x}_{S1})}_{\text{entry}} - \underbrace{\frac{n_L}{N_0}(\bar{x}_L - \bar{x}_{S0})}_{\text{exit}}. \quad (13)$$

The identity is checked in the code, which stops if the three terms do not reproduce the change. Two features of the construction bear on how the table should be read. A member with very few words in a regime cannot be given a mean, and dropping him also moves him out of  $S$  and into  $E$  or  $L$ , which biases the split towards entry; Table OA7 therefore reports three word floors rather than choosing one, and the within share of the Catholic movement across 1894 runs from 66 per cent at the most restrictive floor to 81 per cent at the least. And a member-regime spans between six and thirteen years, so a member who changed party within one carries two labels; the label used is the one under which the member spoke most.

Table OA7: Decomposition of the change in Chamber labour salience

| Party                                                                  | Reform | Before | After | Change | Within | Entry | Exit  | $n_S/n_L/n_E$ |
|------------------------------------------------------------------------|--------|--------|-------|--------|--------|-------|-------|---------------|
| <i>Panel A. Members with at least 300 words in a regime (baseline)</i> |        |        |       |        |        |       |       |               |
| Catholic                                                               | 1894   | 7.90   | 16.28 | 8.39   | 6.51   | 1.30  | 0.57  | 84/76/54      |
| Catholic                                                               | 1900   | 16.28  | 18.46 | 2.17   | 3.33   | 0.25  | -1.41 | 69/69/42      |
| Liberal                                                                | 1894   | 8.90   | 11.82 | 2.92   | 6.37   | -2.54 | -0.92 | 20/107/16     |
| Liberal                                                                | 1900   | 11.82  | 16.44 | 4.63   | 0.70   | 8.16  | -4.23 | 10/26/48      |
| <i>Panel B. Members with at least 1000 words in a regime</i>           |        |        |       |        |        |       |       |               |
| Catholic                                                               | 1894   | 8.17   | 16.29 | 8.11   | 5.33   | 1.97  | 0.81  | 67/74/47      |
| Catholic                                                               | 1900   | 16.29  | 19.00 | 2.72   | 2.77   | 0.88  | -0.93 | 57/57/39      |
| Liberal                                                                | 1894   | 9.09   | 14.66 | 5.57   | 13.10  | -7.64 | 0.11  | 7/100/16      |
| Liberal                                                                | 1900   | 14.66  | 16.53 | 1.87   | -1.01  | 8.04  | -5.16 | 5/18/48       |
| <i>Panel C. Members with at least 100 words in a regime</i>            |        |        |       |        |        |       |       |               |
| Catholic                                                               | 1894   | 7.67   | 15.50 | 7.84   | 6.37   | 1.13  | 0.34  | 93/82/60      |
| Catholic                                                               | 1900   | 15.50  | 18.14 | 2.64   | 3.25   | 0.83  | -1.45 | 81/72/48      |
| Liberal                                                                | 1894   | 8.49   | 13.40 | 4.91   | 8.18   | -2.02 | -1.25 | 33/106/13     |
| Liberal                                                                | 1900   | 13.40  | 16.35 | 2.95   | -0.66  | 4.23  | -0.62 | 16/30/46      |

Labour salience in percentage points of substantive speech, Chamber, Catholic and Liberal members. Cells are member-by-regime; the regimes are the censitary legislatures (1880-92), the plural-suffrage majoritarian legislatures (1894-98) and the first proportional legislatures (1900-06). Writing  $S$  for the members present in both regimes of a comparison,  $L$  for those present only before and  $E$  for those present only after, the change in the mean decomposes exactly as  $(\bar{x}_{S1} - \bar{x}_{S0}) + (n_E/N_1)(\bar{x}_E - \bar{x}_{S1}) - (n_L/N_0)(\bar{x}_L - \bar{x}_{S0})$ , reported as Within, Entry and Exit. Within is the change among the same members against their own baseline. Means are across members, one weight per cell. A member with too few words in a regime cannot be given a mean and is moved out of  $S$ , which biases the split towards Entry; the three panels vary that floor, and the bias falls as the floor falls.

Table OA8: Labour salience of those who stayed, left and arrived

| Party    | Reform | Present in both |       | Leavers | Entrants | $n_S/n_L/n_E$ |
|----------|--------|-----------------|-------|---------|----------|---------------|
|          |        | Before          | After |         |          |               |
| Catholic | 1894   | 8.47            | 14.98 | 7.26    | 18.31    | 84/76/54      |
| Catholic | 1900   | 14.87           | 18.20 | 17.70   | 18.87    | 69/69/42      |
| Liberal  | 1894   | 7.98            | 14.36 | 9.07    | 8.64     | 20/107/16     |
| Liberal  | 1900   | 7.58            | 8.29  | 13.44   | 18.14    | 10/26/48      |

Mean labour salience among the members present in both regimes of a comparison (before and after), among those present only in the earlier regime (leavers) and among those present only in the later one (entrants). Chamber, Catholic and Liberal members, at the baseline word floor. The comparison that carries the selection result is between the leavers and the stayers before a reform.

Table OA8 carries the selection result. At 1894 the members of both parties who kept their seats were more labour-oriented before the reform than those who lost them, and the members who arrived were more labour-oriented than either. Enfranchisement did not select for the conservative wing of the established parties and then move it; it removed the least labour-oriented members, admitted the most, and moved those who remained, all in the same direction. At 1900 the pattern is reversed for the Liberals, whose leavers were more labour-oriented than their stayers, because by then the Liberal survivors of the majoritarian legislatures were a small and unrepresentative group.

## OA9 Differential transcription quality across the two houses

The two chambers' proceedings do not come from the same source. The Chamber corpus is the Plenum.be digitisation; the Senate proceedings we obtained as scans from the Senate's historical archives and processed with our own optical character recognition pipeline. Transcription quality can therefore differ between the two arms of the design, and a reader is entitled to ask whether that difference, rather than electoral exposure, is what the estimates pick up.

Two structural features bound the concern before any test.

A time-invariant quality difference between the houses is a level. It is absorbed by the chamber and politician fixed effects and differences out of the difference-in-exposure coefficient. The only version of the concern that survives the design is Senate transcription quality that *changes at a reform date*, which would manufacture a differential trend where none exists.

And the failure mode is milder for a classifier than for a dictionary. A keyword measure fails outright when garbling breaks a match: the term is either recognised or it is not. A transformer reads subword tokens and degrades gracefully—badly transcribed text produces a weaker contextual representation, which pushes an intervention toward the procedural residual rather than causing a discrete match failure. This is a softer problem, and it is also a testable one in a way the dictionary version was not.

Table OA9: Transcription-noise proxies by chamber and reform period

| Noise proxy                                | Overall mean |        | Senate, by period |          |           |
|--------------------------------------------|--------------|--------|-------------------|----------|-----------|
|                                            | Chamber      | Senate | pre-1894          | 1894–99  | post-1900 |
| Short-token share                          | 0.0041       | 0.0039 | 0.0033            | 0.0044   | 0.0042    |
| Residual-class share                       | 0.5290       | 0.3969 | 0.3704            | 0.3483   | 0.4227    |
| <i>Short-token share as the outcome</i>    |              |        |                   |          |           |
|                                            | Estimate     |        | Std. err.         | <i>t</i> | Obs.      |
| Chamber $\times$ Post-1894                 | 0.00024      |        | (0.00016)         | 1.48     | 208,325   |
| Mining $\times$ Post-1900 $\times$ Chamber | -0.00020     |        | (0.00035)         | -0.55    | 119,448   |

Cross-chamber levels reflect corpus size and speaking practice, not transcription quality. The identification-relevant quantity is whether the Senate proxies move at the reform dates. Note that the Senate, whose text we transcribed ourselves, has the LOWER residual-class share of the two houses: transcription noise is not pushing its interventions out of the substantive categories. The lower panel estimates the paper’s two specifications with the short-token share, a number between zero and one, as the outcome; standard errors clustered by arrondissement.

Table OA9 reports two quantities per chamber and, for the Senate, per reform period: the share of very short tokens, which rises with the fragmentation typical of poor recognition, and the share of probability mass the classifier assigns to the procedural residual.

The first row of Table OA9 shows the houses are close and the levels are tiny. Short tokens are 0.41 per cent of the Chamber corpus and 0.39 per cent of the Senate’s. Within the Senate the share moves from 0.33 per cent before 1894 to 0.44 and 0.42 after: a relative change on a base so small that it cannot plausibly reorganise a topic assignment.

The second row settles the direction of the concern, and it settles it the other way from what a reader might expect. If transcription noise were pushing interventions out of the substantive categories, the Senate—the house whose text we transcribed ourselves—would carry the *higher* residual share. It carries the lower one: 0.397 against 0.529 in the Chamber (second row of the same table). The gap is large and it runs against the hypothesis. Whatever explains the Chamber’s bigger residual—and the obvious candidate is procedural traffic in a larger and busier house—it is not that we did a worse job on the Senate scans.

The identification-relevant test is whether these move with the design rather than across it. Estimating the paper’s own specifications with the short-token share as the outcome gives *t*-statistics of 1.48 for the 1894 difference in differences and  $-0.55$  for the post-1900 triple difference (Table OA9, lower panel). Transcription quality does not shift differentially at either reform date, and it does not shift differentially across mining districts. Whatever noise the two corpora carry, the design does not sort on it.

One coefficient does run in the expected direction and we report it rather than omit it: the residual-class share rises with the short-token share. More fragmented text does receive slightly more residual mass, which is the mechanism the transformer failure mode predicts. But the

effect explains essentially none of the variation in the outcome.

One residual channel remains. Even a Senate-wide change in transcription quality after 1900 could not produce the post-1900 triple difference, because that estimate is identified from variation *across districts within* the Senate, and the proceedings were printed and scanned as chamber-level volumes rather than by the speaker’s constituency. There is no mechanism by which garbling would have hit mining and non-mining Senate districts differentially.

## OA10 Venue versus membership

The chamber variable records where a speech was delivered, not which house the speaker belonged to. Ministers defend legislation before both chambers, so for them the treatment indicator is not constant within politician. We state this rather than let the notation imply otherwise, and bound what it costs.

On the archive’s own person key the estimation sample contains 957 distinct references, of which 149 have recorded speech in both houses (Table OA11, notes). Two discriminants separate ministers from records in which two people have been merged. A minister keeps his constituency when he appears before the other house, and he appears there in the same years he sits in his own; two different men sharing a name usually differ on at least one. Of the 149, most satisfy at least one of the two, and the few that satisfy neither are plainly distinct individuals on inspection: their tenures do not overlap and their constituencies differ.

They matter more than their number suggests. Fifteen per cent of the references produce 51 per cent of the recorded words (Table OA11, notes), which is what one expects when the group is largely ministers: the front bench speaks.

A caution on the person count, since it does not match the 845 politicians reported in Section 4. The two figures do not measure the same object. Section 4 counts politicians with a documented mandate; the figure here counts distinct person references in the corpus, and the name-to-archive matching is fuzzy enough to generate a tail of spurious references: 238 of the 957 carry one or two interventions across thirty-two years (Table OA11, notes). The residual tail has no bearing on any estimate—one- and two-intervention references are singletons within politician and carry no within-person variation—and the counts should not be read as a census of members.

Table OA10 reports four ways of handling dual-house members. Dropping dual-house members’ minority-venue interventions retains the member but assigns him to the house where he mostly sat. Assigning every member to the house holding the majority of his interventions makes the treatment indicator constant within politician by construction. Excluding dual-house members entirely is the most aggressive and the least informative, since ministers are exactly the members whose speech carries most weight.

The baseline is the conservative choice on both margins, and by a clear margin on the second. For the post-1900 triple difference, dropping minority-venue interventions gives 9.68 against a baseline of 9.25 (Table OA10, Panel A), and assigning every member to his majority house gives 9.27. For the 1894 difference in differences the two alternatives give 5.13 and 5.18

against a baseline of 4.46 (Panel B). Every correction for the venue problem moves the estimates *up*. The maximum shift is 0.7 points on the 1894 margin and 0.4 on the 1900 one.

The fourth row is the aggressive version and should be read with its sample size. Excluding dual-house members entirely leaves 56,034 observations of 119,448—these members produce more than half the recorded words—and gives 4.48 with a standard error of 2.35 (Table OA10, Panel A). The estimate falls but stays positive and marginally significant on a sample that has lost the parliament’s most active speakers. That is a bound, not a specification: a design that discards every member who ever addressed both houses is discarding ministers, and ministers are where legislative attention concentrates.

Table OA10: Venue versus membership: bounding the dual-house problem

|                                                     | Estimate | Std. err. | Obs.    |
|-----------------------------------------------------|----------|-----------|---------|
| <i>Panel A. Post-1900 triple difference</i>         |          |           |         |
| Baseline                                            | 9.253*** | (2.351)   | 119,448 |
| Minority-venue speech dropped                       | 9.684*** | (2.310)   | 117,384 |
| Assigned to majority house                          | 9.272*** | (2.295)   | 119,448 |
| Dual-house members excluded                         | 4.480*   | (2.347)   | 56,034  |
| <i>Panel B. Post-1894 difference in differences</i> |          |           |         |
| Baseline                                            | 4.455**  | (2.100)   | 208,325 |
| Minority-venue speech dropped                       | 5.128**  | (2.185)   | 201,558 |
| Assigned to majority house                          | 5.179**  | (2.105)   | 208,325 |

The chamber variable records where a speech was delivered, not the speaker’s house. Ministers defend legislation before both chambers, so for them the treatment indicator is not constant within politician.

The rows bound what that costs. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## OA10.1 The person key

The choice of key bears on every column that absorbs politician effects, and Table OA11 reports both keys for all of them. At the 1900 margin the archive key moves the coefficients away from zero: the triple difference with politician fixed effects goes from  $-2.689$  to  $-3.474$  and the incumbent row from  $-2.783$  to  $-3.989$ . At the 1894 margin it moves them towards zero: column 3 of Table 8 from 3.955 to 2.917, the incumbent subsample with politician effects from 4.075 to 3.193, and the within-person estimate from 4.025 to 3.236. The two margins move in opposite directions, and the reason they move at all is visible in the group counts: on the full sample the archive key resolves 572 name values into 536 members, and within the incumbent subsample it separates 280 name values into 285 members. The name string both merges distinct members and splits single ones, and a politician fixed effect is only as good as the identifier that defines it.

The main tables use the archive key throughout. It is the identifier the source assigns and the one Section 4 adopts, and the objection we previously raised against adopting it—that it

moved these coefficients in the direction that suits the argument—does not survive the 1894 margin, where it does the opposite. A key that helps in one place and costs in another is not a choice made after seeing the results. A reader who prefers the name key can read the second pair of columns; the differences are within a point everywhere, and no conclusion in the paper turns on them.

Table OA11: Within-politician estimates under two person keys

|                                                                                          | Name key          |         | Archive key        |         |
|------------------------------------------------------------------------------------------|-------------------|---------|--------------------|---------|
|                                                                                          | Estimate          | Persons | Estimate           | Persons |
| <i>Plural suffrage, 1880–1899: Chamber <math>\times</math> Post-1894</i>                 |                   |         |                    |         |
| All politicians (Table 8, column 3)                                                      | 3.955<br>(2.389)  | 572     | 2.917<br>(2.507)   | 536     |
| Incumbents (Table 9, Panel A)                                                            | 4.075*<br>(2.381) | 280     | 3.193<br>(2.551)   | 285     |
| Same people (Table 9, Panel B)                                                           | 4.025*<br>(2.380) | 229     | 3.236<br>(2.524)   | 225     |
| <i>Plural suffrage: Mining <math>\times</math> Post-1894 <math>\times</math> Chamber</i> |                   |         |                    |         |
| All politicians (Table 8, column 6)                                                      | -0.551<br>(1.248) | 572     | 0.189<br>(1.360)   | 536     |
| <i>Proportional representation, 1894–1906</i>                                            |                   |         |                    |         |
| Chamber $\times$ Post-1900 (Table 11, column 3)                                          | 2.165<br>(2.331)  | 464     | 2.788<br>(2.386)   | 437     |
| Mining $\times$ Post-1900 $\times$ Chamber (Table 11, column 6)                          | -2.689<br>(2.102) | 464     | -3.474<br>(2.136)  | 437     |
| The same, incumbents (Table 12)                                                          | -2.783<br>(2.098) | 257     | -3.989*<br>(2.187) | 271     |

Politician fixed effects are constructed using either the name string recorded in the proceedings (name key) or the archive’s own person identifier (archive key). Standard errors, in parentheses, are clustered by arrondissement. Persons is the number of distinct values of the key in the estimation sample for that row. The main tables use the archive key. Under that key, the analysis sample contains 957 persons. Of these, 149 have recorded speech in both houses and account for 51 per cent of recorded words, while 238 appear in only one or two interventions. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## OA10.2 Speaker identities and party labels

Section 4 corrects speaker identities and party labels before the analysis sample is drawn. The corrections resolve three problems. A small number of person references resolve to twentieth-century homonyms and are dropped by reference. Party labels vary within a career across the two language versions of the archive, and we assign each person the modal label within each legislature. One socialist member sitting for a coalfield constituency appears under a Liberal label in part of the archive; since a single mislabelled member in a high-exposure district can

drive a mining-interacted estimate, the build halts if any such record survives into the estimation sample. The corrections move interventions in both directions. They take 19,978 interventions out of the Catholic and Liberal group and bring in 19,304 that the record labels otherwise, a net reduction of 674.

Table OA12: The Liberal entrant cell in the coalfields

| Member                                    | District  | First legislature | Interventions | Share of the cell's words (%) | Gradient without him |
|-------------------------------------------|-----------|-------------------|---------------|-------------------------------|----------------------|
| Fulgence Masson                           | Mons      | 1904              | 511           | 36.5                          | 12.20 (1.11)         |
| Paul Trasenster                           | Liège     | 1900              | 228           | 35.3                          | -4.46 (0.87)         |
| Joseph Louis Descamps                     | Mons      | 1904              | 122           | 22.2                          | 12.20 (1.10)         |
| Emile Bertaux                             | Charleroi | 1900              | 23            | 5.4                           | 12.52 (0.90)         |
| Charles Van Marcke de Lummen <sup>†</sup> | Liège     | 1900              | 10            | 0.6                           | 12.41 (1.11)         |
| No member dropped                         |           |                   |               |                               | 12.17 (1.09)         |

Liberal deputies for the four coalfield districts first observed in 1900 or later, in the 1894–1906 window, with the entry legislature built on the parliamentary archive's person identifier. The last column reports the Liberal entrant mining gradient re-estimated after dropping that member, with its cluster-robust standard error. Specifications include arrondissement and year fixed effects, and observations are weighted by intervention length. <sup>†</sup> The interventions recorded under this name carry an archive identifier that points to another person's page; the row is retained so that the shares sum to one hundred.

Table OA13: The entrant gradient by party and without the largest speakers

|                                                                | Estimate  | Std. err. | Obs.   |
|----------------------------------------------------------------|-----------|-----------|--------|
| <i>Entrant mining gradient, cohorts entering 1900 or later</i> |           |           |        |
| All entrants                                                   | 11.073*** | (1.833)   | 14,182 |
| Liberal entrants only                                          | 12.458*** | (1.139)   | 7,945  |
| Catholic entrants only                                         | 2.445     | (2.691)   | 6,234  |
| <i>Person-level concentration</i>                              |           |           |        |
| Most influential coalfield entrant excluded                    | 5.925***  | (1.337)   | 12,060 |
| Two largest coalfield speakers excluded                        | 5.950***  | (1.340)   | 11,549 |

The entrant gradient is a cross-section within the post-reform period, as in the main text. Party splits absorb arrondissement and year fixed effects; the concentration rows exclude individual coalfield entrants ranked by their share of the coalfield entrant cell's recorded words. The named list underlying the exclusions is in the replication materials. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## OA11 Assigning constituencies to senators

The triple difference needs mining exposure for Senate observations, and the Senate’s electoral geography differs from the Chamber’s in two ways. From 1894 roughly a quarter of senators are chosen by the provincial councils rather than elected. And the Senate, holding half the Chamber’s seats, elects in larger constituencies: it paired small arrondissements throughout the period—Arlon–Virton is the most common paired label in the corpus—and from 1900 in broader aggregations such as Charleroi–Thuin and Huy–Waremme, while the 1899 law fixed the Chamber’s 30 post-merger arrondissements. The 1896 census reports employment for the 41 pre-merger arrondissements, so every observation must be placed on that map.

Three rules do it. Directly elected members are assigned to the constituency the proceedings record, in both houses. Provincial senators have no electoral constituency: they carry province-level labels in the proceedings and are excluded from the estimation sample at the build stage, which is the substantively correct treatment—the design measures electoral exposure, and a senator chosen by a provincial council has none. Merged constituencies carry composite labels, and the baseline assigns each to a single component district. In the corpus every composite label is a Senate label—the Chamber proceedings record deputies under single-arrondissement names throughout—and 10.6 per cent of Senate observations in the Panel B window sit on one.

Single-component assignment approximates exposure where the components of a merged constituency differ, so Table OA14 re-estimates under three alternatives: the exposure of every merged constituency replaced by the mean of its components’ mining exposure, by their maximum, and the Senate’s composite constituencies excluded outright. The post-1900 triple difference moves from 9.25 to 9.11, 10.13 and 10.40: less than half a standard error in every case, and upward in two of the three. The estimates do not depend on the assignment rule. The pre-1900 pairings sit where that rule cannot matter: they are Luxembourg and Flemish pairings whose components have negligible and near-identical mining exposure, and the Panel A estimate is accordingly unchanged to the second decimal under the correction (−3.238 against −3.241).

One residual channel is named rather than dismissed. Three labels—Antwerp, Liège and Namur—name both a province and an arrondissement, so the label alone cannot rule out that a provincial senator for one of those provinces enters the sample under the arrondissement’s name. The bound in the final row of the post-1900 panel of Table OA14 requires no senator-level matching: it removes every post-1894 Senate observation carrying one of the three labels—22.5 per cent of the Senate control in the Panel B window, the whole of the Senate–Liège cell among them—and leaves the triple difference at 8.82 (s.e. 2.01,  $t = 4.39$ ). Whatever number of provincial senators sits behind those labels, they are not what the estimate rests on.

Table OA14: Senator geography: the assignment rule and its alternatives

|                                                      | Estimate  | Std. err. | Obs.    |
|------------------------------------------------------|-----------|-----------|---------|
| <i>Post-1900 triple difference, 1894–1906 window</i> |           |           |         |
| Baseline (collapsed constituency coding)             | 9.253***  | (2.351)   | 119,448 |
| Merged constituencies at component mean              | 9.112***  | (2.438)   | 119,448 |
| Merged constituencies at component max               | 10.127*** | (2.129)   | 119,448 |
| Senate composite constituencies excluded             | 10.403*** | (2.025)   | 117,786 |
| Collision-label Senate rows excluded                 | 8.820***  | (2.010)   | 115,904 |
| <i>Post-1894 triple difference, 1880–1899 window</i> |           |           |         |
| Baseline (collapsed constituency coding)             | -3.238*** | (1.031)   | 208,325 |
| Merged constituencies at component mean              | -3.241*** | (1.030)   | 208,325 |

Constituencies merged by the 1899 law (Senate aggregations and the 30 post-1900 Chamber arrondissements) are assigned in the baseline to a single component district; the correction assigns the mean or the maximum of the component districts' mining exposure. Provincial senators, who have no electoral constituency, carry province-level labels and are excluded from the estimation sample at the build stage. Collision labels (Antwerp, Liège, Namur) name both a province and an arrondissement; the corresponding row removes every post-1894 Senate observation carrying one of the three. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table OA15 reports the post-1900 triple difference with standard errors clustered on the 30 post-1900 constituencies instead of the 41 arrondissements, and with the mining exposure averaged within each merged constituency.

Table OA15: Clustering on the post-1900 constituencies

|                                                     | Estimate | Std. err. | Clusters | Obs.    |
|-----------------------------------------------------|----------|-----------|----------|---------|
| Baseline, clustered on the 41 arrondissements       | 9.253    | (2.351)   | 41       | 119,448 |
| Clustered on the post-1900 constituency             | 9.253    | (2.386)   | 30       | 119,448 |
| The same, exposure averaged within the constituency | 9.276    | (2.395)   | 30       | 119,448 |

Post-1900 triple difference, 1894–1906. The 1899 law merged the 41 arrondissements into 30 constituencies. The second row clusters standard errors at the post-1900 constituency level. The third row also assigns every observation within a merged constituency the mean mining exposure of that constituency.

## OA12 Directly elected against provincially elected senators

From 1894 a part of the Senate was not elected by anyone. The constitutional revision created a class of senators chosen by the provincial councils, exempt from the eligibility census that applied to the rest of the house, and holding the same eight-year mandate. Their seats did not depend on appealing to the voters the reform had just admitted, so it is natural to ask whether they moved at all, and whether they moved less than the senators who did face an electorate.

One correction has to come first. The provincially elected senators are not a subgroup of the control group used in the main text; they are not in the estimation sample at all. Having no constituency, they carry no mining exposure, and the build stage drops every observation whose constituency field records a province rather than a district. The Senate control used throughout this paper is therefore already the directly elected senators—the harder of the two comparisons, since from 1894 those men shared the Chamber’s electorate exactly and were insulated only by the census on who could stand.

Identifying them requires a source the main analysis does not use. The archive’s *details* field records a bare province name, and nothing else, for a senator sitting for a province; for every other senator it is either empty or a free note about a replacement or an office held. The marker first appears in the legislature elected in 1894 and is absent from every earlier one, which is what the institutional history says it should be, so it dates a mandate rather than describing a career. A senator is accordingly classified once per legislature: by his constituency where the archive gives him one, by the province marker where it does not, and not at all where neither source speaks. Fourteen senators sat both ways at different times, and the classification follows them. Fifty-seven sat for a province at some point. Their outcome has to be rebuilt from the classifier predictions, since they are outside the estimation file; the same reconstruction applied to the senators who are inside it reproduces that file exactly.

The comparison exists only at the second reform. There were no provincially elected senators before 1894, so nothing about them can be estimated across the first. Across the second, the two groups behave in opposite ways (Table OA16). Among the directly elected senators labour salience is 13.9 per cent of substantive speech in the majoritarian legislatures and 9.9 in 1900–06. Among the provincially elected it rises from 8.3 per cent in the majoritarian legislatures to 16.0 in 1900–06 and 17.5 in 1908–12.

We read this as a correction to the premise rather than as a result about the Senate. Indirect election removed the electoral connection at one remove, not altogether: the provincial councils were themselves elected bodies, and the same 1893 revision widened the provincial franchise. A senator answerable to a council answerable to the new electorate is further from the voters than a directly elected senator, but he is not outside the circuit. The census exemption pulls the same way from the other side. It made the provincial route the open door of the Senate—it is how the first socialist senators entered—so the men who arrived through it were not drawn from the stratum that could meet the tax threshold, even among Catholics and Liberals. And they begin from a low base, below the directly elected senators in the majoritarian legislatures,

so part of the movement is convergence.

The consequence for the design runs against the intuition that prompted the question. The provincially elected senators are not a cleaner control. They respond more than the directly elected ones, not less, so substituting them would widen the Chamber–Senate difference rather than narrow it. The control this paper uses is the conservative one, and it is conservative for a reason different from the one we would have given before looking.

Table OA16: Labour salience by type of senator

| Senator              | Labour salience, per cent |              |         |         |
|----------------------|---------------------------|--------------|---------|---------|
|                      | Censitary                 | Majoritarian | 1900–06 | 1908–12 |
| Directly elected     | 7.79                      | 13.89        | 9.89    | 13.61   |
| Ambiguous            | 11.25                     | 14.58        | 10.32   | 14.08   |
| Provincially elected | –                         | 8.32         | 15.98   | 17.48   |
| <i>Interventions</i> |                           |              |         |         |
| Directly elected     | 15,896                    | 7,794        | 12,374  | 9,379   |
| Ambiguous            | 1,627                     | 1,410        | 2,765   | 1,322   |
| Provincially elected | 0                         | 760          | 1,808   | 933     |

## OA13 Ministerial speech

Cabinet members defend legislation before the Chamber as part of the office. A Minister of Industry and Labour speaking about labour is therefore a portfolio effect, not a response to constituents, and the estimates should be read knowing how much of them ministerial speech carries. This appendix reports that. Every intervention is flagged as delivered in office if its speaker held a cabinet post on that date, matched to the ministerial lists of the Belgian governments from 1880 to 1914; most ministerial spells in the period are placed in the estimation sample, and those that are not belong either to ministers outside it—War and Foreign Affairs were held by men who were not sitting Catholic or Liberal deputies—or to spells during which the minister did not take the floor. Ministers so defined deliver 15.9 per cent of the recorded words (Table OA17).

Table OA17 re-estimates the three headline objects with that speech removed, and the two margins behave differently.

The 1894 margin is not a ministerial artefact. Excluding speech delivered in office leaves 3.08 (s.e. 1.41) against a baseline of 4.46 (Table OA17), and the coefficient is more precisely estimated on the smaller sample rather than less. Allowing ministers their own post-reform shift instead of dropping them leaves the Chamber–Post interaction at 4.15 (s.e. 1.79) while the minister-by-post term is itself null, 2.46 (s.e. 5.58). Within politician the estimate is 3.74 (s.e. 2.49), against the 2.92 the same specification gives on all speakers (Table 8, column 3).

The 1900 margin is substantially ministerial. The triple difference falls from 9.25 to 4.79

(s.e. 1.43) and the entrant gradient from 11.07 to 4.95 (s.e. 1.42); both are more precisely estimated than the baseline, and both are marginal under randomization inference,  $p = 0.057$  and  $p = 0.104$  over 999 permutations of the exposure map (Table 17). Excluding only the nine holders of the Industry and Labour portfolio, rather than the cabinet as a whole, gives 5.24 (s.e. 1.52), so it is that portfolio and not ministerial status in general that carries the difference.

We read this as part of the mechanism rather than as a confound, and the reading is a claim the reader can reject. Proportional representation returned Catholic and Liberal members to the coalfields and some of those members went on to hold the department that legislated on the social question; the labour portfolio was itself held continuously by men elected from or associated with industrial constituencies after 1900. Recomposition operating partly through who obtained the relevant office is a different statement from recomposition operating through backbench speech, and the paper does not need the second to be true. What it cannot claim is that the coalfield gradient is unaffected by who sat in the cabinet: half of it is.

One institutional caveat cannot be tested and is stated instead. The Office du travail was created in November 1894 and the Ministry of Industry and Labour in 1895, so an institution dedicated to the social question appears at almost exactly the moment the reform takes effect. The two cannot be separated by their dates: the election was held on 14 October 1894, the resulting Chamber's first full session ran into 1895, and there is no post-reform sitting time before the office existed. The office was itself a response to the same pressure that produced the reform, so this is a limit on interpretation rather than a confound that could be differenced away.

Who the Liberal entrants were bears on how the recomposition should be read for a different reason. If the deputies carrying the gradient had entered on joint lists with the socialists, recomposition of the establishment would partly be socialist entry under a Liberal label. The coalfield entrant cell contains four Liberal members. They are Fulgence Masson and Joseph Louis Descamps, both first returned for Mons in 1904; Paul Trasenster, first returned for Liège in 1900; and Émile Bertaux, first returned for Charleroi in 1900. Their shares of the cell's recorded words are 36.5, 22.2, 35.3 and 5.4 per cent (Table OA12). The four are a subset of the seven Liberals first returned in the coalfield districts between 1900 and 1906; the other three—Edmond Dewandre and Émile Buisset at Charleroi in 1904, and Pol-Clovis Boël at Soignies in 1904—leave no recorded speech in the estimation sample.

That the cell holds four men is the reason its cluster-robust  $t$  of 10.9 is not evidence, and a leave-one-member-out says how much is at stake. Dropping Bertaux gives 12.52, Descamps 12.20 and Masson 12.20, against 12.17 for the four together (Table OA12). Dropping Trasenster gives  $-4.46$ . The gradient does not attenuate when he is removed; it changes sign. Section 9 reports the Liberal figure on that basis.

Whether any of the four entered on a joint list with the socialists can be settled from the constituency returns rather than member by member. Kollman et al. (2025) codes a joint list under its own party label, *cartel libéral-socialiste*, so a cartel is visible in the data as such. There is none in any district and year at which one of the four entered. At Liège in 1900 the Liberal list polls 25.3 per cent and the Belgian Workers' Party 46.4; at Charleroi in 1900, 13.8 and 58.0;

at Mons in 1904, 27.3 and 42.7. Two separate lists in each case, and the first cartel label in a coalfield district is Soignies in 1908, after the estimation window closes, followed by Charleroi and Thuin in 1912.

A contemporary source agrees, and it comes from the other side: Bertrand's socialist history records that at the election of 27 May 1900 the Parti ouvrier contested twenty-five arrondissements and allied with the radicals in Ghent and Dendermonde only (Bertrand, 1907, vol. 2, p. 564), which excludes both Liège and Charleroi. The sentence is exhaustive rather than partial because the Quaregnon congress had restricted the federations to alliances with the Parti progressiste alone (Bertrand, 1907, vol. 2, p. 552), so an alliance with any other list is not a possibility the source leaves open; and the returns name the same two districts Bertrand does as the only ones carrying a cartel in 1894, Liège and Namur.

Two limits of this exercise remain. The entry legislature used throughout the paper is built on the parliamentary archive's own person identifier, and one identifier in the coalfield Liberal cell is corrupt: it carries two different names across its rows and resolves to the biographical page of a third member. The row is small—ten interventions and 0.6 per cent of the cell's words—and excluding it is what leaves the four members named above, but a key that can carry three identities is a defect of the source rather than of this cell, and we flag it as such. The other limit is that the entry legislature is recorded on the name string rather than the identifier in the file that produced the estimates. Rebuilding it on the identifier moves the Liberal entrant gradient from 12.46 to 12.17 (Tables OA13 and OA12), which is within the range the leave-one-member-out above already spans, so nothing in this appendix turns on the choice; a full rebuild is left for the replication materials.

Table OA17: Headline estimates with ministerial speech excluded

|                                                                         | Estimate  | Std. err. | Obs.    |
|-------------------------------------------------------------------------|-----------|-----------|---------|
| <i>Chamber <math>\times</math> Post-1894</i>                            |           |           |         |
| Baseline                                                                | 4.455**   | (2.100)   | 208,325 |
| Speech in office excluded                                               | 3.076**   | (1.414)   | 170,729 |
| ... within politician                                                   | 3.735     | (2.490)   | 170,595 |
| Minister $\times$ post controlled                                       | 4.146**   | (1.791)   | 208,325 |
| ... coefficient on Minister $\times$ Post-1894                          | 2.462     | (5.579)   | 208,325 |
| <i>Mining <math>\times</math> Post-1900 <math>\times</math> Chamber</i> |           |           |         |
| Baseline                                                                | 9.253***  | (2.351)   | 119,448 |
| Speech in office excluded                                               | 4.791***  | (1.432)   | 97,037  |
| <i>Entrant mining gradient, 1900 cohorts</i>                            |           |           |         |
| Baseline                                                                | 11.073*** | (1.833)   | 14,182  |
| Speech in office excluded                                               | 4.951***  | (1.417)   | 12,007  |
| Labour portfolio only excluded                                          | 5.238***  | (1.522)   | 12,152  |
| <i>Speech delivered in office</i>                                       |           |           |         |
| Share of recorded words, per cent                                       | 15.9      |           | 60,188  |

An intervention is classified as delivered in office if its speaker held a cabinet post on that date, matched to the ministerial lists of the Belgian governments 1880–1914. Cabinet members defend legislation before the Chamber as part of the office, so their labour speech is a portfolio effect rather than a response to constituents. The within-politician row absorbs arrondissement, party, year and politician effects, the last on the parliamentary archive’s person identifier, as in column 3 of Table 8. The last row gives the share of words in the analysis sample delivered in office, with the number of such interventions in the last column. Standard errors clustered by arrondissement. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## OA14 The entrant gradient by entry cohort

Section 9 reports the entrant mining gradient split into five entry cohorts. Cohorts are assigned on the parliamentary archive’s own person identifier rather than on the name string. The choice is not neutral: 209 of the 957 archive identifiers fall in another cohort under at least one of the name strings they carry (Table OA18), and the pooled post-1900 gradient moves from 11.07 on the name key to 10.54 on the identifier. For a pooled coefficient that is a five per cent shift, but a figure resolved by cohort does not attenuate a misassigned member, it plots him in the wrong column, so the identifier is used throughout here.

Figure OA1 plots the five estimates and Table OA18 reports the two within-regime comparisons that distinguish a rule from an accumulating pressure.

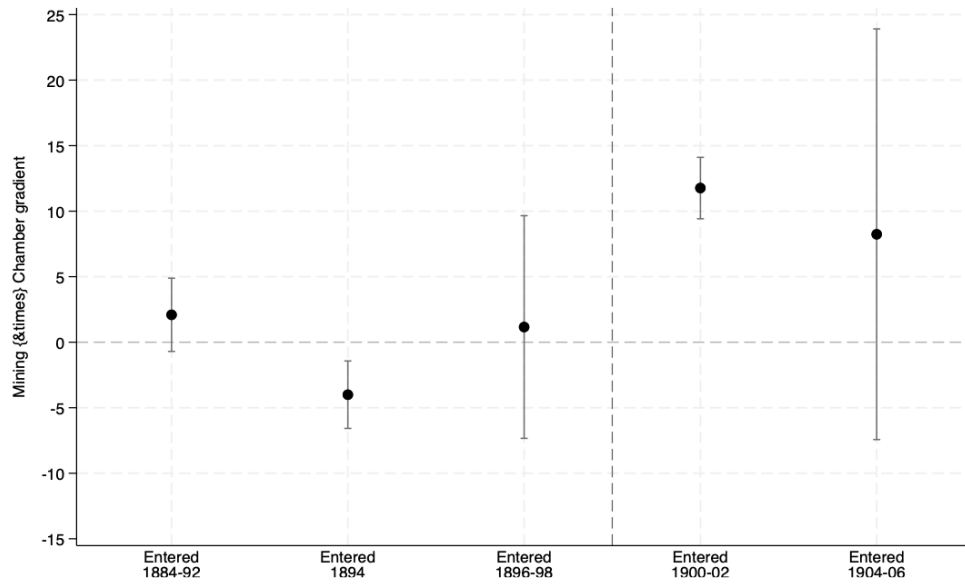


Figure OA1: The entrant mining gradient by entry cohort

*Notes:* Each point is the Mining  $\times$  Chamber gradient estimated on the members whose first legislature falls in the cohort indicated, within the Panel B window, with arrondissement, party and year fixed effects and analytic weights; bars are 95 per cent intervals from standard errors clustered by arrondissement. The vertical rule separates cohorts selected under majoritarian rules from those selected under proportional representation. Entry cohorts are built on the parliamentary archive's person identifier rather than on the name string. The 1896–98 and 1904–06 cells are small, so their intervals are wide and the two within-regime comparisons have limited power; see the text.

Table OA18: Cohort tests: drift within regime, step between regimes

| Entry cohort                                                        | Gradient                  | Std. err. | Obs.   | Members |
|---------------------------------------------------------------------|---------------------------|-----------|--------|---------|
| Before plural suffrage, 1884–92                                     | 2.09                      | (1.43)    | 49,489 | 151     |
| 1894                                                                | -4.00                     | (1.31)    | 14,281 | 67      |
| 1896–98                                                             | 1.16                      | (4.33)    | 8,494  | 62      |
| 1900–02                                                             | 11.77                     | (1.20)    | 15,715 | 101     |
| 1904–06                                                             | 8.24                      | (7.99)    | 1,991  | 55      |
| Equal gradients, 1894 against 1896–98                               | $p = 0.134$               |           |        |         |
| Equal gradients, 1900–02 against 1904–06                            | $p = 0.882$               |           |        |         |
| Step at the reform                                                  | 15.06 (3.23), $p = 0.000$ |           |        |         |
| Members whose first legislature differs between the two person keys | 265 of 957                |           |        |         |
| Members whose entry cohort differs between the two person keys      | 209 of 957                |           |        |         |
| Pooled gradient, entered 1900 or later, archive key                 | 10.54                     |           |        |         |

*Notes:* The upper panel gives the mining gradient of each entry cohort. Below it, the first two rows test equality of the gradient across cohorts selected under the same electoral rule; the third reports the difference between the average proportional and the average majoritarian cohort. An escalating socialist pressure predicts that both equality tests reject and the step is small; a rule that binds on selection predicts that neither rejects and the step is large. The step is reported with its standard error in parentheses. The last two rows give the number of members whose entry cohort differs between the two person keys and the pooled gradient, on the archive key, for members entering in 1900 or later.

Resolving finer than five bins is not possible. Estimated one entry legislature at a time, four of the twelve cohorts between 1884 and 1906 return standard errors above ten and one returns a coefficient of  $-2.8 \times 10^4$ : inside the estimation window a late cohort is observed for one or two legislatures—the 1906 cohort leaves ten recorded interventions per member—and once district, party and year effects are absorbed within a single cohort there is almost nothing left to identify from. The per-legislature estimates are in the replication log for the stage that produces this figure.

## OA15 The share of acts addressing the social question

Section 10 argues from counts of bills deposited rather than from the share of acts addressing the social question, because the share is compositional: it mixes private members' bills, which multiply across the period, with government bills, which do not, and the two carry very different enactment rates. Figure OA2 reports the share series for completeness.

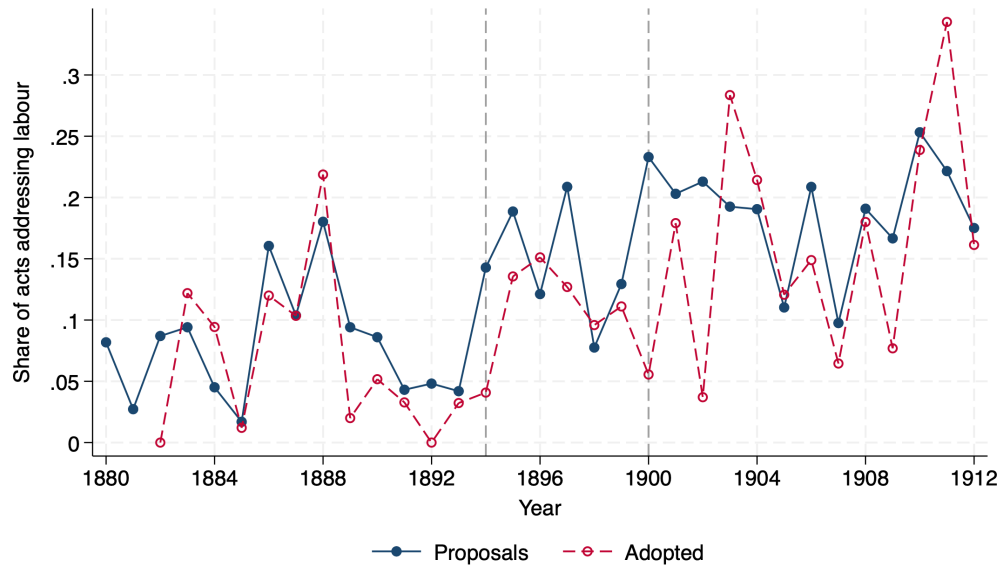


Figure OA2: Share of legislative acts addressing labour and social policy, by year. An act counts if the Genesis archive assigns it the subject tags *Labor & Work*, *Welfare & Social Protection* or *Health*. Proposals are dated by deposit and adopted laws by promulgation; all parties are included, as are bills introduced by the government. The measure is the extensive margin—the share of acts, unweighted—so a long act counts the same as a short one. Vertical lines mark 1894 and 1900. The share mixes private members’ bills with government bills, whose enactment rates differ by a factor of three, which is why Section 10 argues from counts by sponsor rather than from this series.

## The 1886 comparison

The series of members’ labour bills rises after 1886 as well as after 1894. March 1886 saw the Walloon industrial revolt, the largest episode of social unrest in nineteenth-century Belgium, and it was followed directly by the first Belgian social legislation: the 1887 law on the payment of wages and the 1889 law restricting the work of women and children. A rise in labour-related proposals in the years after 1886 is the legislative record doing what the historical literature says it did, and it makes the identification argument of Section 7 concrete. A time-series comparison cannot separate the effect of a reform from the effect of the unrest that produced it: 1886 and 1894 both show movement, and the series alone cannot say which is a response to enfranchisement and which to strikes. This is why the speech analysis does not rely on a time-series comparison, and why Section 10 reports a change in the composition of initiative across the period rather than a jump at a date. The legislative series is corroborative evidence about initiative and is not used for identification.

## OA16 Enfranchisement intensity

Section 11.3 concedes that the 1894 comparison has one treated group, and that a shock common to the whole Chamber and coincident with the reform is not separable from the reform by any variance estimator. The natural remedy is a within-Chamber geography of treatment intensity:

the electorate did not grow by the same factor everywhere, and if the response follows that variation it is not a Chamber-wide shock. This appendix reports that exercise and explains why we do not lean on it.

The measure is the log change in an *arrondissement*’s registered electorate between the elections of 1892 and 1894, taken from the Constituency-Level Elections Archive (Kollman et al., 2025). Two features of the measure matter. It is measured in *votes* rather than electors: under plural suffrage an elector carried one, two or three votes, so the 1894 total of 2.11 million votes corresponds to about 1.37 million electors, while the 1892 total of 136,707 is both. That reading is not an inference. Bertrand (1907, vol. 2, p. 552), writing in 1907, records an 1894 electorate of 1,370,687, of whom 853,628 carried one vote, 293,678 two and 223,381 three; those figures imply 2,111,127 votes, and the archive’s 1894 total is 2,111,127. A contemporary count and a modern compilation agree to the unit, which fixes what the variable measures. And the 1893 revision paired plural suffrage with compulsory voting, so the treatment whose intensity this measures is the package as enacted, which is what the paper claims to estimate throughout.

The measure is not a restatement of mining intensity. Across the 41 *arrondissements* the two are essentially orthogonal, with a correlation of 0.04. What growth does track, mechanically, is how few electors a district had before the reform: its correlation with the log 1892 electorate is  $-0.78$  (Table OA19, notes). Cities already had propertied electors and the countryside had almost none, so Antwerp grows eightfold and Bastogne thirty-twofold.

Table OA19 reports the results, and two of the three are useful. Entered alongside mining intensity, growth carries 1.07 (s.e. 1.47) while the mining interaction is unchanged at  $-3.49$  (s.e. 1.10) against  $-3.24$  on its own: the coalfield reversal of Section 8 is not an artefact of how much the electorate grew in those districts. And on the 1900 window the same measure carries 0.23 (s.e. 1.99), so a predetermined 1894 variable does not predict the margin at which the paper’s triple difference lives.

The third result is the one we do not use. The interaction with the 1894 reform is null— $0.73$  (s.e. 1.61), and  $4.32$  (s.e. 3.02) within politician—but the measure fails its own placebo: moving the cutoff to 1890 on the censitary sample returns 1.76 with a standard error of 0.58 (Table OA19). High-growth districts, which is to say rural ones, were already diverging from urban ones in the Chamber–Senate difference before the reform. A null estimate against a pre-trend of that size is not evidence of no effect, and we make no claim from it in either direction. The table is reported because the exercise is the obvious one to attempt, and a reader who attempts it should find our result rather than reconstruct it.

Table OA19: Enfranchisement intensity at the 1894 margin

|                                                       | Coefficient | (s.e.)  |
|-------------------------------------------------------|-------------|---------|
| Growth $\times$ Post-1894 $\times$ Chamber            | 0.73        | ( 1.61) |
| difference in differences at mean growth              | 4.67        | ( 2.16) |
| politician fixed effects                              | 4.32        | ( 3.02) |
| with mining intensity in the same regression          | 1.07        | ( 1.47) |
| mining intensity, same regression                     | -3.49       | ( 1.10) |
| Placebo, 1890 cutoff, censitary sample                | 1.76        | ( 0.58) |
| Growth $\times$ Post-1900 $\times$ Chamber, 1894–1906 | 0.23        | ( 1.99) |

Growth is the log change in the registered electorate of an arrondissement between the elections of 1892 and 1894, standardised. It is measured in VOTES rather than electors: under plural suffrage an elector carried one, two or three votes, and the 1894 total of 2.11 million votes corresponds to about 1.37 million electors, while the 1892 total of 136,707 is both. Source: CLEA. Across the 41 arrondissements the correlation of growth with mining intensity is 0.04 and with the log of the 1892 electorate -0.78. The politician fixed effects are defined on the parliamentary archive’s person identifier. Panel A window, arrondissement, party and year fixed effects, length weights, clustering by arrondissement. The paper’s estimate at this margin is the difference in differences of Table 8; this table reports its heterogeneity along enfranchisement intensity, the counterpart of the coalfield split in Section 8.

## OA17 Comparison with a dictionary measure

### Why a classifier rather than a dictionary

Attention to a topic in political text has usually been measured with a dictionary: a list of keywords whose frequency stands in for the topic’s salience (Gentzkow et al., 2019; Grimmer & Stewart, 2013). Dictionaries are transparent and cheap, and for many purposes they work well. Here they fail for a reason specific to this corpus. The words that signal the social question in 1890s Belgium—*mine*, *ouvrier*, *industrie*, *travail*—also signal mining concessions, industrial tariffs and public works. A keyword count therefore assigns to labour a good deal of speech about the economy of the coalfields rather than about the people who worked in them, and since our design interacts the outcome with mining intensity, that overlap would write a mechanical relationship between outcome and exposure into the data. Dictionaries also leave out whatever contains none of their keywords, and the share of interventions with any substantive content moves differentially across exactly the chamber-by-period cells the design compares (Appendix A1). A classifier that scores every intervention, procedural material included, avoids both.

Supervised classification into topics is now standard in this literature, first with bag-of-words classifiers trained on hand-coded corpora (Osnabrügge et al., 2023) and increasingly with pretrained transformer language models fine-tuned on a labelled sample of the target corpus,

which reach higher accuracy from far fewer labels (Wang, 2023). We take the second route. The model reads each intervention as a sequence of words in context, so that *mine* in *les accidents de mine* and *mine* in *la concession de la mine* are represented differently, and it learns from labelled examples which patterns of language belong to which domain.

The coverage problem is large. Roughly two-thirds of interventions contain no listed term. Their labour content is zero, so they are not lost information, but they carry no variation, and any specification that drops them selects on the outcome. The vocabulary of the social question also changes across three decades, so a fixed word list mis-measures salience differently at different points in the sample, which is exactly the dimension the design uses. The dictionary measure reported below, on identical rows, agrees with the classifier on the direction of the main result and disagrees sharply on how robust it is; we report the comparison so that a reader can see both.

On identical rows, the post-1900 triple difference is 9.25 ( $t = 3.94$ ) under the classifier and 1.34 ( $t = 1.96$ ) under the dictionary. Leaving out one arrondissement at a time (Table OA20), the classifier estimate remains significant at the 5 per cent level in 40 of 41 exclusions; the dictionary estimate in 16 of 41. Excluding Liège, the classifier gives 3.20 and the dictionary 0.85.

The reason is coverage. In the coalfields, where the Catholic and Liberal presence contracts sharply after 1894 (Table 10), a measure that classifies an intervention only if it contains a listed term is left with too few classified interventions to estimate the gradient precisely. The classifier assigns a probability to every intervention and does not face that constraint.

Table OA20: Leave-one-out, classifier versus dictionary, identical rows

|                              | Classifier       | Dictionary       |
|------------------------------|------------------|------------------|
| Full sample                  | 9.253<br>(2.351) | 1.337<br>(0.681) |
| Excluding Liège              | 3.197<br>(2.543) | 0.851<br>(1.842) |
| Excluding Liège and Mons     | 5.219<br>(1.900) | 1.393<br>(1.984) |
| Exclusions significant at 5% | 40 of 41         | 16 of 41         |
| Range across exclusions      | [3.20, 10.09]    | [0.85, 1.68]     |

Post-1900 triple difference, estimated on identical rows under the two measures. Each is weighted by its own denominator. Standard errors clustered by arrondissement.